

Master's degree in mathematics

Strict univalent universes of an ∞-topos are model independent

Author:
Matteo Francesco Munafò

Supervisor:
Prof. Denis-Charles Cisinski

academic year 2025-26

STRICT UNIVALENT UNIVERSES OF AN ∞-TOPOS ARE MODEL INDEPENDENT

MATTEO FRANCESCO MUNAFÒ

Abstract. In this work we prove that for any ∞-topos the choice of a strict univalent universe for a fixed cardinality coming from a Cisinski model category is essentially unique. More precisely, the groupoid spanned by such universes is a contractible space. Another aim of this work is to simplify Shulman's proof in [Shu19] avoiding simplicial structures and providing a different construction of the universe.

"I was oblivious to the obvious."

Contents

1. Introduction 3
1.1. Context and main results 3
1.2. Organization of the work 3
2. Univalent universes 5
2.1. Homotopy equivalences in model categories 5
2.2. Tribes and clans 14
2.3. Definition of strict univalent object 16
2.4. Embedding of universes 18
3. Shulman's Universe 21
3.1. 2-categorical notions 21
3.2. Construction of the universe 28
3.3. Proof of univalence 35
3.4. Construction of the universe via Giraud's theorem 40
4. Uniqueness of universes up to a contractible space 42
4.1. Calculus of fractions and right approximability 42
4.2. Presentations of ∞-categories 48
4.3. Mapping spaces for model categories 53
4.4. Essential Uniqueness theorem 54
4.5. Generalization to π-tribes 57
4.6. Essential uniqueness of Π and Σ 61
Appendix A. Natural models for homotopy type theory and model indepence for tribes 62
A.1. Natural models for homotopy type theory and tribes 62
A.2. Π and Σ types in tribes 65
A.3. Tribes with universes 67
Acknowledgments 68
References 69

1. Introduction

1.1. **Context and main results.** It is a well known fact that 1-topoi have an internal logic' that is a 'generalized theory of sets'. It is natural then to wonder whether ∞-topoi model some kind of logic in a similar way. Since ∞-categories do not have strict equalities when it comes to compositions and identities we need to find a 'strictification', a 1-category that presents the ∞-topos and that has an 'honest' logic that we can study.
A classic result of higher category theory proves that any locally presentable ∞-category can be presented by a model category. In particular in [Shu19] Michael Shulman proved that any ∞-topos is presented by a model category that models Homotopy type theory. We provide an alternative proof of this fact in Construction 3.57 and Construction 3.4.
A question that has been left unanswered is whether the semantics of HoTT are independent on the model category that interprets it. We propose some results that prove that as long as the choice of the model is reasonable (namely a Cisinski model category or a π-tribe) the strict univalent universes will form a contractible space of choices.
A formal statement we provide is the following:

Theorem 1.1 (Theorem 4.60)**.** Let $\mathcal{T}_\infty$ be an ∞-topos. Let $\mathcal{T}$ be a Cisinski model category. Denote the weak homotopy equivalences of $\mathcal{T}$ with W. Suppose $\mathrm{N}(\mathcal{T})[\mathrm{W}^{-1}] \simeq \mathcal{T}_\infty$. Denote with $\mathrm{W}(\mathrm{Fib}/_X)^\lambda$ the subcategory of $\mathrm{N}(\mathcal{T}/_X)$ spanned by homotopy equivalences between λ-small fibrations. Suppose moreover that $\mathcal{T}$ has a strict univalent universe $\pi : \mathcal{U}_{\lambda\bullet} \twoheadrightarrow \mathcal{U}_\lambda$. Then $\mathcal{T}_\infty(X, \mathcal{U}_\lambda) \simeq ((\mathcal{T}_\infty/_X)^\lambda)^\simeq$, functorially in X.

This result implies, by Yoneda lemma for ∞-categories, the following corollary:

Theorem 1.2 (Theorem 4.61 Essential uniqueness of universes for Cisinski model categories)**.** Let $\mathcal{T}_\infty$ be an ∞-topos. Let $\mathrm{Universes}^\lambda(\mathcal{T}_\infty)$ be the sub-groupoid of $\mathcal{T}_\infty^\simeq$ spanned by strict univalent universes for λ-small fibrations coming from a Cisinksi model category that presents $\mathcal{T}_\infty$. Then $\mathrm{Universes}^\lambda(\mathcal{T}_\infty)$ is a contractible space

What remains to be done is to generalize this result to π-tribes and to prove a similar result for the category of π-tribes that model HoTT and present the given ∞-topos.

1.2. **Organization of the work.** In section 2 we will provide the necessary definitions for the rest of the work. We will define Cisinski model categories (Definition 2.6), study the various notions of homotopy equivalences (Definition 2.18, Definition 2.12), prove when these definitions agree (Lemma 2.25) and when they compute weak equivalences (Lemma 2.30). We will introduce the language of π-tribes (Definition 2.36, Definition 2.38, Definition 2.40) as a natural example that models Homotopy type theory, we will translate some concepts defined for right proper Cisinski model categories in this context and we will define univalent objects there (Definition 2.59). We will define a strict univalent universe for a Cisinski model category (Definition 2.63) and prove that any map between univalent fibrations is a homotopy pullback (Theorem 2.70).
In section 3 we illustrate Shulman's construction of universes (Definition 3.29, Proposition 3.40)for right proper Cisinski model categories with minor deviations from the source material. We will define 'notions of fibered structure' (Definition 3.16) that will be key elements in the proof of Proposition 3.40. We will also provide a second construction of the universe that was not presented in [Shu19] using results from [Gir71] (Construction 3.4).
In section 4 we will illustrate some results to better compute localizations of ∞-categories such as the calculus of fractions (Remark 4.12) or the weak approximability property (Definition 4.15), following [Cis19]. We will use these results to prove that any ∞-topos is presented by a right proper Cisinski's model category (Theorem 4.41), completing the proof of section 3 without using simplicial structures.

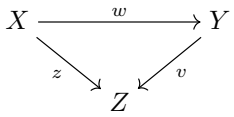
For this task we will mainly follow [GK17]. Lastly we will focus on the proof of the results of uniqueness of universes: Theorem 4.61. As a reference to approach homotopy type theory from a categorical perspective we mostly used [Awo17] which will be elaborated in Appendix A.

2. Univalent universes

2.1. Homotopy equivalences in model categories.

Definition 2.1 ([Cis19], [Rie14])**.** A model category $\mathcal{C}$ is a locally small category endowed with three classes of morphisms Fib (fibrations), Cof (cofibrations) and W (weak equivalences) that satisfy the following conditions:

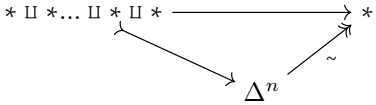
(1) $\mathcal{C}$ has finite limits and colimits
(2) W satisfies the 2-out-of-3 property, ie: for any diagram of the form



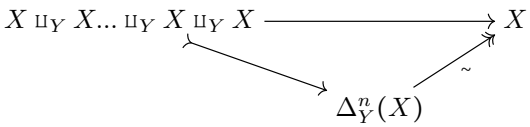
if any two arrows are in W then is the third.
(3) $(\mathrm{Cof} \cap \mathrm{W}, \mathrm{Fib})$ and $(\mathrm{Cof}, \mathrm{W} \cap \mathrm{Fib})$ are weak factorization systems.

We refer to the class $\mathrm{Cof} \cap \mathrm{W}$ as the class of trivial cofibrations and to $\mathrm{W} \cap \mathrm{Fib}$ as the class of trivial fibrations.

Definition 2.2. Let $\mathcal{C}$ be a model category. An n-simplex in $\mathcal{C}$ is a factorization of the map $(id, ..., id) : * \sqcup *... \sqcup * \sqcup * \to *$ of $n+1$ coproducts of the terminal object into a cofibration followed by a trivial fibration:



Let us consider $X \to Y$, a morphism in $\mathcal{C}$. We can analogously define a simplex object on X relatively to Y as a factorization of the following form:



where the domain is the $(n+1)$th iterated pushout of f along itself, the first map is a cofibration and the last map is a trivial fibration. An interval object is defined to be a 1-simplex.

Notation 2.3. Let A be a class of morphisms in a category $\mathcal{C}$. We denote with $r(A)$ the class of maps with the right lifting property against A. We denote with $l(A)$ the class of maps with the left lifting property against A.

Definition 2.4 ([nLa26a])**.** A combinatorial model category $\mathcal{C}$ is a locally presentable model category such that:

(1) there exists a small set of morphisms A such that $l(r(A))$ is the class of cofibrations of $\mathcal{C}$
(2) there exists a small set J of morphisms such that $r(J)$ is the class of fibrations of $\mathcal{C}$

Remark 2.5 ([nLa26a] 3.1 (Jeff Smith's theorem))**.** In a combinatorial model category the class of trivial cofibrations is closed under transfinite composition and pushout.

Definition 2.6. A Cisinski model category is a a Grothendieck topos together with a combinatorial model structure whose cofibrations coincide with the monomorphisms.

Example 2.7. [Cis14] If $\mathcal{E}$ is a Cisinski model category we can choose the subobject classifier Ω as an interval object.

Proof. For any X in $\mathcal{E}$ we claim that the projection $X \times \Omega \to X$ is a trivial fibration. For any lifting problem of the form:

$$\begin{array}{ccccc} Y & & \longrightarrow & & \Omega \times X & \longrightarrow & \Omega \\ {\scriptstyle i}\downarrow & & & & \downarrow{\scriptstyle pr} & & \\ Z & & \xrightarrow[q]{} & & X & & \end{array}$$

where i is a cofibration. If we consider the induced map $Y \to \Omega$ we obtain the following pullback square:

$$\begin{array}{ccc} W & \longrightarrow & * \\ \downarrow & \lrcorner & \downarrow{\scriptstyle T} \\ Y & \xrightarrow{u} & \Omega \end{array}$$

By definition of sub-object classifier we have a pullback square of the form:

$$\begin{array}{ccc} W & \longrightarrow & * \\ \downarrow & \lrcorner & \\ Y & & \downarrow{\scriptstyle T} \\ \downarrow & & \\ Z & \xrightarrow[\chi_W]{} & \Omega \end{array}$$

which allows us to construct a lift for the problem we started with:

$$\begin{array}{ccc} Y & \longrightarrow & \Omega \times X \\ \downarrow & \nearrow_{(\chi_W, q)} & \downarrow{\scriptstyle pr} \\ Z & \xrightarrow[q]{} & X \end{array}$$

The top triangle commutes because of the following diagram:

$$\begin{array}{ccc} W & \longrightarrow & * \\ \downarrow & \lrcorner & \\ Y & & \downarrow \\ \downarrow & \searrow & \\ Z & \xrightarrow[\chi_W]{} & \Omega \end{array}$$

We define the map 0 through the definition of sub-object classifier as

$$\begin{array}{ccc} \varnothing & \longrightarrow & * \\ \downarrow & \lrcorner & \downarrow{\scriptstyle t} \\ * & \xrightarrow[0]{} & \Omega \end{array}$$

This defines a map $\partial\Omega := \{0,1\} = * \sqcup * \to \Omega$ via:

$$\begin{array}{ccc} \partial\Omega & \longleftarrow & * \\ \uparrow & \searrow & \downarrow{\scriptstyle T} \\ * & \xrightarrow[0]{} & \Omega \end{array}$$

We rename $1 := T : * \to \Omega$. Notice that $pr : X \times \Omega \to X$ has two sections, $(id_X, 1)$ and $(id_X, 0)$, which are also weak equivalences by 2-out-of-3. Now take $X = *$ to conclude. □

Definition 2.8. Let $s : Y \hookrightarrow X$ a map with a retraction $f : X \to Y$, we say $Y \hookrightarrow X$ is a deformation retract of X if there exists a map $H : X \times \Omega \to X$ such that for $i \in \{0, 1\}$

$$\begin{array}{ccc} \Omega \times X & \xrightarrow{H} & Y \\ {\scriptstyle i}\uparrow & \nearrow_{H_i} & \\ * & & \end{array}$$

satisfies $H_1 = s \circ f$ and $H_0 = id$.

Remark 2.9. In a Cisinski model category, $\mathcal{E}$, we can use a small object argument to construct a functorial factorization into cofibrations and trivial fibrations. This implies that we can define a functor:

$$\mathbb{\Delta} \to \mathcal{E}$$

given by the following assignment: $[n] \mapsto \Delta^n$, with Δ^n is given by the functorial factorization.

Definition 2.10 ([Cis02])**.** Let $\mathcal{E}$ be Cisinski model category with an interval object $\mathcal{I}$. Let $f_0, f_1 : X \to Y$ be two morphisms in $\mathcal{C}$. We say that f_0 and f_1 are $\mathcal{I}$-homotopic if and only if there exists an $H : X \times \mathcal{I} \to Y$ such that $H \circ (id_X \times \partial\mathcal{I}) : X \sqcup X \hookrightarrow X \times \mathcal{I} \to Y$ is equal to (f_0, f_1). We write $f_0 \sim_{\mathcal{I}} f_1$

Discussion 2.11. Being $\mathcal{I}$-homotopic is oftentimes not an equivalence relation however, there are way to define notions of homotopy that are equivalence relations and that coincide with the one of Definition 2.10 in most cases that we will encounter.

Definition 2.12. Let $\mathcal{C}$ be a model category. Let X be an object in $\mathcal{C}$. We define a path object for X as any object X^{Δ^1} such that there exists a factorization of $\Delta : X \to X \times X$ of the form:

$$X \xrightarrow{\sim} X^{\Delta^1} \twoheadrightarrow X \times X$$

where the first map is a trivial cofibration and the second is a fibration.

Example 2.13. Let us assume that $\mathcal{E}$ is a Cisinski model category. Since $\mathcal{E}$ is cartesian closed we can define an internal Hom-object $\mathcal{E}(-,-)$. Let us consider an interval object $\mathcal{I}$ (for example $\Omega_{\mathcal{E}}$). Then $\mathcal{E}(\mathcal{I}, X)$ is a path object for any fibrant object X.

Proof. We know by functoriality of $\mathcal{E}(\mathcal{I}, -)$ that the following diagram is commutative:

$$\begin{array}{ccc} X & \xrightarrow{\Delta} & X \times X \\ & {\scriptstyle id}\searrow \quad \nearrow{\scriptstyle (\partial_0, \partial_1)_*} & \\ & \mathcal{E}(\mathcal{I}, X) & \end{array}$$

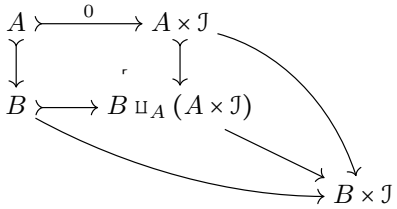
We are left to prove that any $(\partial_i)_*$ is a trivial fibration for $i = 0, 1$. If that is the case we conclude by 2-out-of-3 and the fact that any morphism that has a retraction is a monomorphism. Let us consider any diagram of the following form:

$$\begin{array}{ccc} A & \longrightarrow & \mathcal{E}(\mathcal{I}, X) \\ \downarrow & & \downarrow{\scriptstyle \partial_0} \\ B & \longrightarrow & X \end{array}$$

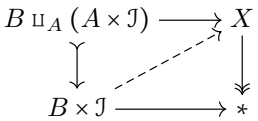
We form the intersection in $B \times \mathcal{I}$ of B and $A \times \mathcal{I}$:

$$\begin{array}{ccc} A & \xrightarrow{0} & A \times \mathcal{I} \\ \downarrow & \lrcorner & \downarrow \\ B & \xrightarrow[0]{} & B \times \mathcal{I} \end{array}$$

If we consider the union of $A \times \mathcal{I}$ and B in $B \times \mathcal{I}$ we notice that its inclusion is a trivial cofibration (trivial cofibrations are stable under pushouts, topoi are adhesive and we conclude by 2-out-of-3):

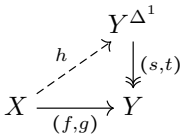


We therefore deduce that there is a solution to the following lifting problem:



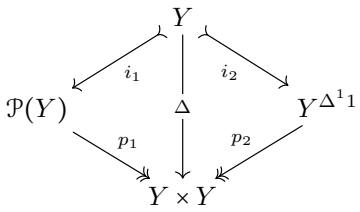
Since, by construction, this solution agrees with $A \times \mathcal{I}$ and $0 : B \to B \times \mathcal{I}$, under the adjunction $- \times \mathcal{I} \dashv \mathcal{E}(\mathcal{I}, -)$ it corresponds to a solution of the starting lifting problem. □

Definition 2.14. Let $\mathcal{C}$ be a model category. Let $f, g : X \to Y$ be two morphisms in $\mathcal{C}$ we say that f is right-homotopic to g through $h : X \to Y^{\Delta^1}$ (and we write $h : f \sim_{\mathrm{R}} g$) if the following diagram commutes:

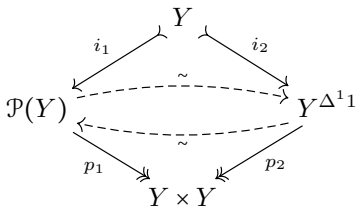


Proposition 2.15. Let $\mathcal{C}$ be a model category. Let $\mathcal{P}_Y, Y^{\Delta^1}$ be two path objects of an object Y. Then there exist comparison maps between $\mathcal{P} \to Y^{\Delta^1}$ that are weak equivalences on $Y \times Y$.

Proof. The following diagram is granted by the definition of path object:



Since both i_1, i_2 are trivial cofibrations and p_1, p_2 are fibrations we can complete as follows:



Notice also that since i_1 and i_2 are weak equivalences, by 2-out-of-3, so are the dotted maps. □

Remark 2.16. Proposition 2.15 also implies that being right-homotopic does not depend on the path object chosen.

Discussion 2.17. In a model category there is also a factorization into a cofibration followed by a trivial fibration We can provide a dual definition of path object (that has already been partially given in Definition 2.2). This also another yields a definition of homotopy between maps. Fortunately these two definitions coincide under good hypotheses.

Definition 2.18. Let $\mathcal{C}$ be a model category. Let X be an object of $\mathcal{C}$. A cylinder object for X is a factorization of the map $\nabla : X \amalg X \to X$ as follows:

$$X \amalg X \xrightarrowtail[(\partial_0,\partial_1)]{} \mathcal{I}(X) \xrightarrow[\pi]{} X$$

where $\partial = (\partial_0, \partial_1)$ is a cofibration and π is a trivial fibration.

Example 2.19. Let $\mathcal{E}$ be a Cisinski model category. Let X be an object of $\mathcal{E}$. Then for any interval object $\mathcal{I}$ $X \times \mathcal{I}$ is a cylinder object.

Proof. The following diagram proves that pr_2 is a weak equivalence:

$$\begin{array}{ccc} X \amalg X & & \\ {\scriptstyle (\partial_0 \times id, \partial_1 \times id)}\downarrow & \searrow^{\nabla} & \\ \mathcal{I} \times X & \xrightarrow[pr_2]{} & X \\ \downarrow & \lrcorner & \downarrow \\ \mathcal{I} & \xrightarrow{\sim} & * \end{array}$$

∂ is a cofibration because it is the composition of monomorphisms ($\partial_i \times id$ have a retraction for $i = 0, 1$). □

Definition 2.20. Let $\mathcal{C}$ be a model category. Let $f, g : X \to Y$ be two morphisms in $\mathcal{C}$. We say that $f \sim_{\mathrm{L}} g$ if (f, g) factors through a cylinder object $\mathcal{I}(X)$ as follows:

$$\begin{array}{ccc} X \amalg X & \xrightarrow{(f,g)} & Y \\ {\scriptstyle (\partial_0 \times id, \partial_1 \times id)}\downarrow & \nearrow_{H} & \\ \mathcal{I}(X) & & \end{array}$$

Remark 2.21. Dually to the proof for path objects, the notion of right-homotopy does not depend on the cylinder object.

Remark 2.22. In a Cisinski model category, by Example 2.19, this definition is equivalent to the definition of $\mathcal{I}$-homotopy for any interval object $\mathcal{I}$.

Proposition 2.23. Let $\mathcal{C}$ be a model category. Let Y be a fibrant object. Let X be a cofibrant object. Let $f : X \to Y$ be two morphisms in $\mathcal{C}$. Then $f \sim_{\mathrm{L}} g$ if and only if $f \sim_{\mathrm{R}} g$.

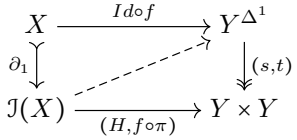
Proof. Since the statement is self dual we suppose $f \sim_{\mathrm{L}} g$:

$$\begin{array}{ccc} X \amalg X & \xrightarrow{(f,g)} & Y \\ {\scriptstyle (\partial_0 \times id, \partial_1 \times id)}\downarrow & \nearrow_{H} & \\ \mathcal{I}(X) & & \end{array}$$

Since X is cofibrant, the following diagram proves that ∂_1 is a cofibration:

$$\begin{array}{ccccc} \varnothing & \rightarrowtail & X & = & X \\ \downarrow & \ulcorner & \downarrow & & \downarrow{\scriptstyle \partial_1} \\ X & \rightarrowtail & X \amalg X & \rightarrowtail & \mathcal{I}(X) \end{array}$$

By 2-out-of-3 it is also a weak equivalence. Let us now consider the following lifting problem:



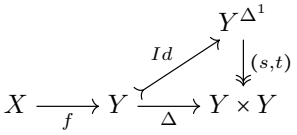
It has a solution since (s,t) is a fibration and ∂_1 is a trivial cofibration. □

Remark 2.24. From previous proof it follows that in a Cisinski model category any object is cofibrant therefore any left-homotopic maps are also right homotopic.

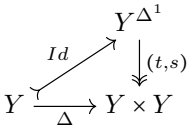
Lemma 2.25. Let $\mathcal{E}$ be a Cisinski model category. Let Y be a fibrant object. Let $f, g : X \to Y$ be any morphism in $\mathcal{E}$. Then, for any interval object $\mathcal{I}$, being $\mathcal{I}$-homotopic is an equivalence relation on $\mathrm{Hom}_{\mathcal{E}}(X, Y)$. Moreover, the following are equivalent:

(1) f is right homotopic to g
(2) f is left homotopic to g
(3) f is $\mathcal{I}$-homotopic to g

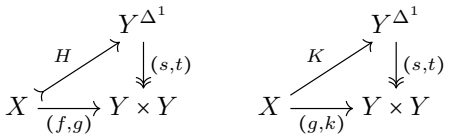
Proof. All objects in $\mathcal{E}$ are cofibrant. The equivalence between (1) and (2) has already been proven in Proposition 2.23. (2) is equivalent to (3) by Example 2.19. Now we prove that being right homotopic is an equivalence relation. Reflexivity follows by definition:



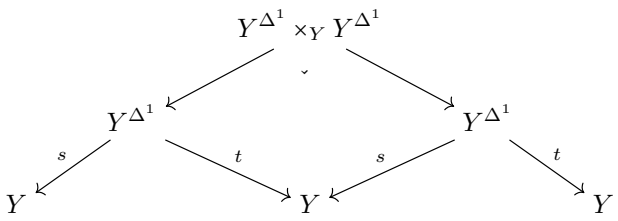
Regarding symmetry the following is again a path object:



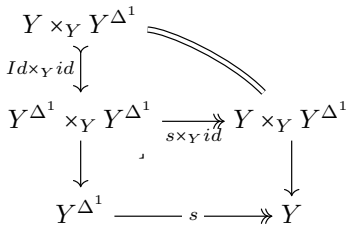
But Proposition 2.15 shows that being right homotopic does not depend on the path object. For transitivity we resort again to Proposition 2.15 to reduce to the following case:



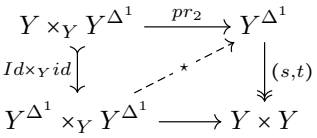
Let us consider the following pullback:



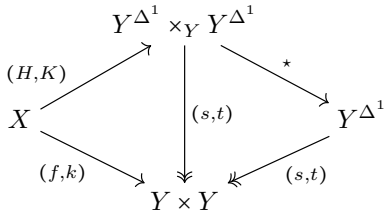
The following diagram proves that $Id \times_Y id$ is a trivial cofibration since s is a trivial fibration:



Therefore the following lifting problem has a solution:



We can construct the following diagram that provides a right homotopy from f to k



□

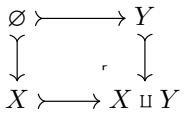
Remark 2.26. Notice that in the previous proof we proved that being right homotopic is an equivalence relation without any assumption on domain or codomain. Dually being left-homotopic is an equivalence relation. This provides us with the following definition:

Definition 2.27 ([Cis02] 2.2.14)**.** Let $\mathcal{C}$ be a model category. Let X, Y be two objects in $\mathcal{C}$. We denote with $[X,Y] := \mathcal{C}(X,Y)/\sim_{\mathrm{L}}$. We can consider the category $\mathrm{h}(\mathcal{C})$ whose objects are the objects of $\mathcal{C}$ and whose Hom-sets are $[-,-]$. We clearly have a canonical functor $Q : \mathcal{C} \to \mathrm{h}(\mathcal{C})$. A morphism f in $\mathcal{C}$ is a homotopy equivalence if $Q(f)$ is an isomorphism in $\mathrm{h}(\mathcal{C})$

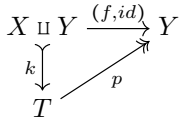
Discussion 2.28. We adapt the next lemma to our context to make the theory more concise even though we cite the original result that inspired this version.

Lemma 2.29 ([Cis19] 2.2.7 (Ken Brown's lemma))**.** Let $\mathcal{C}$ be a model category. Let $F : \mathcal{C} \to \mathcal{D}$ be a functor. Assume that $\mathcal{D}$ is endowed with a class of weak equivalences that contains isomorphisms and has the 2-out-of-3 property. If F sends trivial cofibrations between cofibrant objects to weak equivalences then F sends weak equivalences between cofibrant objects to weak equivalences.

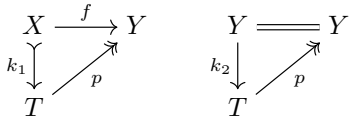
Proof. Let $f : X \to Y$ be a weak equivalence between cofibrant objects. We may form the following pushout:



Since X and Y are cofibrant, the maps into the coproduct are cofibrations. Let us consider the following factorization of (f, id) into a cofibration followed by a trivial fibration:



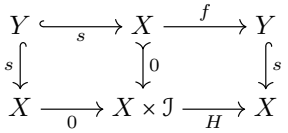
This produces the following commutative triangles:



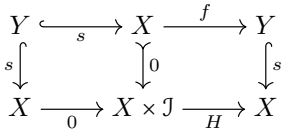
F sends k_2 to a weak equivalence by hypothesis. We deduce that $F(p)$ is a weak equivalence by 2-out-of-3. F sends k_i to a weak equivalence for the same reason as for k_2. By 2-out-of-3 $F(f)$ is a weak equivalence. □

Lemma 2.30 ([Cis02] 2.24)**.** Let $\mathcal{E}$ be a Cisinski model category with interval object $\mathfrak{I}$. Let X and Y be two fibrant objects. Let $s : X \to Y$ a morphism in $\mathcal{E}$. Then f is a weak equivalence if and only if f is an $\mathfrak{I}$-homotopy equivalence.

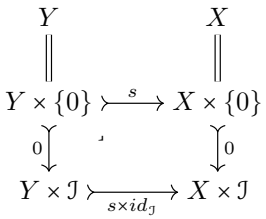
Proof. Let us consider $\mathcal{E}_f$ the category of fibrant objects of $\mathcal{E}$. We say that a morphism $s : Y \hookrightarrow X$ is a deformation retract if and only if there exists an $f : X \to Y$ and an $H : X \times \mathfrak{I} \to X$ such that $f \circ s = id_Y$ and H is an $\mathfrak{I}$-homotopy from $s \circ f$ and id_X. As a diagram:



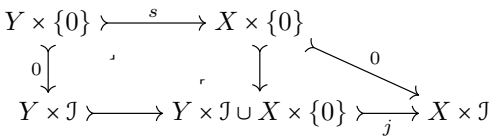
We prove the following claim: If $Y \to X$ is a deformation retract then it is a trivial cofibration.
Proof of Claim 1: the following diagram exhibits $s : Y \hookrightarrow X$ as a retract of a trivial cofibration:



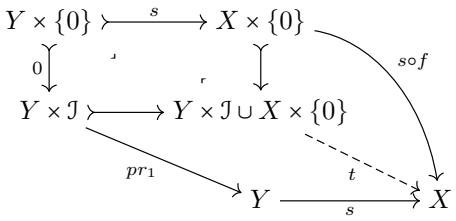
Secondly we claim that a trivial cofibration between fibrant objects is a deformation retract.
Proof of Claim 2: s has a retraction since the following lifting problem has a solution. Let us consider the following pullback:



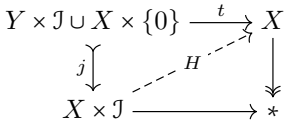
Since trivial cofibrations are stable under pushouts, and 1-topoi are adhesive, we deduce that the induced map j in the following diagram is a cofibration:



it also is a weak equivalence by 2-out-of-3. We can also construct a map t as in the following diagram through the universal property of pushouts:



Let us now consider the following lifting problem:



Which has a solution by fibrancy of X. The obtained $\mathcal{I}$-homotopy exhibits s as deformation retract by construction. Let us now consider $id_{\mathcal{E}_f} : \mathcal{E}_f \to \mathcal{E}_f$.By the previous claims we know that it sends trivial cofibrations between cofibrant objects of $\mathcal{E}_f$ to $\mathcal{I}$-homotopy equivalences between cofibrant objects of $\mathcal{E}_f$ (in a Cisinski model category every object is cofibrant). By virtue of Lemma 2.29 we deduce that it send weak equivalences to $\mathcal{I}$-homotopy equivalences. It is trivial to see that $\mathcal{I}$-homotopy equivalences are weak equivalences. □

Remark 2.31. We provided a proof in the context of Cisinski model categories because we believe that this proof reduces the computations. We could generalize it by restricting to the case when the X is cofibrant and considering a cylinder object instead of $X \times \mathcal{I}$. Another generalization resides in endowing Set with a class of weak equivalences given by bijections and proving that $[-, X]$ sends trivial cofibrations between cofibrant objects to bijections. The interesting computations remain the same.

Discussion 2.32. We adapt the following lemma to the case of Cisinski model categories for two main reasons: we illustrated Lemma 2.30 in the case of Cisinski model categories; to facilitate the proof, we use the fact that Cisinski model categories have a functorial factorization (which will be a core result in the next section). Since thanks to Theorem 4.41 all the model categories that we will use will be Cisinski model categories this reduction will not impact any further result.

Lemma 2.33 ([Cis19] 2.2.17)**.** Let $\mathcal{E}$ be Cisinski model category with an interval object $\mathcal{I}$. Let X be any object in $\mathcal{E}$ and let Y be fibrant. Then $[X, Y] \cong \mathrm{Hom}_{\mathrm{ho}(\mathcal{E})}(X, Y)$.

Proof. If we restrict to $\mathrm{ho}(\mathcal{E}_f)$ we know that isomorphisms in $\mathrm{ho}(\mathcal{E}_f)$ coincide with isomorphisms in $\mathrm{h}(\mathcal{E}_f)$ by Lemma 2.30. Since both are localizations of $\mathcal{E}_f$ they must be equivalent as categories. Lastly $\mathrm{ho}(\mathcal{E}) \simeq \mathrm{ho}(\mathcal{E}_f)$. Since the small object argument gives us a functorial factorization on Cisinski model categories we have a well defined functor that sends any object to its functorial fibrant replacement. Let us denote this factorization with R. For any X, when we apply R we obtain a trivial cofibration followed by a fibration:

$$X \xrightarrow{\sim} R(X) \twoheadrightarrow *$$

The trivial cofibration is an isomoprhism in $\mathrm{ho}(\mathcal{E})$, so $\mathrm{Hom}_{\mathrm{ho}(\mathcal{E})}(X, Y) \cong \mathrm{Hom}_{\mathrm{ho}(\mathcal{E})}(R(X), R(Y)) = \mathrm{Hom}_{\mathrm{ho}(\mathcal{E}_f)}(R(X), R(Y))$ and R is essentially surjective. □

Remark 2.34. For a more complete proof see [Qui67](Chap. I, 1.13, Theorem 1).

2.2. **Tribes and clans.**

Discussion 2.35. We will illustrate what a univalent universe is in the language of tribes, a more generalized framework for homotopy type theory, better suited for comprehensive definition of strict univalent universes. This decision has been motivated by reading [AP26].

Definition 2.36 ([Joy17] Definition 1.1.1)**.** Let $\mathcal{C}$ be a category with terminal object $*$. A clan structure Fib on $\mathcal{C}$ is a class of maps (called fibrations) in $\mathcal{C}$ that satisfies the following properties:

(1) Pullbacks along fibrations exist.
(2) Fib contains all isomorphisms
(3) Fib is stable under composition and base change
(4) Fib contains all maps $X \to *$ for any X in $\mathcal{C}$

A clan is a category $\mathcal{C}$ with terminal object and a fixed clan structure.

Definition 2.37 ([Joy17] 3.1.1)**.** Let $\mathcal{C}$ be a clan. An anodyne map in $\mathcal{C}$ is a morphism i in $\mathcal{C}$ that has the left lifting property with respect to any fibration.

Definition 2.38 ([Joy17] 3.1.6)**.** Let $\mathcal{C}$ be a clan. We say that $\mathcal{C}$ is a tribe if it satisfies the following properties:

(1) any map can be factored into an anodyne map followed by a fibration
(2) anodyne maps are stable under base change along fibrations

Example 2.39. Let $\mathcal{E}$ be a right proper Cisinski model category. Let us denote with $\mathcal{E}_f$ the full subcategory of fibrant objects of $\mathcal{E}$. Then $\mathcal{E}_f$ is a tribe with the clan structure given by the fibrations of $\mathcal{E}$. We can observe that anodyne maps are trivial cofibrations.

Definition 2.40. A tribe $\mathcal{C}$ is a π-tribe if, for any fibration $p : X \to Y$ in $\mathcal{C}$, $p^* : \mathcal{C}/_Y \to \mathcal{C}/_X$ has a right adjoint p_* which preserves fibrations.

Remark 2.41. Any π-tribe $\mathcal{C}$ is cartesian closed since $- \times X$ is just the base change along $X \to *$. We denote its internal Hom-functor with $\mathcal{C}(-,-)$.

Example 2.42. The tribe defined in Example 2.39 is also a π-tribe.

Definition 2.43. Let $\mathcal{C}$ be a tribe. Let X be an object in $\mathcal{C}$. We define a path object for X as any object X^{Δ^1} such that there exists a factorization of $\Delta : X \to X \times X$ of the form:

$$X \xrightarrow{\sim} X^{\Delta^1} \twoheadrightarrow X \times X$$

where the first map is an anodyne map and and the second is a fibration.

Definition 2.44. Let $\mathcal{C}$ be a tribe. Let $f, g : X \to Y$ be two morphisms in $\mathcal{C}$. We say that f is homotopic to g through $h : X \to Y^{\Delta^1}$ (and we write $h : f \sim_{\mathrm{R}} g$) if the following diagram commutes:

$$\begin{array}{ccc} & & Y^{\Delta^1} \\ & \overset{h}{\nearrow} & \downarrow{\scriptstyle (s,t)} \\ X & \xrightarrow[(f,g)]{} & Y \end{array}$$

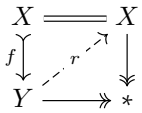
This defines an equivalence relation on any $\mathrm{Hom}_{\mathcal{C}}(-,-)$. We define $\mathrm{h}(\mathcal{C})$ to be the category with objects the objects of $\mathcal{C}$ and Hom-sets $[-,-] := \mathrm{Hom}_{\mathcal{C}}(-,-)/_{\sim_{\mathrm{R}}}$ with $\sim_{\mathrm{R}}$ the equivalence relation defined above. We say that f is a homotopy equivalence in $\mathcal{C}$ if the canonical functor $\mathcal{C} \to \mathrm{h}(\mathcal{C})$ sends f to an isomorphism.

Remark 2.45. Let $\mathcal{C}$ be a tribe. If we define W to be the class of homotopy equivalence, then $\mathcal{C}[\mathrm{W}^{-1}] \simeq \mathrm{h}(\mathcal{C})$.

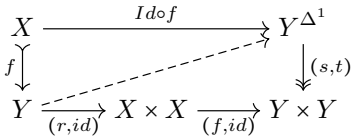
Remark 2.46. By Example 2.13, for a right proper Cisinski model category $\mathcal{E}$ and for any interval object $\mathbb{I}$, two maps in $\mathcal{E}_f$ are homotopic (in the sense of tribes) if and only if they are $\mathbb{I}$-homotopic in the sense of Definition 2.10 .

Lemma 2.47. Let $\mathcal{C}$ be a tribe. Then anodyne maps are homotopy equivalences

Proof. Let $f : X \to Y$ be an anodyne map. We can consider the following lifting problem:



Since f is anodyne it has a solution r. Now we can consider another lifting problem that provides a homotopy between $f \circ r$ and id_Y:



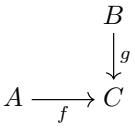
□

Proposition 2.48. Let $\mathcal{C}$ be a tribe. Let $\mathcal{P}_Y, Y^{\Delta^1}$ be two path objects of Y. Then there exist comparison maps between $\mathcal{P}(Y)$ and Y^{Δ^1} which are homotopy equivalences on $Y \times Y$.

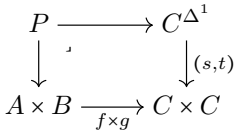
Proof. We construct comparison maps as in the proof of Proposition 2.15. Notice also that since i_1 and i_2 are anodyne maps the dotted maps are homotopy equivalences. □

Remark 2.49. The previous proposition also proves that the relation of being homotopic does not depend on the choice of path object: the comparison maps commute with the path object structure.

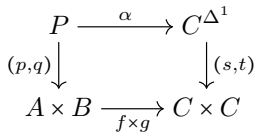
Definition 2.50. Let $\mathcal{C}$ be a tribe. Let us consider the following diagram in $\mathcal{C}$:



We define its homotopy pullback P as the following cartesian square (depending on the choice of path object of $\mathcal{C}$):



Definition 2.51. Let $\mathcal{C}$ be a tribe. A homotopy square is the datum of a commutative square



2.3. **Definition of strict univalent object.**

Construction 2.52. Let $\mathcal{C}$ be a π-tribe. Let X, Y be two objects in $\mathcal{C}$. Following the notation of the previous example we define Rinv as the following pullback:

$$\begin{array}{ccc} \mathrm{Rinv}(X,Y) & \longrightarrow & \mathcal{C}(Y, Y^{\Delta^1}) \\ \downarrow & \lrcorner & \downarrow{\scriptstyle ((pr_1)_*,(pr_2)_*)} \\ \mathcal{C}(X,Y)\times\mathcal{C}(Y,X) & \xrightarrow[(\circ,id)]{} & \mathcal{C}(Y,Y)\times\mathcal{C}(Y,Y) \end{array}$$

Analogously for Linv:

$$\begin{array}{ccc} \mathrm{Linv}(X,Y) & \longrightarrow & \mathcal{C}(X, X^{\Delta^1}) \\ \downarrow & \lrcorner & \downarrow{\scriptstyle ((pr_1)_*,(pr_2)_*)} \\ \mathcal{C}(X,Y)\times\mathcal{C}(Y,X) & \xrightarrow[(\circ,id)]{} & \mathcal{C}(X,X)\times\mathcal{C}(X,X) \end{array}$$

Definition 2.53. Let $\mathcal{C}$ be π-tribe. Let X, Y be two objects in $\mathcal{C}$. We define $\mathrm{Equiv}(X,Y)$ as the following pullback:

$$\begin{array}{ccc} \mathrm{Equiv}(X,Y) & \twoheadrightarrow & \mathrm{Rinv}(X,Y) \\ \downarrow & \lrcorner & \downarrow \\ \mathrm{Linv}(X,Y) & \twoheadrightarrow & \mathcal{C}(X,Y) \end{array}$$

Remark 2.54. Since all π-tribes are cartesian closed we can define $\mathrm{Equiv}_Z(X,Y)$ as $\mathrm{Equiv}(X\times Z, Y\times Z)$ in $\mathcal{C}/_Z$.

Lemma 2.55. Let $\mathcal{C}$ be a π-tribe. Let $X, Y.Z, W$ be objects of $\mathcal{C}$. Let us denote with $\mathcal{C}_Z(-,-)$ the internal Hom-set of $\mathcal{C}/_Z$. The,

$$\mathcal{C}_W(X,Y)\times_W Z \cong \mathcal{C}_Z(X\times_W Z, Y\times_W Z)$$

Proof. Without loss of generality we can assume $W = *$. Let $f : V \to Z$ be an object of $\mathcal{C}/_Z$. We have the following pullback square:

$$\begin{array}{ccc} \mathrm{Hom}_{\mathcal{C}/_Z}(V, \mathcal{C}_Z(X\times Z, Y\times Z)) & & \\ \| & & \\ \mathrm{Hom}_{\mathcal{C}/_Z}(V\times_Z (X\times Z), Y\times Z) & \longrightarrow & \mathrm{Hom}_{\mathcal{C}}(V\times_Z (X\times Z), Y\times Z) \\ \downarrow & \lrcorner & \downarrow \\ * & \longrightarrow & \mathrm{Hom}_{\mathcal{C}}(V\times_Z (X\times Z), Z) \end{array}$$

We can replace the left vertical map as follows:

$$\begin{array}{ccc} \mathrm{Hom}_{\mathcal{C}}(V\times_Z (X\times Z), Y\times Z) & = & \mathrm{Hom}_{\mathcal{C}}(V\times_Z (X\times Z), Y)\times\mathrm{Hom}_{\mathcal{C}}(V\times_Z (X\times Z), Z) \\ \downarrow & & \swarrow_{pr_2} \\ \mathrm{Hom}_{\mathcal{C}}(V\times_Z (X\times Z), Z) & & \end{array}$$

Notice that the following pullback squares provide a description of $V\times_Z (X\times Z)$ as $V\times X$:

$$\begin{array}{ccccc} V\times_Z (X\times Z) & \longrightarrow & X\times Z & \longrightarrow & X \\ \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow \\ V & \longrightarrow & Z & \longrightarrow & * \end{array}$$

So we can consider the following pasting of pullback squares:

$$\begin{array}{ccccc} \mathrm{Hom}_{\mathcal{C}/Z}(V\times X, Y\times Z) & \longrightarrow & \mathrm{Hom}_{\mathcal{C}}(V\times X, Y\times Z) & \longrightarrow & \mathrm{Hom}_{\mathcal{C}}(V\times X, Y) \\ \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow \\ * & \longrightarrow & \mathrm{Hom}_{\mathcal{C}}(V\times X, Z) & \longrightarrow & * \end{array}$$

The bottom composite map is the identity by the universal property of the terminal object. We deduce that the top map is a bijection. Now we try to relate this description to $\mathrm{Hom}_{\mathcal{C}/Z}(V, \mathcal{C}(X,Y)\times Z)$. The following squares are cartesian by definition and by the universal property of the product:

$$\begin{array}{ccccc} \mathrm{Hom}_{\mathcal{C}/Z}(V, \mathcal{C}(X,Y)\times Z) & \longrightarrow & \mathrm{Hom}_{\mathcal{C}}(V, \mathcal{C}(X,Y)\times Z) & \longrightarrow & \mathrm{Hom}_{\mathcal{C}}(V, \mathcal{C}(X,Y)) \\ \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow \\ * & \longrightarrow & \mathrm{Hom}_{\mathcal{C}}(V, Z) & \longrightarrow & * \end{array}$$

This provides a bijection $\mathrm{Hom}_{\mathcal{C}/Z}(V, \mathcal{C}(X,Y)\times Z) \cong \mathrm{Hom}_{\mathcal{C}}(V, \mathcal{C}(X,Y)) \cong \mathrm{Hom}_{\mathcal{C}}(V\times X, Y)$. We conclude using the Yoneda Lemma. □

Remark 2.56. $\mathrm{Hom}_{\mathcal{C}/Z}(Z, \mathcal{C}_W(X,Y)\times_W Z) \cong \mathrm{Hom}_{\mathcal{C}/W}(W, \mathcal{C}_W(X,Y))$. If we consider the case $W = *$ we get:

$$\mathrm{Hom}_{\mathcal{C}/Z}(Z\times X, Z\times Y) \cong \mathrm{Hom}(Z, \mathcal{C}(X,Y))$$

Lemma 2.57 ([AP26] 4.3)**.** Let X, A, B be objects in a π-tribe $\mathcal{C}$. A map $e : X \to \mathrm{Equiv}(A,B)$ in $\mathcal{C}$ corresponds to the following data:

(1) A map $f : X\times A \to X\times B$
(2) Two maps $g, h : X\times B \to X\times A$
(3) $H : g\circ f \sim id$, $K : f\circ h \sim id$

Proof. Note that the base change of a path object remains a path object. We therefore deduce that $X\times A^{\Delta^1}$ and $X\times B^{\Delta^1}$ are path objects for $X\times A$ and $X\times B$ respectively. The rest follows by the various universal properties as in Remark 2.56. □

Definition 2.58. Let $\mathcal{C}$ be a tribe. Let $f : X\to Y$ in $\mathcal{C}$. There is a canonical homotopy square

$$\begin{array}{ccc} X & \xrightarrow{id\circ\Delta} & (X\times_Y X)^{\Delta^1} \\ {\scriptstyle\Delta}\downarrow & & \downarrow{\scriptstyle(s,t)} \\ X\times X & \xrightarrow[\Delta\times\Delta]{} & (X\times_Y X)\times(X\times_Y X) \end{array}$$

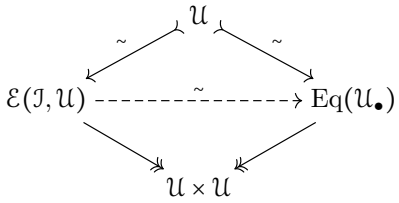
We say that f is a homotopy monomorphism if the induced map $X \to X\times^{\mathrm{h}}_Y X$ is a homotopy equivalence.

Definition 2.59. Let us consider a fibration $\mathcal{U}_\bullet \to \mathcal{U}$ in a Cisinski model category. We define $\mathrm{Eq}(\mathcal{U}_\bullet) := \mathrm{Equiv}_{\mathcal{U}\times\mathcal{U}}(\mathcal{U}_\bullet\times\mathcal{U}, \mathcal{U}\times\mathcal{U}_\bullet)$.

Definition 2.60. A univalent morphism for a π-tribe $\mathcal{C}$ is a fibration $\mathcal{U}_\bullet \to \mathcal{U}$ in $\mathcal{C}$ such that $\mathrm{Eq}(\mathcal{U}_\bullet)$ is a path object for $\mathcal{U}$.

Theorem 2.61. Let $\mathcal{U}_\bullet \twoheadrightarrow \mathcal{U}$ be a univalent morphism for the tribe associated to a right proper Cisinski model category $\mathcal{E}$.Let $\mathcal{I}$ be an interval object for $\mathcal{E}$. Then there is a weak equivalence $\mathcal{E}(\mathcal{I}, \mathcal{U}) \to$

$\mathrm{Eq}(\mathcal{U}_\bullet)$ as in the following diagram:



Proof. This theorem is a special case of Proposition 2.15. □

Definition 2.62. Let λ be a cardinal. Let $\mathcal{C}$ be any category. Let $f : X \to Y$ be a map in $\mathcal{C}$. We say that f has λ-small fibers if, for any pullback of the form,

$$\begin{array}{ccc} A & \longrightarrow & X \\ \downarrow & \lrcorner & \downarrow \\ B & \longrightarrow & Y \end{array}$$

if A is λ-compact then B is.

Definition 2.63. A strict univalent universe for a right proper Cisinski model category (or for a π-tibe) $\mathcal{E}$ is a univalent morphism $\pi : \mathcal{U}_\bullet \twoheadrightarrow \mathcal{U}$ such that there exists a cardinal λ for which any λ-small fibration $A \twoheadrightarrow B$ there exists a cartesian square of the form:

$$\begin{array}{ccc} A & \longrightarrow & \mathcal{U}_\bullet \\ \downarrow & \lrcorner & \downarrow \\ B & \longrightarrow & \mathcal{U} \end{array}$$

2.4. **Embedding of universes.**

Discussion 2.64. We would like to prove Theorem 2.70 which states that any morphism between univalent fibrations in a tribe is a homotopy embedding. To prove it we recall some technical lemmas for homotopy pullbacks in a tribe.

Lemma 2.65. Let us consider the following homotopy pullback:

$$\begin{array}{ccc} A \times^{\mathrm{h}}_C B & \longrightarrow & B \\ \downarrow & \lrcorner & \downarrow f \\ A & \xrightarrow[g]{} & C \end{array}$$

Let f be a fibration. Then the canonical map from the strict pullback $A \times_C B \to A \times^{\mathrm{h}}_C B$ is a homotopy equivalence.

Lemma 2.66. Let us consider a homotopy square as follows:

$$\begin{array}{ccc} P & \xrightarrow{\alpha} & C^{\Delta^1} \\ {\scriptstyle (g,u)}\downarrow & & \downarrow{\scriptstyle (s,t)} \\ A \times B & \xrightarrow[v\times f]{} & C \times C \end{array}$$

Let f be a homotopy equivalence. Then the canonical map $P \to A \times^{\mathrm{h}}_C B$ is a homotopy equivalence if and only if g is a homotopy equivalence.

Proposition 2.67. Given the datum of two homotopy squares:

$$\begin{array}{ccc} P & \xrightarrow{\alpha} & C^{\Delta^1} \\ {\scriptstyle (p,q)}\downarrow & & \downarrow{\scriptstyle (s,t)} \\ A \times B & \xrightarrow[f\times g]{} & C \times C \end{array} \qquad \begin{array}{ccc} B & \xrightarrow{\beta} & D^{\Delta^1} \\ {\scriptstyle (u,v)}\downarrow & & \downarrow{\scriptstyle (s,t)} \\ C \times E & \xrightarrow[h\times k]{} & D \times D \end{array}$$

Then we can construct the datum of a homotopy square as follows:

$$\begin{array}{ccc} P & \xrightarrow{\gamma} & D^{\Delta^1} \\ {\scriptstyle (p,q)}\downarrow & & \downarrow{\scriptstyle (s,t)} \\ A \times E & \xrightarrow[(h\circ f,k)]{} & D \times D \end{array}$$

We will refer to this homotopy square as a composite of the two starting homotopy squares.

Lemma 2.68. Under the hypothesis of the previous lemma, if the left homotopy square is a homotopy pullback then the right one is a homotopy pullback if and only if the composite square is.

Lemma 2.69. Let $i : A \to B$ be a homotopy equivalence in a tribe. If i has a retraction r then i is an anodyne map.

Proof. We need to provide a solution for any lifting problem

$$\begin{array}{ccc} A & \xrightarrow{x} & X \\ {\scriptstyle i}\downarrow & & \downarrow{\scriptstyle f} \\ B & \xrightarrow[y]{} & Y \end{array}$$

where f is a fibration. Let us consider the following lifting problem:

$$\begin{array}{ccccccc} B & \xrightarrow{r} & A & \xrightarrow{i} & B & \xrightarrow{y} & Y \\ {\scriptstyle Id}\downarrow & & & \nearrow_{H} & & & \downarrow \\ B^{\Delta^1} & & & & & \longrightarrow & * \end{array}$$

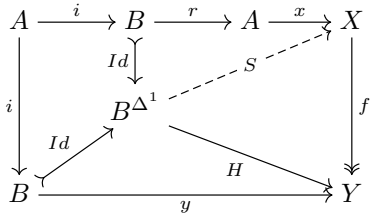
It has a solution since Id is an anodyne by definition and Y is fibrant. We can produce the following commutative diagram:

$$\begin{array}{ccccccc} B & & \xrightarrow{r} & & A & \xrightarrow{x} & X \\ & \searrow & & & \downarrow{\scriptstyle i} & & \\ {\scriptstyle Id}\downarrow & & & & B & \searrow^{y} & \downarrow{\scriptstyle f} \\ & \nearrow & & & & & \\ B^{\Delta^1} & & & \xrightarrow[H]{} & & & Y \end{array}$$

This yields the following lifting problem:

$$\begin{array}{ccccc} B & \xrightarrow{r} & A & \xrightarrow{x} & X \\ {\scriptstyle Id}\downarrow & & & \nearrow_{S} & \downarrow{\scriptstyle f} \\ B^{\Delta^1} & & \xrightarrow[H]{} & & Y \end{array}$$

It has a solution since f is a fibration and Id is an anodyne map. We can upgrade it to the following diagram:



This clearly provides a solution for our starting lifting problem. □

Theorem 2.70 ([AP26] 4.18)**.** Let $\mathcal{U}_\bullet \twoheadrightarrow \mathcal{U}$ be a univalent fibration in a π-tribe $\mathcal{C}$. Let $\mathcal{V} \to \mathcal{U}$ be a morphism in $\mathcal{C}$. Define $\mathcal{V}_\bullet$ as: $\mathcal{V}_\bullet := \mathcal{V} \times_{\mathcal{U}} \mathcal{U}_\bullet$. Then $\mathcal{V}_\bullet \to \mathcal{V}$ is univalent if and only if $\mathcal{V} \to \mathcal{U}$ is an embedding.

Proof. Let us begin with the following strict pullback:

$$\begin{array}{ccc} \mathcal{V}_\bullet & \longrightarrow & \mathcal{U}_\bullet \\ {\scriptstyle\delta}\downarrow & \lrcorner & \downarrow{\scriptstyle\pi} \\ \mathcal{V} & \xrightarrow[f]{} & \mathcal{U} \end{array}$$

Since Equiv is preserved by base change, the following square is a strict pullback:

$$\begin{array}{ccc} \mathrm{Eq}(\mathcal{V}_\bullet) & \longrightarrow & \mathrm{Eq}(\mathcal{U}_\bullet) \\ \downarrow & \lrcorner & \downarrow \\ \mathcal{V} \times \mathcal{V} & \xrightarrow[f\times f]{} & \mathcal{U} \times \mathcal{U} \end{array}$$

Let us consider the following diagram:

$$\begin{array}{ccc} \mathcal{V} & \xrightarrow{f} & \mathcal{U} \\ \downarrow & & \downarrow{\scriptstyle\sim} \\ \mathrm{Eq}(\mathcal{V}_\bullet) & \longrightarrow & \mathrm{Eq}(\mathcal{U}_\bullet) \\ \downarrow & \lrcorner & \downarrow \\ \mathcal{V} \times \mathcal{V} & \xrightarrow[f\times f]{} & \mathcal{U} \times \mathcal{U} \end{array}$$

The top square is a homotopy pullback if and only the whole square is thanks to Lemma 2.65 and Lemma 2.68. This is equivalent to the map $\mathcal{V} \to \mathrm{Eq}(\mathcal{V}_\bullet)$ being a homotopy equivalence by Lemma 2.66. By Lemma 2.69, this is equivalent to it being an anodyne map, which is equivalent to $\mathrm{Eq}(\mathcal{V}_\bullet)$ being a path object. Notice that $\mathcal{V} \to \mathrm{Eq}(V_\bullet)$ is a homotopy equivalence if and only if $\mathcal{V}$ is equivalent to the homotopy pullback of $\Delta : \mathcal{V} \to \mathcal{V} \times_{\mathcal{U}} \mathcal{V}$ along itself, since $\mathrm{Eq}(\mathcal{U}_\bullet)$ is a path object. □

3. Shulman's Universe

Definition 3.1. Let λ, k be two cardinals. We say $k \rhd \lambda$ if $k \geq \lambda$ and any λ-accessible category is also k-accessible. For more equivalent conditions to this property see [Shu19](4.3).

Remark 3.2. $\lambda \rhd k$ is necessary for the proof of Proposition 3.26 as discussed in [Shu19].

3.1. 2-categorical notions.

Notation 3.3. Throughout this subsection, let $\mathcal{H}$ be any 2-category and let $\mathcal{E}$ be any large, locally small category. Following Shulman's notation, we denote with $\mathcal{E}(-,-)$ the usual Hom-set in $\mathcal{E}$.

Notation 3.4. We denote by Gpd the large category of 1-groupoids. We denote by $\mathrm{Psh}(\mathcal{E})$ the category $\mathrm{PsFun}(\mathcal{E}^{\mathrm{op}}, \mathrm{Gpd})$, the very large 2-category of pseudo-functors from $\mathcal{E}^{\mathrm{op}}$ to Gpd. (i.e.: functorial identities are defined up to isomorphism)

Remark 3.5. In the next section we denote ∞-groupoids or spaces with An in order to avoid confusion with 'honest' 1-groupoids.

Definition 3.6. Let X, Y be two objects of $\mathcal{H}$. Let $f : X \to Y$ be a morphism between them. f is an *internal fibration* if, for any object Z of $\mathcal{H}$, $f \circ - : \mathcal{H}(Z,X) \to \mathcal{H}(Z,Y)$ is a cartesian fibration. Explicitly: Let $g : Z \to X$, $h : Z \to Y$. Let $\alpha : h \to f \circ g$ in $\mathcal{H}(Z,Y)$.

$$\begin{array}{ccc} & & g \\ & & \top \\ & & \vdots \\ & & \downarrow{\scriptstyle f\circ -} \\ h & \xrightarrow{\alpha} & f \circ g \end{array}$$

Then there exists a cartesian lift of α, namely: $l_\alpha : \alpha^*(h) \to g$ in $\mathcal{H}(Z,X)$ such that $f \circ \alpha^*(h) = h$, $f \circ -(l_\alpha) = \alpha$ and it is universal with this property up to an equivalence of 2-categories.

$$\begin{array}{ccc} \alpha^*(h) & \xrightarrow{l_\alpha} & g \\ \top & \lrcorner & \top \\ \vdots & & \vdots \\ \downarrow & & \downarrow{\scriptstyle f\circ -} \\ h & \xrightarrow{\alpha} & f \circ g \end{array}$$

We say f is discrete if the lift is unique.

Definition 3.7. A strict fibration in $\mathrm{Psh}(\mathcal{E})$ is a strictly natural transformation $\alpha : \mathbb{X} \to \mathbb{Y}$ such that, for any object of $\mathcal{E}$ A, $\alpha_A : \mathbb{X}(A) \to \mathbb{Y}(A)$ is a (cartesian) fibration of groupoids. Explicitly: for any object of $\mathbb{X}(A)$ x, for any object of $\mathbb{Y}(A)$ y and for any $l : y \to \alpha_A(x)$ (which is an isomorphism since it is an arrow in a groupoid) there exists a cartesian lift

$$\begin{array}{ccc} \mathbb{X}(A) & & \exists l^*(y) & \xrightarrow{\sim} & x \\ & & \top & \lrcorner & \top \\ \downarrow{\scriptstyle \alpha_A} & & \vdots & & \vdots \\ & & \downarrow & & \downarrow \\ \mathbb{Y}(A) & & y & \xrightarrow{\sim} & \alpha_A(x) \end{array}$$

We say that α is discrete if the cartesian lift is unique (α is objectwise discrete).

Remark 3.8. Cartesian fibrations of groupoids coincide with isofibrations, this implies that in Definition 3.7 we can just ask for α to be an objectwise isofibration.

Notation 3.9. Let X, Y, Z be objects of $\mathcal{H}$. Let $f : X \to Z$ and $g : Y \to Z$ be morphisms in $\mathcal{H}$. We denote with $X \times_Z^h Y$ the 2-categorical weak pullback.

Lemma 3.10. Let the following square in $\mathcal{H}$ be commutative up to a 2-isomorphism α and let f be an internal fibration ($\mathcal{H}(Z, f)$ being an isofibration for all Z in $\mathcal{H}$ is enough):

$$\begin{array}{ccc} A' & \xrightarrow{u} & A \\ {\scriptstyle f'}\downarrow & \overset{\alpha}{\swarrow} & \downarrow{\scriptstyle f} \\ B' & \xrightarrow[v]{} & B \end{array}$$

i.e.: $\alpha : v \circ f' \cong f \circ u$. Then u is equivalent to a morphism $\tilde{u} : A' \to A$ such that the following square is strictly commutative:

$$\begin{array}{ccc} A' & \xrightarrow{\tilde{u}} & A \\ {\scriptstyle f'}\downarrow & = & \downarrow{\scriptstyle f} \\ B' & \xrightarrow[v]{} & B \end{array}$$

Proof. The condition on α can be rewritten as follows:

$$\begin{array}{ccc} \mathcal{H}(A', A) & & u \\ \downarrow{\scriptstyle f\circ -} & & \top \\ \mathcal{H}(A', B) & & \vdots\ {\scriptstyle f\circ -} \\ & & \downarrow \\ & v \circ f' \xrightarrow[\alpha]{} & f \circ u \end{array}$$

Since $\mathcal{H}(A', f)$ is an isofibration (in fact more is true: f is an internal fibration) there exists a lift $can : \tilde{u} \cong u$ completing the diagram:

$$\begin{array}{ccc} \tilde{u} & \xrightarrow{can} & u \\ \top & & \top \\ \vdots & & \vdots\ {\scriptstyle f\circ -} \\ \downarrow & & \downarrow \\ v \circ f' & \xrightarrow[\alpha]{} & f \circ u \end{array}$$

such that $f \circ \tilde{u} = f \circ -(\tilde{u}) = v \circ f'$. □

Remark 3.11. An internal fibration of groupoids is just an isofibration between groupoids: one implication is just $\mathrm{Fun}(*, -) \cong id_{\mathrm{Gpd}}$, the other follows by the functoriality of the lift.

Lemma 3.12. Let $\mathbb{Z}, \mathbb{Y}$ be two objects of $\mathrm{Psh}(\mathcal{E})$. Let $f : \mathbb{Z} \to \mathbb{Y}$ be a morphism in $\mathrm{Psh}(\mathcal{E})$. Then f is equivalent to a strict fibration over $\mathbb{Y}$. Moreover: such a fibration is discrete if f is faithful (i.e.: $\mathrm{Psh}(\mathcal{E})(\mathbb{Z}, -)$ sends f to a faithful functor for all presheaves $\mathbb{Z}$).

Proof. For any A in $\mathcal{E}$ we may consider

$$\begin{array}{ccccc} \mathbb{Z}(A) & \xrightarrow{(id_{\mathbb{Z}(A)}, f_A)} & \mathbb{Z}(A) \times^h_{\mathbb{Y}(A)} \mathbb{Y}(A) & \longrightarrow & Iso(\mathbb{Y}(A)) \\ \downarrow{\scriptstyle f_A} & & \downarrow & \lrcorner & \downarrow{\scriptstyle (pr_1, pr_2)} \\ & & \mathbb{Z}(A) \times_{\mathbb{Y}(A)} \mathbb{Y}(A) & & \\ & & \downarrow & & \\ \mathbb{Y}(A) & \xleftarrow{pr_2} & \mathbb{Z}(A) \times \mathbb{Y}(A) & \xrightarrow[f_A \times id_{\mathbb{Y}(A)}]{} & \mathbb{Y}(A) \times \mathbb{Y}(A) \end{array}$$

where $(id_{\mathbb{Z}(A)}, f_A)$ is section of an equivalence and the map from $\mathbb{Z}(A) \times^h_{\mathbb{Y}(A)} \mathbb{Y}(A)$ to $\mathbb{Y}(A)$ is the composition of two isofibrations of groupoids (one being a base change of an isofibration). This shows that we can write f_A as the composition of an equivalence with an isofibration p. Now we may define $\mathbb{T} := \mathbb{Z} \times^h_{\mathbb{Y}} \mathbb{Y}$ by picking for any A in $\mathcal{E}$ a representative of $\mathbb{Z}(A) \times^h_{\mathbb{Y}(A)} \mathbb{Y}(A)$. For any morphism $B \to A$

in $\mathcal{E}$ we can consider

$$\begin{array}{ccc} \mathbb{T}(A) & \xrightarrow{p_A} & \mathbb{Y}(A) \\ \downarrow & & \downarrow \\ \mathbb{T}(B) & \xrightarrow[p_B]{} & \mathbb{Y}(B) \end{array}$$

Since p_B is an isofibration due to Lemma 3.10, we may impose (up to natural isomorphism) that this square commute strictly and that we can assemble the p_A's into a $p : \mathbb{T} \to \mathbb{Y}$ that is an objectwise isofibration, making it an internal fibration. If f is discrete, after evaluating at any A, the uniqueness of the lift imposes the faithfulness condition (and vice-versa). □

Lemma 3.13. Let $\mathbb{X}, \mathbb{Y}$ be two elements of $\mathrm{Psh}(\mathcal{E})$. Let $f : \mathbb{X} \to \mathbb{Y}$ be a morphism between them. f is an internal fibration if and only if it is a discrete fibration.

Proof. If f is an internal fibration we can find a lift that is objectwise cartesian. For the converse, let us assume that f is a discrete fibration. Let us consider two morphisms $g : \mathbb{Z} \to \mathbb{X}$, $h : \mathbb{Z} \to \mathbb{X}$ two morphisms of $\mathrm{Psh}(\mathcal{E})$. Let $\alpha : h \Rightarrow f \circ g$ be a natural transformation. If we evaluate at any A in $\mathcal{E}$ we observe that α_A is a natural transformation of functors between groupoids, hence objectwise invertible. Thusα_A is invertible, therefore α is invertible to begin with. Objectwise, we can find a (cartesian) lift β_A of α_A, namely: $\beta_A : k_A \cong g_A$ such that $f_A \circ k_A = h_A$ and $f_A \circ -(\beta_A) = \alpha_A$. We would like to assemble all the objectwise data to an actual global cartesian lift. This procedure goes through by the proof of Lemma 3.10. □

Definition 3.14. Let $\phi : \mathbb{X} \to \mathbb{Y}$ be a morphism of pseudo-presheaves. We say it is locally representable if, for any object Z in $\mathcal{E}$ and for any cartesian square

$$\begin{array}{ccc} \mathbb{P} & \longrightarrow & \mathbb{X} \\ \downarrow & \lrcorner & \downarrow{\scriptstyle \phi} \\ \mathcal{E}(-, Z) & \longrightarrow & \mathbb{Y} \end{array}$$

Then $\mathbb{P}$ is representable.

Definition 3.15. Let $\mathcal{E}$ be a category. We denote with $\mathbb{E} : \mathcal{E}^{\mathrm{op}} \to \mathrm{Gpd}$ the core of the self-indexing of $\mathcal{E}$. Namely, it is the pseudo-presheaf on $\mathcal{E}$ that acts on objects as $Y \mapsto (\mathcal{E}/_Y)^{\cong}$ and sends arrows to the corresponding base change functor between the slices.

Definition 3.16 ([Shu19] 3.1)**.** Let $\mathcal{E}$ be a category. A notion of fibered structure on $\mathcal{E}$ is a morphism of pseudo-presheaves $\mathbb{F} \to \mathbb{E}$ that is a strict discrete fibration and has small fibers.

Definition 3.17. Let $\mathcal{E}$ be a category and $\mathbb{F} \to \mathbb{E}$ a notion of fibered structure on $\mathcal{E}$. Let f be a morphism in $\mathcal{E}$. The $\mathbb{F}$-structure on f is the fiber of $\mathbb{F}$ at f. We refer to objects of the $\mathbb{F}$-structure on f as $\mathbb{F}$-algebras. Let us consider the following cartesian square:

$$\begin{array}{ccc} X' & \xrightarrow{u} & X \\ {\scriptstyle f'}\downarrow & \lrcorner & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[v]{} & Y \end{array}$$

If we fix an $\mathbb{F}$-algebra on both f and g we say that the square is an $\mathbb{F}$-morphism if $\mathbb{F}(u)$ sends the $\mathbb{F}$-algebra on f to the $\mathbb{F}$-algebra on g. Sometimes we will say that f is an $\mathbb{F}$-algebra if there is a choice of an $\mathbb{F}$-algebra on f.

Definition 3.18. Let $\mathcal{E}$ be any category. If we consider, a full subcategory of $(\mathcal{E}/_y)^{\cong}$ for each Y in $\mathcal{E}$, that is stable under base change, this defines a notion of fibered structure $\mathbb{F} \hookrightarrow \mathbb{E}$. We refer to such notions of a fibered structure as full notions of a fibered structure.

Example 3.19. Let $\mathcal{E}$ be any category. A functorial factorization is a section of the composition functor $\circ : \mathcal{E}^{0\to 1} \times_{\mathcal{E}} \mathcal{E}^{0\to 1} \to \mathcal{E}^{0\to 1}$. Let us consider a functorial factorization E. We can define a notion of fibered structure $\mathbb{R}_E$: for any arrow $f : X \to Y$ in $\mathcal{E}$ an $\mathbb{R}_E$ algebra is given by any solution to the following lifting problem:

$$\begin{array}{ccc} X & = & X \\ \downarrow & \overset{r_f}{\nearrow} & \downarrow f \\ E_f & \longrightarrow & Y \end{array}$$

This assembles to a notion of fibered structure on $\mathcal{E}$.

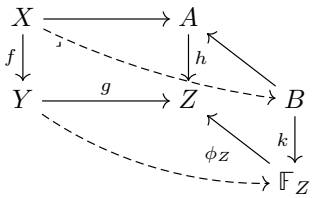
Example 3.20. Let $\mathbb{F}, \mathbb{G}$ be two notions of a fibered structure on $\mathcal{E}$. Then $\mathbb{F} \times_{\mathbb{E}} \mathbb{G}$ is again a notion of fibered structure on $\mathcal{E}$.

Definition 3.21. A morphism of pseudo-presheaves $\phi : \mathbb{F} \to \mathbb{E}$ is a locally representable notion of fibered structure if it is both a notion of fibered structure and a locally representable.

Remark 3.22. Let us better elaborate previous definition. We consider the following cartesian square:

$$\begin{array}{ccc} \mathbb{P} & \longrightarrow & \mathbb{F} \\ \downarrow & \lrcorner & \downarrow \\ \mathcal{E}(-, Z) & \longrightarrow & \mathbb{E} \end{array}$$

The universal property is already witnessed by representable presheaves since limits are computed objectwise. By the Yoneda lemma, $\mathbb{P}$ is representable if and only if there exist an object $\mathbb{F}_Z$ and a $\mathbb{F}$ $\phi_Z : \mathbb{F}_Z \to Z$ such that for any Y in $\mathcal{E}$, any map $g : Y \to Z$ and any $\mathbb{F}$-algebra $f : X \to Y$ as in:



there exist a pair of maps (as the dotted ones) such that all squares in the diagram are cartesian and the following square is an $\mathbb{F}$-morphism:

$$\begin{array}{ccc} X & \dashrightarrow & B \\ {\scriptstyle f}\downarrow & \lrcorner & \downarrow{\scriptstyle k} \\ Y & \dashrightarrow & \mathbb{F}_Z \end{array}$$

Definition 3.23. Let E be a functorial factorization. We say that E is cartesian if for any cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{u} & X \\ {\scriptstyle f'}\downarrow & \lrcorner & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[v]{} & Y \end{array}$$

its factorization via E is such that all the following squares are cartesian:

$$\begin{array}{ccc} X' & \xrightarrow{u} & X \\ \downarrow & \lrcorner & \downarrow \\ E_{f'} & \longrightarrow & E_f \\ \downarrow & \lrcorner & \downarrow \\ Y' & \xrightarrow[v]{} & Y \end{array}$$

Example 3.24. Let us consider a cartesian functorial factorization E. If $\mathcal{E}$ is locally cartesian closed, then its associated notion of fibered structure $\mathbb{R}_E$ as in Example 3.19 is locally representable if $\mathcal{E}$ is locally cartesian closed. If $\mathbb{F}$ and $\mathbb{G}$ are two locally representable notions of a fibered structure, then $\mathbb{F} \times_{\mathbb{E}} \mathbb{G}$ is as well.

Proof. We prove the equivalent condition illustrated in Remark 3.22 for $\mathbb{F} = \mathbb{R}_E$. Let us consider the following cartesian square where f has an $\mathbb{R}_E$ algebra r_f:

$$\begin{array}{ccc} X & \longrightarrow & A \\ {\scriptstyle f}\downarrow & \lrcorner & \downarrow{\scriptstyle h} \\ Y & \xrightarrow[g]{} & Z \end{array}$$

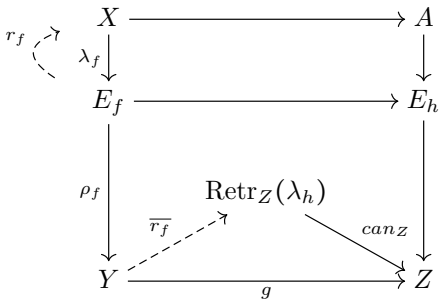
When we apply the cartesian factorization E to it, we get:

$$\begin{array}{ccccc} & & X & \longrightarrow & A \\ & {\scriptstyle r_f}\circlearrowleft & {\scriptstyle \lambda_f}\downarrow & \lrcorner & \downarrow{\scriptstyle \lambda_h} \\ & & E_f & \longrightarrow & E_h \\ & & {\scriptstyle \rho_f}\downarrow & \lrcorner & \downarrow{\scriptstyle \rho_h} \\ & & Y & \xrightarrow[g]{} & Z \end{array}$$

Since $\mathcal{E}$ is locally cartesian closed we can construct the following pullback in $\mathcal{E}/_Z$:

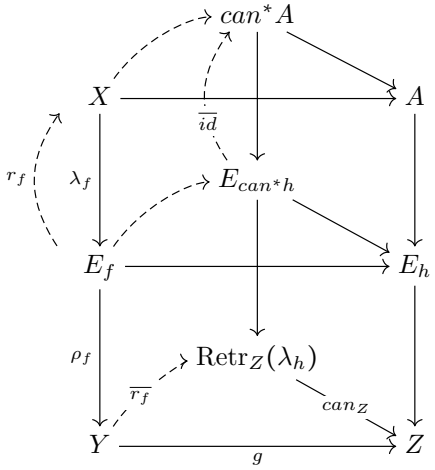
$$\begin{array}{ccc} \mathrm{Retr}_Z(\lambda_h) & \longrightarrow & \mathcal{E}_Z(A, E_h) \\ {\scriptstyle can_Z}\downarrow & \lrcorner & \downarrow{\scriptstyle -\circ\lambda_h} \\ Z & \xrightarrow[id_{\rho_h}]{} & \mathcal{E}_Z(E_h, E_h) \end{array}$$

$can_Z : \mathrm{Retr}_Z(\lambda_h) \to Z$ is such that the lifts of g to can_Z correspond to the retractions of $g^*\lambda_h = \lambda_f$. Therefore r_f corresponds to a lift $\overline{r_f}$ of g:



If we pull back $\mathrm{Retr}_Z(\lambda_h)$ along $A \to Z$ we notice that $id_{\mathrm{Retr}_Z(\lambda_h)}$ induces a canonical retraction $\overline{id}$ of $can_Z^*\lambda_h$ by definition of $\mathrm{Retr}_Z(\lambda_h)$. Since E is cartesian $can_Z^*\lambda_h = \lambda_{can^*h}$. This defines an $\mathbb{R}_E$-algebra

on can^*h. By the universal property of pullbacks we can complete the diagram with dotted maps:



All squares are cartesian by the pullback pasting lemma. Since the lower triangle is commutative, by the universal property of $\mathrm{Repr}_Z(\lambda_h)$ the upper dotted square commutes. We deduce that the square we constructed is an $\mathbb{R}_E$-morphism. For the second example let us consider the following cartesian square:

$$\begin{array}{ccc} \mathbb{P}_2 & \longrightarrow & \mathbb{F} \times_{\mathbb{E}} \mathbb{G} \\ \downarrow & \lrcorner & \downarrow \\ \mathcal{E}(-, Z) & \longrightarrow & \mathbb{E} \end{array}$$

By the pullback pasting lemma, we can decompose it into two cartesian squares as follows:

$$\begin{array}{ccccc} \mathbb{P}_2 & \longrightarrow & \mathbb{F} \times_{\mathbb{E}} \mathbb{G} & \longrightarrow & \mathbb{G} \\ \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow \\ \mathbb{P}_1 & \longrightarrow & \mathbb{F} & \longrightarrow & \mathbb{E} \\ \downarrow & \lrcorner & \downarrow & & \\ \mathcal{E}(-, Z) & \longrightarrow & \mathbb{E} & & \end{array}$$

Since $\mathbb{F}$ is locally representable we deduce that $\mathbb{P}_1$ is representable; using the local representability of $\mathbb{G}$, we conclude that $\mathbb{P}_2$ is representable. □

Lemma 3.25. The following two statements are equivalent in a locally presentable category $\mathcal{E}$:

(1) There is a unique (up to natural isomorphism) solution to the following lifting problem:

$$\begin{array}{ccc} \mathrm{colim}_{i:I} \mathcal{E}(-, y_i) & \longrightarrow & \mathbb{F} \\ \downarrow & \nearrow & \downarrow \\ \mathcal{E}(-, \mathrm{colim}_{i:I} y_i) & \longrightarrow & \mathbb{E} \end{array}$$

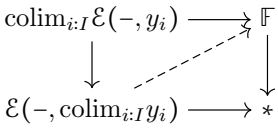
for any $I \to \mathcal{E}$ with I small category.

(2) $\phi : \mathbb{F} \to \mathbb{E}$ is locally representable.

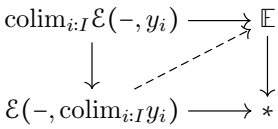
Proof. Let I be any small category. First we claim that if $\mathbb{E}$ preserves I-shaped colimits then $\mathbb{F}$ has the same property if and only if the following lifting problem

$$\begin{array}{ccc} \mathrm{colim}_{i:I} \mathcal{E}(-, y_i) & \longrightarrow & \mathbb{F} \\ \downarrow & \nearrow & \downarrow \\ \mathcal{E}(-, \mathrm{colim}_{i:I} y_i) & \longrightarrow & \mathbb{E} \end{array}$$

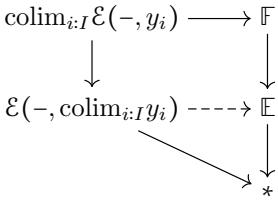
has an unique solution (up to natural isomorphisms) $\mathbb{F}$ preserves I-shaped colimits if and only if the following lifting problem has a unique solution:



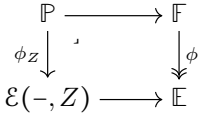
We know that $\mathbb{E}$ preserves them so the following lifting problem has a unique solution:



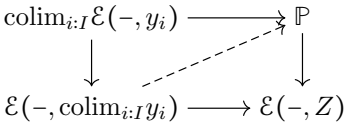
We can construct the following diagram:



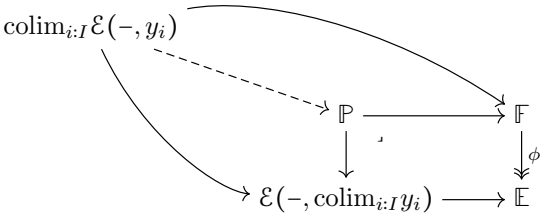
which shows that the lift to $\mathbb{F} \to *$ coincides with the lift to $\mathbb{F} \to \mathbb{E}$. For any Z in $\mathcal{E}$ let us assume that the following square is cartesian:



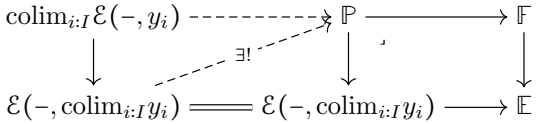
$\mathbb{P}$ has values in Set since $\mathbb{F}$ is a notion of fibered structure and $\mathcal{E}(-, Z)$ has values in Set. Since $\mathcal{E}$ is locally presentable $\mathbb{P}$ preserves colimits if and only if it is representable. Since $\mathcal{E}(-, Z)$ is representable, it preserves colimits. We conclude that $\mathbb{P}$ is representable if and only if the following lifting problem has a unique solution for any small category I:



If we assume (2) then ϕ has a unique solution to the lifting problem and so does ϕ_Z and, therefore $\mathbb{P}$ is representable by the previous line of arguments. Viceversa, $\mathcal{E}(-, \mathrm{colim}_{i:I} y_i)$ is representable, therefore:



Since $\mathbb{P}$ is representable by hypothesis we know that the lifting problem on the right has an essentially unique solution:



which provides an essentially unique solution for the starting problem. □

3.2. **Construction of the universe.**

Proposition 3.26 ([Shu19] 4.8)**.** Let $\mathcal{E}$ be a locally presentable and locally cartesian closed category. Then there exists a cardinal k such that for any $\lambda \triangleright k$ the full notion of λ-small morphisms inside $\mathbb{E}$ form a locally representable notion of fibered structure on $\mathcal{E}$. We denote it with $\mathbb{E}^\lambda$

Definition 3.27. Let $\mathbb{X} \to \mathbb{Y}$ be a morphism in $\mathrm{Psh}(\mathcal{E})$. We say it is an acyclic fibration if there exists a solution to any lifting problem of the form

$$\begin{array}{ccc} \mathcal{E}(-,A) & \longrightarrow & \mathbb{X} \\ {\scriptstyle \mathcal{E}(-,u)}\downarrow & \nearrow & \downarrow \\ \mathcal{E}(-,B) & \longrightarrow & \mathbb{Y} \end{array}$$

when u is a cofibration.

Example 3.28. Let $\mathbb{X} \to \mathbb{Y}$ be a locally representable morphism in $\mathrm{Psh}(\mathcal{E})$. Then it is an acyclic fibration if and only if, for any pullback square:

$$\begin{array}{ccc} \mathcal{E}(-,A) & \longrightarrow & \mathbb{X} \\ {\scriptstyle \mathcal{E}(-,u)}\downarrow & \lrcorner & \downarrow \\ \mathcal{E}(-,B) & \longrightarrow & \mathbb{Y} \end{array}$$

u is a trivial fibration.

Proof. For any cofibration $j : W \to Z$ in $\mathcal{E}$ we can construct the following diagram:

$$\begin{array}{ccccc} \mathcal{E}(-,W) & \dashrightarrow & \mathbb{P} & \longrightarrow & \mathbb{F} \\ {\scriptstyle \mathcal{E}(-,j)}\downarrow & \nearrow & \downarrow{\scriptstyle u_*} & \lrcorner & \downarrow{\scriptstyle \phi} \\ \mathcal{E}(-,Z) & = & \mathcal{E}(-,Z) & \longrightarrow & \mathbb{E} \end{array}$$

The diagonal dotted map is obtained from the solution of the lifting problem in $\mathcal{E}$. □

Definition 3.29. Let $\mathbb{F} \to \mathbb{E}$ be a notion of fibered structure on $\mathcal{E}$. A universe for $\mathbb{F}$ is a cofibrant object $\mathcal{U} \in \mathcal{E}$ and an acyclic fibration $\mathcal{E}(-,\mathcal{U}) \to \mathbb{F}$ in $\mathrm{Psh}(\mathcal{E})$.

Remark 3.30. Let us elaborate on Definition 3.29 more concretely. $\mathcal{E}(-,\mathcal{U}) \to \mathbb{F}$ corresponds, by Yoneda, to $s \in \mathbb{F}(\mathcal{U})$, let us apply $\mathbb{F} \to \mathbb{E}$ to s:

$$s \mapsto \pi : \mathcal{U}_\bullet \to \mathcal{U} \in (\mathcal{E}/_{\mathcal{U}})^{\simeq}$$

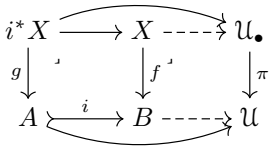
Since $\mathcal{E}(-,\mathcal{U}) \to \mathbb{F}$ is an acyclic fibration we know that for any lifting problem

$$\begin{array}{ccc} \mathcal{E}(-,A) & \xrightarrow{\mathcal{E}(-,u)} & \mathcal{E}(-,\mathcal{U}) \\ {\scriptstyle \mathcal{E}(-,i)}\downarrow & \nearrow & \downarrow{\scriptstyle \pi} \\ \mathcal{E}(-,B) & \longrightarrow & \mathbb{F} \end{array}$$

where i is a cofibration, there exists a lift in the form of the dotted arrow. The map $\mathcal{E}(-,A) \to \mathbb{F}$ corresponds to an $\mathbb{F}$-algebra over A. By a computation with Yoneda, the map $\mathcal{E}(-,B) \to \mathbb{F}$ provides us with an $\mathbb{F}$-algebra on B which pulls back along i to the one on Asince the diagram is commutative. Analogously we deduce that the canonical $\mathbb{F}$-algebra on $\mathcal{U}$ pulls back to the $\mathbb{F}$-algebra on A. This yields the following diagram:

$$\begin{array}{ccccc} i^*X & \longrightarrow & X & \longrightarrow & \mathcal{U}_\bullet \\ {\scriptstyle g}\downarrow & \lrcorner & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle \pi} \\ A & \xrightarrow{i} & B & \longrightarrow & \mathcal{U} \end{array}$$

where all vertical maps are $\mathbb{F}$-algebras, i is a cofibration and all squares are $\mathbb{F}$ morphisms. A solution to the lifting problem is equivalent (via an analogous line of reasoning) to the existence of a morphism $B \to \mathcal{U}$ that pulls back to $X \to B$ along π. Explicitly, we can complete the following diagram with dotted arrows



Where all squares are $\mathbb{F}$-morphisms.

Definition 3.31. Let $\mathbb{X} \in \mathrm{Psh}(\mathcal{E})$ be a presheaf on $\mathcal{E}$. We say $\mathbb{X}$ is a stack of cell complexes if it preserves coproducts as well as pushouts of cofibrations and transfinite compositions of cofibrations. In this definition we consider the model structure on Gpd where:

(1) weak equivalences are equivalences of categories
(2) fibrations are isofibrations
(3) cofibrations are functors injective on objects

Remark 3.32. There is some ambiguity in the original work regarding the meaning of 'preserves' in the previous definition. We believe that a better way to express this property is the following: A pseudo presheaf $\mathbb{F}$ is a stack of cell complexes if and only if $\mathbb{F} \to *$ has the right lifting property against any map $\mathrm{colim}_{\mathcal{I}}(-, X_i) \to \mathcal{E}(-\mathrm{colim}_{\mathcal{I}} X_i)$ where $\mathcal{I}$ defines a colimit listed in the previous definition and the lift is unique up to natural isomorphism. This second formulation is closer to 'sending colimits in $\mathcal{E}$ to a groupoid canonically equivalent to the limit in Gpd'.

Example 3.33. Let $\mathcal{E}$ be a Grothendieck 1-topos where cofibrations are also monomorphisms. Then $\mathbb{E}$ is a stack of cell complexes.

(1) For coproducts, we just notice that coproducts are universal.
(2) For pushouts of cofibrations, we know that a Grothendieck topos is adhesive (pushout squares of monomorphisms are also pullback squares).
(3) For transfinite compositions of cofibrations we use the fact that Grothendieck topoi are exhaustive i.e.: transfinite compositions of monomorphisms exist and are stable under pullback.

Lemma 3.34 ([Shu19] 5.7)**.** Let $\mathcal{E}$ be such that $\mathbb{E}$ is a stack of cell complexes. Let $\phi : \mathbb{F} \to \mathbb{E}$ be any locally representable notion of fibered structure on $\mathcal{E}$. Then $\mathbb{F}$ is a stack of cell complexes.

Proof. Let $\mathcal{D}$ be a class of morphisms of the form:

$$\mathrm{colim}_{i:I} \mathcal{E}(-, y_i) \to \mathcal{E}(-, \mathrm{colim}_{i:I} y_i)$$

where $\mathrm{colim}_{i:I}$ could be either a pushout of cofibrations, a coproduct or a transfinite composition of cofibrations. By hypothesis we know that all maps of $\mathcal{D}$ lift uniquely (up to natural isomorphism) against $\mathbb{E} \to *$ then, by virtue of Lemma 3.25 all maps of $\mathcal{D}$ lift uniquely (up to natural isomoprhism) against $\mathbb{F} \to \mathbb{E}$. □

Theorem 3.35 ([Shu19] 5.9)**.** Let $\mathcal{E}$ be a combinatorial model category. Let $\mathbb{Z}$ be a small-valued stack for cell complexes. Then any morphism $\mathcal{E}(X, -) \to \mathbb{Z}$ in $\mathrm{Psh}(\mathcal{E})$ factors as

$$\mathcal{E}(-, X) \xrightarrow{j_*} \mathcal{E}(-, Y) \xrightarrow{p} \mathbb{Z}$$

where j is a cofibration and p is an acyclic fibration

Proof. The proof of this statement is essentially the 2-categorical small object argument. We will construct a cell complex sequence by transfinite induction on n ordinal: Let us denote with $\mathcal{J}$ a set of generating trivial cofibrations for $\mathcal{E}$.

0) $X_0 = X,\ f_0 = f$.

Limit) $X_n = \mathrm{colim}_{m<n} X_m$ $\mathrm{colim}_{m<n}\mathcal{E}(-, X_m) \to \mathbb{Z}$. It exists since $\mathbb{Z}$ is a stack of cell complexes, IE:

$$\begin{array}{ccc} \mathrm{colim}_{m<n}\mathcal{E}(-, X_m) & \dashrightarrow & \mathbb{Z} \\ \downarrow & \nearrow_{\exists!} & \\ \mathcal{E}(-, \mathrm{colim}_{m<n} X_m) & & \end{array}$$

n+1) Let us consider the class of diagrams of the following form:

$$\begin{array}{ccc} \mathcal{E}(-, A) & \longrightarrow & \mathcal{E}(-, X_n) \\ {\scriptstyle j_*}\downarrow & & \downarrow \\ \mathcal{E}(-, B) & \longrightarrow & \mathbb{Z} \end{array}$$

where j is in $\mathcal{J}$ and the square commutes up to natural isomorphism. Let us pick a class of representatives up to isomorphism. We denote them with $\mathcal{S}_n$. We identify diagrams that have isomorphic evaluations at every object of $\mathcal{E}$. Since $\mathbb{Z}$ is small-valued and $\mathcal{E}$ is locally small such diagrams are small-valued. We deduce that $\mathcal{S}_n$ is a set. We define X_{n+1} as the following pushout:

$$\begin{array}{ccc} \amalg_{s:\mathcal{S}_n} A_s & \longrightarrow & X_n \\ \downarrow & \ulcorner & \downarrow \\ \amalg_{s:\mathcal{S}_n} B_s & \longrightarrow & X_{n+1} \end{array}$$

We already have a diagram commuting up to natural isomorphism of the form:

$$\begin{array}{ccc} \amalg_{s\in\mathcal{S}_n}\mathcal{E}(-, A_s) & \longrightarrow & \mathcal{E}(-, X_n) \\ \downarrow & & \downarrow \\ \amalg_{s\in\mathcal{S}_n}\mathcal{E}(-, B_s) & \longrightarrow & \mathbb{Z} \end{array}$$

Since $\mathbb{Z}$ is a stack of cell complexes the map from the pushout in $\mathrm{Psh}(\mathcal{E})$ extends to $\mathcal{E}(-, X_{n+1})$:

$$\begin{array}{ccccc} \mathcal{E}(-, \amalg_{s:\mathcal{S}_n} A_s) & \longrightarrow & \mathcal{E}(-, X_n) & & \\ \downarrow & & \downarrow & & \\ \mathcal{E}(-, \amalg_{s:\mathcal{S}_n} B_s) & \longrightarrow & Q & \dashrightarrow & \mathbb{Z} \\ & & \downarrow & \nearrow & \\ & & \mathcal{E}(-, X_{n+1}) & & \end{array}$$

which defines an unique map (up to natural isomoprhism) $\mathcal{E}(-, X_{n+1}) \to \mathbb{Z}$.

Since $\mathcal{E}$ is locally presentable (by definition of combinatorial) there exists a regular cardinal λ such that all domains of maps in J are λ-presented. Therefore, $\mathcal{E}(A, -)$ commutes with λ-small colimits. Hence there exists a limit ordinal n such that $A \to X_\lambda$ factors as $A \to X_n \to X_\lambda$:

$$\begin{array}{ccccc} \mathcal{E}(-, A) & \longrightarrow & \mathcal{E}(-, X_n) & \longrightarrow & \mathcal{E}(-, X_\lambda) \\ {\scriptstyle j_*}\downarrow & & & & \\ \mathcal{E}(-, B) & & & & \end{array}$$

By definition of X_{n+1} and the fact that λ is a regular cardinal:

$$\begin{array}{ccccccc} \mathcal{E}(-, A) & \longrightarrow & \mathcal{E}(-, X_n) & \longrightarrow & \mathcal{E}(-, X_\lambda) & \longrightarrow & \mathbb{Z} \\ {\scriptstyle j_*}\downarrow & & \downarrow & \nearrow & & & \\ \mathcal{E}(-, B) & \dashrightarrow & \mathcal{E}(-, X_{n+1}) & & & & \end{array}$$

This yields to the following lifting solution:

$$\begin{array}{ccccc} \mathcal{E}(-,A) & \longrightarrow & \mathcal{E}(-,X_n) & \longrightarrow & \mathcal{E}(-,X_\lambda) \longrightarrow \mathbb{Z} \\ {\scriptstyle j_*}\downarrow & & \downarrow & \nearrow & \\ \mathcal{E}(-,B) & \dashrightarrow & \mathcal{E}(-,X_{n+1}) & & \end{array}$$

We deduce that f_λ is an acyclic fibration. Notice that $i : X \to X_\lambda$ is obtained through operations that preserve cofibrations by definition of combinatorial model structure. □

Corollary 3.36. Let $\mathcal{E}$ be a Grothendieck topos with a combinatorial model structure where fibrations are monomorphisms. Let $\mathbb{F}$ be a locally representable notion of fibered structure on $\mathcal{E}$. Then $\mathbb{F}$ has a universe.

Proof. We know that $\mathbb{F}$ is a stack of cell complexes by virtue of Lemma 3.34. We can apply Theorem 3.35 to $\mathcal{E}(-,\varnothing) \to \mathbb{F}$:

$$\mathcal{E}(-,\varnothing) \xrightarrow{j_*} \mathcal{E}(-,\mathcal{U}) \xrightarrow{p} \mathbb{Z}$$

Since $j : \varnothing \to \mathcal{U}$ is a cofibration, $\mathcal{U}$ is indeed an universe for $\mathbb{F}$. □

Remark 3.37. We now need to relax the assumptions of this theorem to allow notions of a fibered structure $\mathbb{F} \to \mathbb{E}$ such that $|\mathbb{F}| := \mathrm{Img}(\mathbb{F}) \to \mathbb{E}$ is given by fibrations with λ-small fibers.

Definition 3.38. Let $\mathbb{F}$ be a notion of fibered structure on a model category $\mathcal{E}$. We say that $\mathbb{F}$ is relatively acyclic if $\mathbb{F} \to |\mathbb{F}|$ is an acyclic fibration.

Example 3.39. Any full notion of fibered structure is relatively acyclic.

Proposition 3.40. Let $\mathcal{E}$ be a Grothendieck 1-topos with a combinatorial model structure whose cofibrations are also monomorphisms. Let $\mathbb{F}$ be a small valued, locally representable and relatively acyclic notion of fibered structure on $\mathcal{E}$. Then $|\mathbb{F}|$ has a universe

Proof. The composition of acyclic fibrations is an acyclic fibration. □

Proposition 3.41 ([Shu19] 5.18)**.** Let $\mathcal{E}$ be a Cisinski model category. Let $\mathbb{F}$ be a relatively acyclic notion of fibered structure on $\mathcal{E}$. Let E be a cartesian factorization on $\mathcal{E}$ that factors any $\mathbb{F}$-algebra as an acyclic cofibration followed by a fibration. Then $\mathbb{F} \times_{\mathbb{E}} \mathbb{R}_E$ is relatively acyclic for $\mathbb{R}_E$ as in Example 3.19.

Proof. For any notion of fibered structure $\mathbb{Z} \to \mathbb{E}$, $\mathbb{Z}$ is relatively acyclic if there exists a solution to any lifting problem as the following:

$$\begin{array}{ccc} \mathcal{E}(-,Y') & \longrightarrow & \mathbb{Z} \\ {\scriptstyle i_*}\downarrow & \nearrow & \downarrow{\scriptstyle \phi} \\ \mathcal{E}(-,Y) & \longrightarrow & |\mathbb{Z}| \end{array}$$

when i is a cofibration. By the Yoneda lemma, this is equivalent to the following property for $\mathbb{F}$: for any cartesian square of the form

$$\begin{array}{ccc} X' & \xrightarrow{g} & X \\ {\scriptstyle f'}\downarrow & \lrcorner & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[i]{} & Y \end{array}$$

where f and f' are equipped with a $\mathbb{Z}$-algebra; then there exists a choice of a $\mathbb{Z}$-algebra on f such that the square is a $\mathbb{Z}$-morphism. We prove this second formulation for $\mathbb{Z} = \mathbb{R}_E \times_{\mathbb{E}} \mathbb{F}$. Consider a square

$$\begin{array}{ccc} X' & \xrightarrow{g} & X \\ {\scriptstyle f'}\downarrow & \lrcorner & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[i]{} & Y \end{array}$$

where i is a cofibration and both f and f' are $\mathbb{F}$-algebras as well as $\mathbb{R}_E$-algebras. Let us apply the functorial factorization E to it:

$$\begin{array}{ccc} X' & \xrightarrow{g} & X \\ {\scriptstyle \lambda_{f'}}\downarrow & \lrcorner & \downarrow{\scriptstyle \lambda_f} \\ E_{f'} & \longrightarrow & E_f \\ {\scriptstyle \rho_{f'}}\downarrow & \lrcorner & \downarrow{\scriptstyle \rho_f} \\ Y' & \xrightarrow[i]{} & Y \end{array}$$

By definition of $\mathbb{R}_E$ in Example 3.19, the $\mathbb{R}_E$ algebras on f and f' are, respectively, solutions to the following lifting problems:

$$\begin{array}{ccc} X & = & X \\ \downarrow & \overline{r_f} \nearrow & \downarrow{\scriptstyle f} \\ E_f & \longrightarrow & Y \end{array} \qquad \begin{array}{ccc} X' & = & X' \\ \downarrow & r_{f'} \nearrow & \downarrow{\scriptstyle f'} \\ E_{f'} & \longrightarrow & Y' \end{array}$$

Since $\mathbb{F}$ is relatively acyclic we can find an $\mathbb{F}$-algebra on f such that the previous square is an $\mathbb{F}$-morphism. We are left to find a map r_f as in the following diagram such that the starting square is an $\mathbb{R}_E$-morphism

$$\begin{array}{ccc} X & = & X \\ \downarrow & r_f \nearrow & \downarrow{\scriptstyle f} \\ E_f & \longrightarrow & Y \end{array}$$

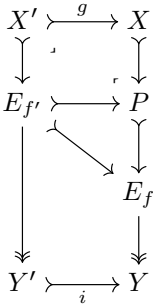
The original square is an $\mathbb{R}_E$-morphism if and only if the following square commutes:

$$\begin{array}{ccc} E_{f'} & \longrightarrow & E_f \\ {\scriptstyle r_{f'}}\downarrow & & \downarrow{\scriptstyle r_f} \\ X' & \longrightarrow & X \end{array}$$

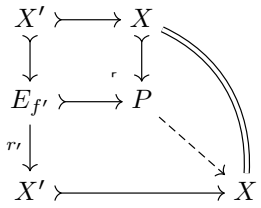
First we notice that f is a fibration since it is a retract of ρ_f:

$$\begin{array}{ccccc} X & \xrightarrow{\lambda_f} & E_f & \xrightarrow{r_f} & X \\ {\scriptstyle f}\downarrow & & {\scriptstyle \rho_f}\downarrow & & \downarrow{\scriptstyle f} \\ Y & = & Y & = & Y \end{array}$$

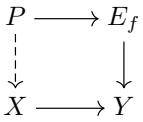
Let us consider the pushout of $E_{f'} \leftarrow X' \rightarrow X$. Since cofibrations coincide with monomorphisms and $\mathcal{E}$ is a topos, the pushout square is also cartesian:



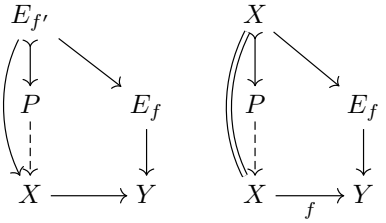
Since $\mathcal{E}$ is a topos, $P \rightarrow E_f$ is a monomorphism and a weak equivalence by 2-out-of-3. We can define a map $P \rightarrow X$ by the universal property of the pushout:



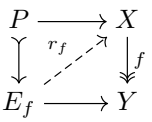
We want to verify that the following square commutes:



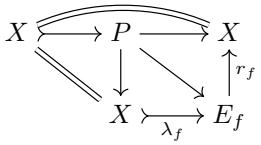
This follows by the uniqueness of the induced map from a pushout and the commutativity of the two following diagrams:



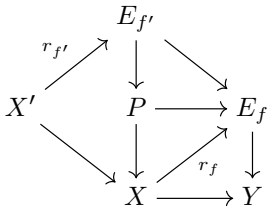
We deduce that the following lifting problem has a solution since f is a fibration:



The following diagram proves that r_f is a retraction of λ_f:



The following diagram commutes by construction:



This proves that r_f is an $\mathbb{R}_E$-algebra and that it is compatible with $r_{f'}$. □

Remark 3.42. The previous proof hints towards the fact that if there is a functorial factorization E such that an arrow f factors into a trivial cofibration followed by a fibration, then f admits an $\mathbb{R}_E$-algebra if and only if f is a fibration. Therefore we can consider $\mathbb{F} = \mathbb{E}^\lambda$ and we deduce that $|\mathbb{F} \times_{\mathbb{E}} \mathbb{R}_E|$ is relatively acyclic whenever there is such factorization.

Remark 3.43. If there exists a cartesian functorial factorization as in Remark 3.42 then by Example 3.24 we know that $\mathbb{R}_E \times_{\mathbb{E}} \mathbb{E}^\lambda$ is locally representable. We denote this notion of fibered structure with $\mathbb{F}\mathrm{ib}^\lambda$.

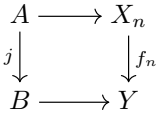
Proposition 3.44. Any Cisinski model category $\mathcal{E}$ admits a functorial factorization as in Remark 3.42.

Proof. We prove this similarly to Theorem 3.35 since the two results are analogous (they are both instances of the small object argument). We know there exists a small set of morphisms J such that $r(J) = \mathrm{Fib}$. We deduce that $l(r(J))$ is the class of trivial cofibrations of $\mathcal{E}$. Let $f : X \to Y$ be any morphism. We construct a cell complex sequence by transfinite induction.

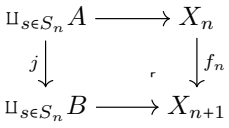
0) $X_0 = X$, $f_0 = f$.

Limit) $X_n = \mathrm{colim}_{m<n} X_m$. This defines a map $f_n : X_n \to Y$ by the universal property of colimits.

n+1) Let S_n be the set of squares of the form

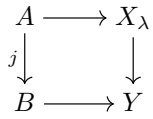


It is a set since J is a set. Therefore we can consider the pushout

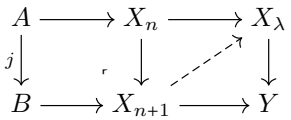


which naturally provides a map $X_{n+1} \to Y$

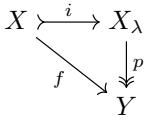
Since $\mathcal{E}$ is locally presentable any domain of a map in J is λ-presentable for some regular cardinal λ. Therefore, for any square of the form



where j is in J. The map $A \to X_\lambda$ factors: $A \to X_n \to X_\lambda$ for some $n \leq \lambda$. Since λ is regular X_{n+1} also gives a map into X_λ. We observe from the following diagram that $X_\lambda \to Y$ lifts against all maps in J:



We deduce that f_λ is a fibration. We have obtained a factorization



By Remark 2.5 we deduce that i is a trivial cofibration. So far we have not used the fact that $\mathcal{E}$ is a right proper Cisinski model category; however, since $\mathcal{E}$ is a topos, pullbacks preserve colimits, in particular the colimits that lead to this factorization. We deduce that with these additional assumptions the factorization is a cartesian functorial factorization. □

Theorem 3.45 ([Shu19] 6.3)**.** In a Cisinski model category $\mathcal{E}$ the full notion of fibered structure spanned by λ-small fibrations admits a universe as in Definition 3.29.

Proof. By virtue of Proposition 3.44 we can construct a cartesian functorial factorization E that factors all arrows into a trivial cofibration followed by a fibration. We know by Example 3.24 that $\mathbb{E}^\lambda \times_{\mathbb{E}} \mathbb{R}_E$ is locally representable. We know that $\mathbb{E}^\lambda$ is relatively acyclic since it is a full notion of fibered structure. By Proposition 3.41 we deduce that $\mathbb{E}^\lambda \times_{\mathbb{E}} \mathbb{R}_E$ is relatively acyclic. By Proposition 3.40 we know that it has a universe. Lastly by Remark 3.42 we know that $|\mathbb{E}^\lambda \times_{\mathbb{E}} \mathbb{R}_E|$ is the full notion of fibered structure spanned by λ-small fibrations. □

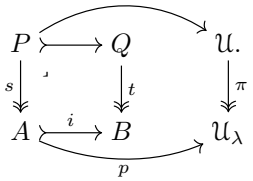
3.3. **Proof of univalence.**

Discussion 3.46. Our recipe to prove that a certain fibration is a strict univalent universe follows [KLV18] and the notation in [Shu15] which is the following:

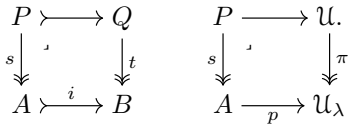
1) Build a fibration $\mathcal{U}_\bullet \to \mathcal{U}$
2) Prove that all λ-small fibrations come from a strict pullback of π
3) Prove that $\mathcal{U}$ is fibrant
4) Prove that $\mathrm{Eq}(\mathcal{U}_\bullet) \to \mathcal{U}$ is a trivial fibration

Notice that 2), 3) and 4) can be modified to

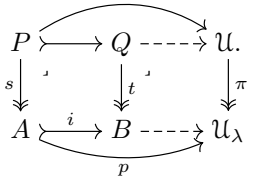
2') For any diagram



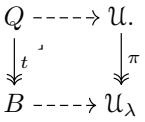
where i is a cofibration, s, t are fibrations with λ-small fibers and the following squares are cartesian:



then we need to complete the first diagram to

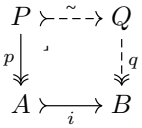


in such a way that the square

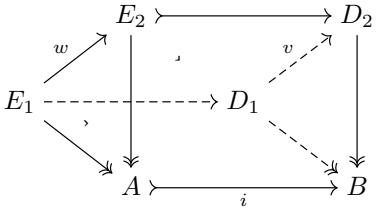


is cartesian.

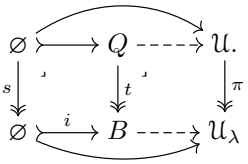
3') For any fibration $P \to A$ and any trivial cofibration $i : A \to B$ there exists a fibration q on B such that the following diagram is cartesian:



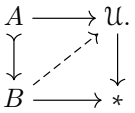
4') Let $i : A \to B$ be a cofibration. Let $E_1 \to A$ and $D_2 \to B$ be two λ-small fibrations. Let $w : E_1 \to E_2$ be a weak equivalence over A. Denote by $E_2 \to A$ the pullback of $D_2 \to B$ along i. Then there exists a λ-small fibration $D_1 \to B$ and a weak equivalence $v : D_1 \to D_2$ that pulls back to w along i:



Proof. If all objects are cofibrant (as in a Cisinski model category) 2') implies 2):



Assume 2'). 3) is equivalent to the existence of a solution to the following lifting problem:



By 2) this is equivalent to 3'). By definition of $\mathrm{Eq}(\mathcal{U}_\bullet)$ and 2) it is obvious that 4) is equivalent to 4'). □

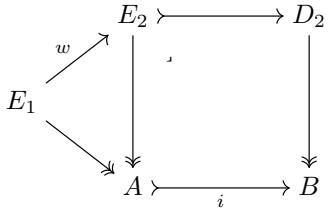
Remark 3.47. Notice that any universe for the notion of fibered structure $|\mathbb{F}\mathrm{ib}^\lambda|$ satisfies 2') by Remark 3.30.

Remark 3.48. Notice that while 3) and 4) depend on $\mathcal{U}$, 3') and 4') are properties of the model structure in general so if a category has a strict univalent universe, any other object satisfying 2') or 2) is a strict univalent universe.

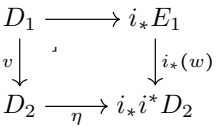
Discussion 3.49. The proof of Theorem 3.50 follows the approach of [KLV18] and [Shu15] adapted to right proper Cisinski model categories. The first reference was suggested in a remark of [Cis14]. In particular, it was implied that there is no need to use simplicial structures.

Theorem 3.50. Let $\mathcal{E}$ be a right proper Cisinski model category. Then $\mathcal{E}$ satisfies property 4') as in Discussion 3.46.

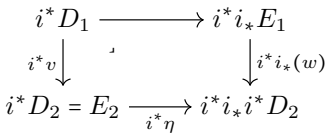
Proof. Let $i : A \to B$ be a cofibration. Let $E_1 \to A$ and $D_2 \to B$ two λ-small fibrations. Let $w : E_1 \to E_2$ be a weak equivalence over A. Denote with $E_2 \to A$ the pullback of $D_2 \to B$ along i:



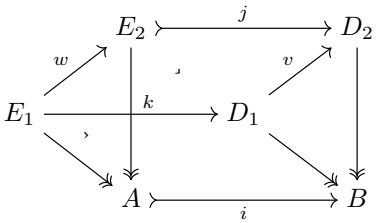
We define D_1 and v as the following base change:



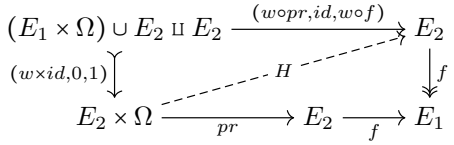
Since i is a monomorphism i_* is full and faithful and $i^*\eta$ is an isomorphism. Since i^* is a right adjoint of $i_!$, it preserves pullbacks, which implies that the following diagram is cartesian:



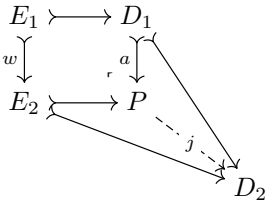
This proves that $i^*D_1 \cong i^*i_*E_1 \cong E_1$ where the last isomorphism is the inverse of the counit. We obtain the following diagram:



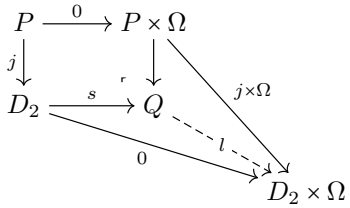
We are left to prove that v is a weak equivalence. By virtue of Example 2.7 we can pick the subobject classifier, Ω, as an interval object. By Lemma 2.30 we are left to prove that $v : D_1 \hookrightarrow D_2$ is a deformation retract. Now we adapt the proof of [KLV18](3.4.1) as suggested in [Cis14](2.19). We can factor w as a trivial cofibration followed by a fibration. Since w was a weak equivalence to begin with, both factors are weak equivalences by 2-out-of-3. We reduce to study the case in which w is a trivial fibration or a trivial cofibration. If w is a fibration, we know that i_* preserves trivial fibrations since i^* preserves cofibrations. Hence v is a trivial fibration, since it is a base change of a trivial fibration. Let us now suppose that w is a trivial cofibration. We shall exhibit $v : D_1 \to D_2$ as a deformation retract; by virtue of Lemma 2.30 it suffices to show that it is a trivial cofibration. First we can exhibit $E_1 \to E_2$ as a deformation retract by Lemma 2.30 with a homotopy of the form



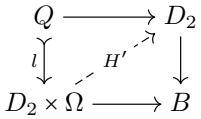
Let us now consider



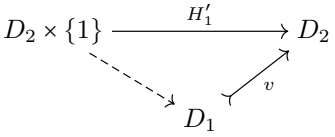
Observe that in the following pushout



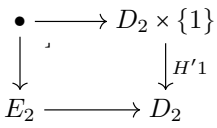
l is a trivial cofibration by 2-out-of-3 (s is trivial because it is a pushout of a trivial cofibration). Let us now construct a map $Q \to D_2$ compatible with l, which assembles naturally by choice of the conditions imposed on H.The following lifting problem has a solution:



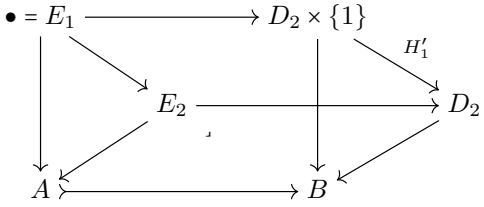
We claim that H' exhibits $D_1 \to D_2$ as a deformation retract. We thus want to prove that



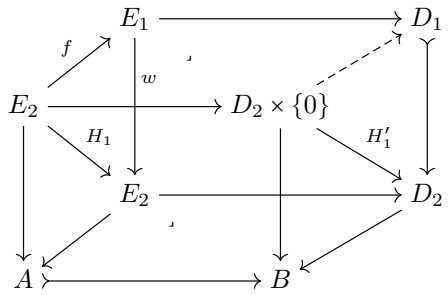
Consider the following pullback:



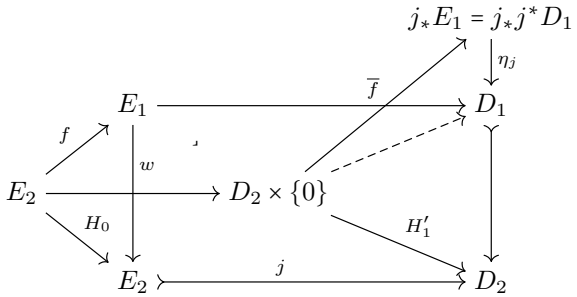
By pullback pasting we can compute $\bullet$:



Now, by definition of $H_1 : D_2 \to D_2$ and there is a factorization of the following kind:



By the adjunction $j^* \dashv j_*$ we obtain a map $\overline{f}$ that, when composed with η_j, gives

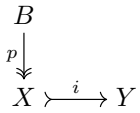


as desired. □

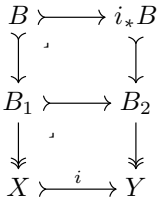
Remark 3.51. In the previous proof the choice of an interval object is arbitrary as already stated in section 2. The author of this work used the sub-object classifier when he wrote this proof for the first time.

Theorem 3.52. Let $\mathcal{E}$ be a right proper Cisinski model category. Then $\mathcal{E}$ satisfies 3') as in Discussion 3.46.

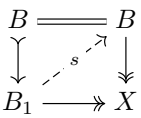
Proof. Let $B \twoheadrightarrow X$ be a fibration and $X \to Y$ be a trivial cofibration.



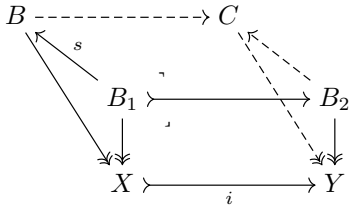
Since i is a monomorphism, i_* is full and faithful. Since $\mathcal{E}$ is a right proper Cisinski model category, we can construct a cartesian functorial factorization into a trivial cofibration followed by a fibration. This produces the following diagram:



We can construct a lifting problem that clearly has a solution:



s is a weak equivalence by 2-out-of-3. We can apply Theorem 3.50 as follows:



which provides a fibration as desired. □

3.4. **Construction of the universe via Giraud's theorem.**

Discussion 3.53. We have illustrated Shulman's proof of the existence of the universe through the 2-categorical small object argument and relatively acyclic notions of fibered structures. We would however like to provide a more abstract proof of the existence of the universe using 'very old' technology that would simplify the proof.

Notation 3.54. Let $\mathcal{E}$ denote a right proper Cisinski model category. Let $\mathcal{E}_k$ denote the full subcategory of $\mathcal{E}$ spanned by k-compact objects. Since $\mathcal{E}$ is a topos, there exists a cardinal k such that $\mathcal{E} \simeq \mathrm{Sh}(\mathcal{E}_k)$ if we endow $\mathcal{E}_k$ with the canonical topology. Let us denote a universe of small 1-categories with Cat_1.

Remark 3.55. Let $F : \mathcal{C}^{\mathrm{op}} \to \mathrm{Cat}_1$ be a functor of categories. Let X be an object of $\mathcal{C}$. Let us consider the Grothendieck construction:

$$\begin{array}{ccccc} \int F_X & \longrightarrow & \int F & \longrightarrow & (\mathrm{Cat}_1)_\bullet \\ \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow \\ * & \xrightarrow[X]{} & \mathcal{C}^{\mathrm{op}} & \xrightarrow[F]{} & \mathrm{Cat}_1 \end{array}$$

Since the Grothendieck construction is an equivalence of categories, $\int F_X$ is isomorphic to $F(X)$.

Theorem 3.56 ([Gir71] 2.2.3.2-2.2.4)**.** Let E be a small site. Let $F \to E$ be a cartesian fibration. The following hold:

2.2.4) there exists a 2-functor $A : \mathrm{cocart}(E) \to \mathrm{Stacks}(E)$ together with a natural transformation $a : id_{\mathrm{Cocart}(E)} \to \int \mathrm{A}$.

2.2.3.2) $A(F)$ is a sheaf of categories and $a : F \to \int \mathrm{A}(F)$ is an equivalence of categories

By the Grothendieck construction, there exists an f such as follows, making the following square cartesian:

$$\begin{array}{ccc} F & \longrightarrow & (\mathrm{Cat}_1)_\bullet \\ \downarrow & \lrcorner & \downarrow \\ E & \xrightarrow[f]{} & \mathrm{Cat}_1 \end{array}$$

and the natural transformation $a : F \Rightarrow \int \mathrm{A}(F)$ is an equivalence of categories objectwise. Moreover, $\mathrm{A}(F)$ is an sheaf of categories on E.

Construction 3.57. Let us consider a cardinal: $\lambda \rhd k$. We provide a universal fibration $\pi : \mathcal{U}_{\lambda\bullet} \twoheadrightarrow \mathcal{U}_\lambda$ for λ-small fibrations as in the following sense: for any λ-small fibration $f : X \twoheadrightarrow Y$, there exists a pullback square of the form:

$$\begin{array}{ccc} X & \longrightarrow & \mathcal{U}_{\lambda\bullet} \\ {\scriptstyle f}\downarrow & \lrcorner & \downarrow{\scriptstyle \pi} \\ Y & \longrightarrow & \mathcal{U}_\lambda \end{array}$$

Let us consider the notion of fibered structure $\mathbb{F}\mathrm{ib}^\lambda$ defined in Remark 3.43. $\mathbb{F}\mathrm{ib}^\lambda \to \mathbb{E}$ is locally representable and $\mathbb{E} \to *$ is also locally representable. By Lemma 3.25, we deduce that $\mathbb{F}\mathrm{ib}^\lambda$ commutes with k-small colimits. This implies it is a stack on $\mathcal{E}_k$ for the canonical topology. By Theorem 3.56, we deduce there exists a natural transformation $a : \mathbb{F}\mathrm{ib}^\lambda \to \mathrm{A}(\int \mathbb{F}\mathrm{ib}^\lambda)$ which is an equivalence when evaluated at any object. $\mathrm{Ob}(\mathrm{A}(F))$ is a sheaf of sets for the canonical topology on $\mathcal{E}_k$. Since $\mathcal{E}_k$ is endowed with the canonical topology, we deduce that$\mathrm{Ob}(\mathrm{A}(\mathbb{F}\mathrm{ib}^\lambda))$ is represented by an object $\mathcal{U}_\lambda$. Let us consider $id_{\mathcal{U}_\lambda} \in \mathcal{E}(\mathcal{U}_\lambda, \mathcal{U}_\lambda)$. By construction, $\mathcal{E}(\mathcal{U}_\lambda, \mathcal{U}_\lambda) \cong \mathrm{Ob} \int \mathrm{A}(\mathbb{F}\mathrm{ib}^\lambda)(\mathcal{U}_\lambda)$. Via this isomorphism, $id_{\mathcal{U}_\lambda}$ corresponds to an object of $\mathrm{A} \int \mathbb{F}\mathrm{ib}^\lambda(\mathcal{U}_\lambda)$. Since $a : \int \mathrm{A}(\mathbb{F}\mathrm{ib}^\lambda)(\mathcal{U}_\lambda) \to \mathbb{F}\mathrm{ib}^\lambda(\mathcal{U}_\lambda)$ is an equivalence, such object is isomorphic to the image of an object of $\mathbb{F}\mathrm{ib}^\lambda(\mathcal{U}_\lambda)$. If we forget the $\mathbb{F}\mathrm{ib}^\lambda$-algebra structure, we obtain a morphism $\pi : \mathcal{U}_{\lambda\bullet} \to \mathcal{U}_\lambda$ that is a fibration by Remark 3.42.

Theorem 3.58. The fibration $\pi : \mathcal{U}_{\lambda\bullet} \twoheadrightarrow \mathcal{U}_\lambda$ constructed in Construction 3.57 is a strict univalent universe for λ-small fibrations in $\mathcal{E}$.

Proof. We already know that 3') and 4') do not depend on the fibration chosen. Therefore we reduce to show that π is universal for λ-small fibrations in the sense of Construction 3.57. Let us consider a λ-small fibration $p : X \twoheadrightarrow Y$ in $\mathcal{E}$. We can always consider a $\mathbb{F}\mathrm{ib}^\lambda$-algebra structure on it by Remark 3.42. This element corresponds to a quasi-unique object of $\int \mathrm{A}(\mathbb{F}\mathrm{ib}^\lambda)(Y)$. By construction, this defines a unique element $f \in \mathcal{E}(Y, \mathcal{U}_\lambda)$. The fact that $\pi^* f = p$ is a consequence of the Yoneda lemma: let us consider a diagram

$$id_{\mathcal{U}_\lambda} \longmapsto \overline{\pi}$$

$$\begin{array}{ccc} \mathcal{E}(\mathcal{U}_\lambda, \mathcal{U}_\lambda) & \xrightarrow{\cong} & \mathrm{Ob} \int \mathrm{A}(\mathbb{F}\mathrm{ib}^\lambda)(\mathcal{U}_\lambda) \\ {\scriptstyle -\circ f}\downarrow & & \downarrow \\ \mathcal{E}(Y, \mathcal{U}_\lambda) & \xrightarrow{\cong} & \mathrm{Ob} \int \mathrm{A}(\mathbb{F}\mathrm{ib}^\lambda)(Y) \end{array}$$

$$f \longmapsto \overline{f}$$

Up to canonical equivalence,

$$\overline{\pi} \longmapsto \pi$$

$$\begin{array}{ccccc} \int \mathrm{A}(\mathbb{F}\mathrm{ib}^\lambda)(\mathcal{U}_\lambda) & \xrightarrow{a_{\mathcal{U}_\lambda}} & \mathbb{F}\mathrm{ib}^\lambda(\mathcal{U}_\lambda) & \longrightarrow & |\mathbb{F}\mathrm{ib}^\lambda|(\mathcal{U}_\lambda) \\ \downarrow & & \downarrow{\scriptstyle f^*} & & \downarrow{\scriptstyle f^*} \\ \int \mathrm{A}(\mathbb{F}\mathrm{ib}^\lambda)(Y) & \xrightarrow{a_Y} & \mathbb{F}\mathrm{ib}^\lambda(Y) & \longrightarrow & |\mathbb{F}\mathrm{ib}^\lambda|(Y) \end{array}$$

$$\overline{f} \longmapsto p$$

which translates to:

$$\begin{array}{ccc} X & \longrightarrow & \mathcal{U}_{\lambda\bullet} \\ {\scriptstyle p}\downarrow & \lrcorner & \downarrow{\scriptstyle \pi} \\ Y & \xrightarrow[f]{} & \mathcal{U}_\lambda \end{array}$$

□

4. Uniqueness of universes up to a contractible space

Notation 4.1. In this section, the terms 'category' or '∞-category' will denote $(\infty,1)$-categories unless otherwise specified. lim and colim will denote limits and colimits in ∞-categories unless specified otherwise. Model categories will always be 'honest' 1-categories. Psh will denote the ∞-category of ∞-presheaves (in contrast with previous sections). The ∞-category of spaces will be denoted by An (animae) to distinguish them from 1-groupoids of Section 3.

Discussion 4.2. For some results in this section we decided to reference [CCNW26] because the author of this work is more familiar with the synthetic approach. The proofs should remain valid and can be found also in [Cis19].

4.1. Calculus of fractions and right approximability.

Notation 4.3. Let $\mathcal{C}$ be any ∞-category. Whenever we specify a subcategory $\mathrm{W} \subseteq \mathcal{C}$ to which we refer as 'weak equivalences', then $\gamma : \mathcal{C} \to \mathrm{L}(\mathcal{C})$ denotes the localization functor $\mathcal{C} \to \mathcal{C}[\mathrm{W}^{-1}]$.

Lemma 4.4 ([CCNW26] 8.1.19)**.** Let $u : A \to B$ be a functor between locally small categories, then the induced functor:

$$u^* : \mathrm{Fun}(A^{\mathrm{op}}, \mathrm{An}) \to \mathrm{Fun}(B^{\mathrm{op}}, \mathrm{An})$$

has a left adjoint $u_!$.

Proof. Let us consider a universe of animae $\pi : \mathrm{An}_\bullet \twoheadrightarrow \mathrm{An}$. Let $F : A^{\mathrm{op}} \to \mathrm{An}$ be a functor. This corresponds to a right fibration $p : E \to A$. Consider a factorization $u \circ p = q \circ j$ into a final functor followed by a right fibration. There exists a classifying functor of q^{op} as follows:

$$\begin{array}{ccc} \mathcal{D}^{\mathrm{op}} & \longrightarrow & \mathrm{An}_\bullet \\ {\scriptstyle q}\downarrow & \lrcorner & \downarrow \\ B^{\mathrm{op}} & \xrightarrow[u_!(F)]{} & \mathrm{An} \end{array}$$

For any $G : B^{\mathrm{op}} \to \mathrm{An}$, we have the following equivalences: $\mathrm{Hom}_{\mathrm{Fun}(B^{\mathrm{op}},\mathrm{An})}(u_!(F), G) \simeq \mathrm{Map}_B(q, \pi^* G)$ by straightening and unstraightening. Since j is final, the right-hand side is equivalent to $\mathrm{Map}_B(q \circ j, \pi^* G) \simeq \mathrm{Map}_B(u \circ p, \pi^* G) \simeq \mathrm{Map}_A(p, u^* \pi^* G)$ where the last equivalence holds by the adjunction $u_! \dashv u^*$ for slices. By applying unstraightening once again, we notice that the right term is equivalent to $\mathrm{Hom}_{\mathrm{Fun}(A^{\mathrm{op}},\mathrm{An})}(F, G \circ u)$. By the Yoneda lemma, we can assemble all the data of $u_!(F)$ into a functor that is automatically an adjoint to u^*. □

Remark 4.5. Under the assumptions of the previous lemma, if u^* is an equivalence, by straightening and unstraightening, it is equivalent to u being fully faithful.

Definition 4.6 ([Cis19] 7.4.12)**.** Let $\mathcal{C}$ be an ∞-category together with two classes of morphisms Fib and W. We say that the triple $(\mathcal{C}, \mathrm{Fib}, \mathrm{W})$ is a category with weak equivalences and fibrations if it satisfies the following properties:

(1) $\mathcal{C}$ has a terminal object
(2) W satisfies the 2-out-of-3 property
(3) Fib is closed under composition
(4) for any cartesian square as below, in which p is a fibration between fibrant objects and y' is fibrant

$$\begin{array}{ccc} x' & \longrightarrow & x \\ {\scriptstyle p'}\downarrow & \lrcorner & \downarrow{\scriptstyle p} \\ y' & \longrightarrow & y \end{array}$$

if p is in W, p' is in W.
(5) any map with fibrant codomain factors as a weak equivalence followed by a fibration.

Definition 4.7. Let $\mathcal{C}$ be an ∞-category with a class of weak equivalences W and fibrations Fib. We say that W is saturated if any f that is sent to an invertible morphism by $\gamma : \mathcal{C} \to \mathrm{L}(\mathcal{C})$ is in W.

Example 4.8. Let $\mathcal{M}$ be a model category. Then $\mathrm{N}(\mathcal{M})$ is an ∞-category with weak equivalences and fibrations.

Definition 4.9 ([CCNW26] 10.3.1)**.** Let $\mathcal{C}$ be an ∞-category with a class of weak equivalences W that contains equivalences and has the 2-out-of-3 property. Let us denote by $\gamma : \mathcal{C} \to \mathrm{L}(\mathcal{C})$ the localization of $\mathcal{C}$ at W. A putative right calculus of fractions at an object x in $\mathcal{C}$ is a functor

$$p : \mathrm{W}(x) \to \mathcal{C}/_x$$

that satisfies the following properties:

(1) for any object w in $\mathrm{W}(x)$, $p(w)$ is an object of $\mathcal{C}/_x$ that is sent to an invertible map via γ
(2) there is an object i_x in $\mathrm{W}(x)$ such that $p(i_x) = id_x$
(3) $\mathrm{W}(x)$ is weakly contractible

Definition 4.10. Let $\mathcal{C}$ be an ∞-category with a class of weak equivalences W that contains equivalences and has the 2-out-of-3 property. Let us denote by $\gamma : \mathcal{C} \to \mathrm{L}(\mathcal{C})$ the localization of $\mathcal{C}$ at W. Let x be an object in $\mathcal{C}$. A right calculus of fractions at x is a putative right calculus of fractions

$$p : \mathrm{W}(x) \to \mathcal{C}/_x$$

such that the functor $y \mapsto \mathrm{colim}_{z:\mathrm{W}(x)^{\mathrm{op}}} \mathrm{Hom}_{\mathcal{C}}(z, y)$ sends every map in W to an equivalence of animae.

Proposition 4.11 ([CCNW26] 10.3.5)**.** Let $p : \mathrm{W}(x) \to \mathcal{C}/_x$ be a putative right calculus of fractions at x. Then it is a right calculus of fractions if and only if there is a map

$$\mathrm{can}_y : \mathrm{colim}_{z:\mathrm{W}(x)^{\mathrm{op}}} \mathrm{Hom}_{\mathcal{C}}(z, y) \simeq \mathrm{Hom}_{\mathrm{L}(\mathcal{C})}(x, y)$$

functorial in y, that is an equivalence for any y.

Proof. Clearly $\mathrm{Hom}_{\mathrm{L}(\mathcal{C})}(x, -)$ sends morphisms in W to equivalences. We conclude that if there exists a natural equivalence $\mathrm{can} : \mathrm{colim}_{z:\mathrm{W}(x)^{\mathrm{op}}} \mathrm{Hom}_{\mathcal{C}}(z, -) \simeq \mathrm{Hom}_{\mathrm{L}(\mathcal{C})}(x, -)$, then $\mathrm{colim}_{z:\mathrm{W}(x)^{\mathrm{op}}} \mathrm{Hom}_{\mathcal{C}}(z, -)$ sends W to equivalences in $\mathrm{L}(\mathcal{C})$. We are left to prove that if p is a calculus of fractions, can_y is an equivalence for every y. We have an adjunction obtained in Lemma 4.4:

$$\mathrm{Fun}(\mathcal{C}, \mathrm{An}) \underset{\gamma^*}{\overset{\gamma_!}{\rightleftarrows}} \mathrm{Fun}(\mathrm{L}(\mathcal{C}), \mathrm{An}) \quad (\gamma_! \dashv \gamma^*)$$

We know that γ^* is fully faithful. Let us consider the map $\eta : \mathrm{Hom}_{\mathcal{C}}(x, -) \to \gamma^* \gamma_! \mathrm{Hom}_{\mathcal{C}}(x, -) = \mathrm{Hom}_{\mathrm{L}(\mathcal{C})}(\gamma(x), \gamma(-))$. For $F = \mathrm{colim}_{z:\mathrm{W}(x)^{\mathrm{op}}} \mathrm{Hom}_{\mathcal{C}}(z, -)$ we obtain $\eta_F : \mathrm{colim}_{z:\mathrm{W}(x)^{\mathrm{op}}} \mathrm{Hom}_{\mathcal{C}}(z, -) \to \mathrm{colim}_{z:\mathrm{W}(x)^{\mathrm{op}}} \mathrm{Hom}_{\mathrm{L}(\mathcal{C})}(\gamma(z), \gamma(-))$ which coincides with the colimit of the unit maps of $\mathrm{Hom}_{\mathcal{C}}(z, -)$ for each z in $\mathrm{W}(x)$. η_F is an isomorphism if and only if F sends W to equivalences, since its right adjoint is fully faithful. This is exactly the condition defining a right calculus of fractions. $\mathrm{Hom}_{\mathrm{L}(\mathcal{C})}(x, y) \simeq \mathrm{colim}_{z:\mathrm{W}(x)^{\mathrm{op}}} \mathrm{Hom}(z, y)$ since $\mathrm{W}(x)$, is sent to a contractible anima in $\mathrm{L}(\mathcal{C})$ by definition of putative calculus of fractions. We conclude that the colimit is equivalent to the evaluation at i_x. We assemble these equivalences into the following diagram:

$$\begin{array}{ccc} \mathrm{colim}_{z:\mathrm{W}(x)^{\mathrm{op}}} \mathrm{Hom}_{\mathcal{C}}(z, -) & \xrightarrow{\sim} & \mathrm{colim}_{z:\mathrm{W}(x)^{\mathrm{op}}} \mathrm{Hom}_{\mathrm{L}(\mathcal{C})}(\gamma(z), \gamma(-)) \\ & \searrow^{\mathrm{can}} & \downarrow{\scriptstyle \sim} \\ & & \mathrm{Hom}_{\mathrm{L}(\mathcal{C})}(\gamma(x), \gamma(-)) \end{array}$$

as desired. □

Remark 4.12. We have the following equivalence of ∞-categories: $\operatorname{colim}_{z:\mathrm{W}(x)^{\mathrm{op}}} \mathrm{Hom}_{\mathcal{C}}(z,y) \simeq \pi_\infty \mathrm{W}(x)/_y$ since we have the following pullbacks of ∞-categories:

$$\begin{array}{ccccccc} \mathrm{W}(x)/_y^{\mathrm{op}} & \longrightarrow & (\mathcal{C}/_x)/_y^{\mathrm{op}} & \longrightarrow & \mathcal{C}^{\mathrm{op}}/_y & \longrightarrow & \mathrm{Cat}_\bullet \\ \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow{\scriptstyle \text{forget}} \\ \mathrm{W}(x)^{\mathrm{op}} & \longrightarrow & \mathcal{C}/_x^{\mathrm{op}} & \longrightarrow & \mathcal{C}^{\mathrm{op}} & \xrightarrow[\mathrm{Hom}_{\mathcal{C}}(-,y)]{} & \mathrm{Cat} \end{array}$$

Theorem 4.13 ([Cis19] 7.5.6)**.** Let $\mathcal{C}$ be an ∞-category. $\gamma_f : \mathcal{C}_f \to \mathrm{L}(\mathcal{C}_f)$ is left exact. Moreover, for any ∞-category $\mathcal{D}$ with finite limits, for any left exact functor $F : \mathcal{C}_f \to \mathcal{D}$, the induced functor $\overline{F} : \mathrm{L}(\mathcal{C}_f) \to \mathcal{D}$ (if it exists) is left exact.

Theorem 4.14 ([Cis19] 7.5.18)**.** Let $\mathcal{C}$ be an ∞-category with weak equivalences and fibrations. Then the canonical functor $\bar{i} : \mathrm{L}(\mathcal{C}_f) \to \mathrm{L}(\mathcal{C})$ is an equivalence of ∞-categories. In particular, $\mathrm{L}(\mathcal{C})$ has finite limits and $\gamma : \mathcal{C} \to \mathrm{L}(\mathcal{C})$ is left exact.

Definition 4.15. Let $\mathcal{C}$ and $\mathcal{D}$ be two ∞-categories with weak equivalences W and W′. Let $f : \mathcal{C} \to \mathcal{D}$ be a functor of ∞-categories. We say that f satisfies the right approximation property if the following hold:

(1) a morphism w belongs to W if and only if $f(w)$ belongs in W′
(2) for any object y_0 in $\mathcal{D}$, for any object x_1 in $\mathcal{C}$, for any morphism $\psi : y_0 \to f(x_1)$, there exist an object x_0 and a morphism $\varphi : x_0 \to x_1$ in $\mathcal{C}$ and a weak equivalence $u : y_0 \to f(x_0)$ such that the following triangle commutes:

$$\begin{array}{ccc} y_0 & \xrightarrow{\psi} & f(x_1) \\ {\scriptstyle u}\downarrow & \nearrow_{f(\varphi)} & \\ f(x_0) & & \end{array}$$

Remark 4.16. Let $f : \mathcal{C} \to \mathcal{D}$ be a functor with the right approximation property. For any object X, the induced functor $\mathcal{C}/_X \to \mathcal{D}/_{f(X)}$ has the right approximation property

Remark 4.17 ([Cis19] 7.6.3)**.** Let $\mathcal{C}$ be an ∞-category with weak equivalences and fibrations. Then $\mathcal{C}_f \to \mathcal{C}$ has the right approximation property.

Remark 4.18 ([Cis19] 7.6.4)**.** Let $\mathcal{C}$ be an ∞-category with weak equivalences and fibrations. Then $\mathcal{C} \to \mathrm{L}(\mathcal{C})$ has the right approximation property.

Lemma 4.19 ([Cis19] 7.6.7)**.** Let $f : \mathcal{C} \to \mathcal{C}'$ be a left exact functor of ∞-categories with weak equivalences and fibrations. If f has the right approximation property, then the induced functor between weak equivalences is final.

Corollary 4.20 ([Cis19] 7.6.9)**.** Let $\mathcal{C}$ be an ∞-category with weak equivalences and fibrations. Let us denote by W its weak equivalences. Let us suppose that W is saturated. Then $\mathrm{W} \to \mathrm{L}(\mathcal{C})^{\simeq}$ is final (hence a weak homotopy equivalence).

Proof. We have a diagram as follows:

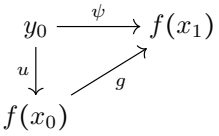
$$\begin{array}{ccc} \mathrm{W}_f & \longrightarrow & \mathrm{W} \\ \downarrow & & \downarrow \\ \mathrm{L}(\mathcal{C}_f)^{\simeq} & \longrightarrow & \mathrm{L}(\mathcal{C})^{\simeq} \end{array}$$

The lower horizontal map is an equivalence by Theorem 4.14. The top horizontal map is final by Lemma 4.19. By Theorem 4.13, $\mathcal{C}_f \to \mathrm{L}(\mathcal{C}_f)$ is left exact. Taking $\mathrm{L}(\mathcal{C}_f)^{\simeq}$ as the class of weak equivalences for $\mathrm{L}(\mathcal{C}_f)$, it also has the right approximation property. We deduce that the left vertical functor is final. We conclude that the right vertical functor is final. □

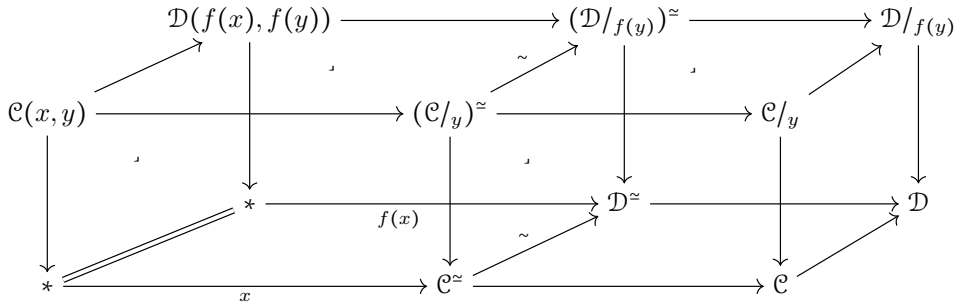
Theorem 4.21 ([Cis19] 7.6.10)**.** Let $\mathcal{C}$ and $\mathcal{D}$ be two ∞-categories with finite limits. Suppose that $f : \mathcal{C} \to \mathcal{D}$ commutes with finite limits. then the following are equivalent:

(1) f is an equivalence of ∞-categories
(2) $\mathrm{ho}(f) : \mathrm{ho}(\mathcal{C}) \to \mathrm{ho}(\mathcal{D})$ is an equivalence of 1-categories
(3) f has the right approximation property

Proof. Let us consider invertible maps as the classes of weak equivalences for both $\mathcal{C}$ and $\mathcal{D}$. (1)⇒(2) is trivial. (2)⇒(3): for any map $\psi : y_0 \to f(x_1)$, we can find an invertible map $u : y_0 \to f(x_0)$ such that the following diagram commutes:



Since $\mathrm{ho}(f)$ is full, we can find a φ such that $f(\varphi) = g$. We prove that (3) ⇒ (1). Let us consider an object y in $\mathcal{D}$. Let e be the terminal object of $\mathcal{C}$. Since f is left exact, $f(e)$ is the terminal object in $\mathcal{D}$. We have a unique arrow $y \to f(e)$. By the right approximation property, there exists an invertible morphism $u : y \to f(x)$ for some x in $\mathcal{C}$. This proves essential surjectivity. Let y be an object in $\mathcal{C}$. Thanks to Lemma 4.19, we deduce that $f^{\simeq} : \mathcal{C}^{\simeq} \to \mathcal{D}^{\simeq}$ is an equivalence. By Remark 4.16, $f/_y : \mathcal{C}/_y \to \mathcal{D}/_{f(y)}$ is left exact and has the right approximation property. Applying once again Lemma 4.19, we deduce the following identity: $(f/_y)^{\simeq} : (\mathcal{C}/_y)^{\simeq} \to (\mathcal{D}/_{f(y)})^{\simeq}$. We know that $\mathrm{forg} : \mathcal{C}/_y \to \mathcal{C}$ is conservative. All the diagrams assemble into the following:



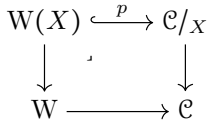
By the pullback pasting lemma, we deduce that the upper left arrow is an equivalence, as desired. □

Lemma 4.22 ([CCNW26] 13.1.2)**.** Let $\mathcal{C}$ be an ∞-category. Let F be a family of morphisms closed under pullbacks. For any $v : y \to y'$ in F, for any x in $\mathcal{C}$, the following functor has a right adjoint:

$$v_! \times_{\mathcal{C}} id : \mathcal{C}/_y \times_{\mathcal{C}} \mathcal{C}/_x \longrightarrow \mathcal{C}/_{y'} \times_{\mathcal{C}} \mathcal{C}/_x$$

Proof. Since the Yoneda embedding is fully faithful and commutes with slices and limits, up to replacing $\mathcal{C}$ with $\mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathrm{An})$, we can assume that $\mathcal{C}$ has pullbacks and a terminal object. Then the statement becomes trivial, since $\mathcal{C}/_y \times_{\mathcal{C}} \mathcal{C}/_x \simeq \mathcal{C}/_{y \times x}$ and $(v \times id)_!$ has a right adjoint $(v \times id)^*$. □

Theorem 4.23 ([CCNW26] 13.1.3)**.** Let $\mathcal{C}$ be an ∞-category. Let W be a class of morphisms closed under composition and pullbacks. Then, for any object X of $\mathcal{C}$, $\mathcal{C} \to \mathcal{C}[\mathrm{W}^{-1}]$ admits a right calculus of fractions given by the base change of $\mathrm{W} \hookrightarrow \mathcal{C}$ along $\mathrm{forg} : \mathcal{C}/_X \to \mathcal{C}$:



Proof. It is trivial to see that $\mathrm{W}(X)$ is a putative calculus of fractions. By Lemma 4.22, for any w in W, $w \times id : \mathcal{C}/_Y \times_{\mathcal{C}} \mathcal{C}/_X \to \mathcal{C}/_{Y'} \times_{\mathcal{C}} \mathcal{C}/_X$ has a right adjoint. Since W is stable under pullback and composition, the adjunction restricts to $w \times id : \mathcal{C}/_Y \times_{\mathcal{C}} \mathrm{W}(X) \to \mathcal{C}/_{Y'} \times_{\mathcal{C}} \mathrm{W}(X)$. This proves that $\pi_\infty(W(X)/_Y) \simeq \pi_\infty(\mathrm{W}(X)/_{Y'})$. □

Definition 4.24. Let W be a class of morphisms in an ∞-category $\mathcal{C}$. We define the saturation of W as the following pullback:

$$\begin{array}{ccc} \overline{\mathrm{W}} & \longrightarrow & \mathcal{C} \\ \downarrow & \lrcorner & \downarrow \\ (\mathcal{C}[\mathrm{W}^{-1}])^{\simeq} & \longrightarrow & \mathcal{C}[\mathrm{W}^{-1}] \end{array}$$

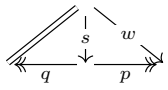
Corollary 4.25 ([Cis19] 7.2.18)**.** Let $\mathcal{C}$ be a category with a class of weak equivalences W. If there exists $\mathrm{W}' \subseteq \mathrm{W}$ closed under composition and pullbacks such that $\mathrm{W} \subseteq \overline{\mathrm{W}'}$, then there exists a right calculus of fractions for $\mathcal{C} \to \mathcal{C}[\mathrm{W}^{-1}]$.

Proof. The previous theorem provides a right calculus of fractions for $\mathcal{C} \to \mathcal{C}[\mathrm{W}'^{-1}]$. By the universal property of the localization, we have the following canonical equivalences: $\mathcal{C}[\mathrm{W}^{-1}] \simeq \mathcal{C}[\overline{\mathrm{W}}^{-1}] \simeq \mathcal{C}[\overline{\mathrm{W}'}^{-1}] \simeq \mathcal{C}[\mathrm{W}'^{-1}]$. This implies that a calculus of fractions for W' which is also a calculus of fractions for W. □

Remark 4.26. Let $\mathcal{C}$ be a category with weak equivalences and fibrations. There exists a right calculus of fractions $\mathrm{W}/_X \hookrightarrow \mathcal{C}/_X$ for X fibrant, where $\mathrm{W}/_X$ is given by the following pullback:

$$\begin{array}{ccc} \mathrm{W}/_X & \hookrightarrow & \mathcal{C}/_X \\ \downarrow & \lrcorner & \downarrow \\ \mathrm{W} & \hookrightarrow & \mathcal{C} \end{array}$$

Any weak equivalence w with fibrant codomain can be factored as follows:



where p is a fibration, s is a weak equivalence and q is both a weak equivalence and a fibration. This follows from the fact that $\mathcal{C}/_X$ inherits the structure of category with weak equivalences and fibrations when X is fibrant. This implies that we can define a notion of path object and mimic the proof of Lemma 3.10. The restriction to fibrant objects is necessary since the factorization and pullback stability properties are only granted when codomains are fibrant. The saturation of the class of maps that are both weak equivalences and fibrations contains all weak equivalences. By Corollary 4.25, there exists a right calculus of fractions. This generality is unnecessary when $\mathcal{C} = \mathrm{N}(\mathcal{M})$ for a model category $\mathcal{M}$.

Theorem 4.27 ([Cis19] 7.6.13)**.** Let $\mathcal{C}$ be a category with weak equivalences and fibrations. Let us denote by W the class of weak equivalences. Let $\gamma : \mathcal{C} \to \mathrm{L}(\mathcal{C})$ be the localization functor at the class of weak equivalences. Let X be a fibrant object in $\mathcal{C}$. Then $\mathcal{C}/_X \to \mathrm{L}(\mathcal{C})/_{\gamma(X)}$ induces an equivalence of ∞-categories:

$$\mathrm{L}(\mathcal{C}/_X) \simeq \mathrm{L}(\mathcal{C})/_{\gamma(X)}$$

where $\mathrm{L}(\mathcal{C}/_X) := \mathcal{C}/_X[(\mathrm{W}/_X)^{-1}]$ with

$$\begin{array}{ccc} \mathrm{W}/_X & \hookrightarrow & \mathcal{C}/_X \\ \downarrow & \lrcorner & \downarrow \\ \mathrm{W} & \hookrightarrow & \mathcal{C} \end{array}$$

Proof. Let us assume that W is saturated (up to replacing W with its saturation). To prove the first property of right approximability, we show that $\mathrm{L}(C/_X) \to \mathrm{L}(C)/_X$ is conservative. By Theorem 4.14, we can assume that $\mathcal{C}/_X = (\mathcal{C}/_X)_f$. Since $\mathrm{L}(\mathcal{C})/_X \to \mathrm{L}(\mathcal{C})$ is always conservative, we are left to prove it for $\mathrm{L}(\mathcal{C}/_X) \to \mathrm{L}(\mathcal{C})$. The weak equivalences of $\mathcal{C}/_X$ are given by the following pullback:

$$\begin{array}{ccc} \mathrm{W}/_X & \hookrightarrow & \mathcal{C}/_X \\ \downarrow & \lrcorner & \downarrow \\ \mathrm{W} & \hookrightarrow & \mathcal{C} \end{array}$$

By Remark 4.26, there exists a right calculus of fractions on $\mathcal{C}/_X$ given by the map:

$$\mathrm{W}/_X \hookrightarrow \mathcal{C}/_X$$

We have the following equivalence $\mathrm{Hom}_{\mathrm{L}(\mathcal{C}/_X)}(A,B) \simeq \mathrm{colim}_{Z\to A, Z:(\mathrm{W}/_X)^{\mathrm{op}}} \mathrm{Hom}_{\mathcal{C}/_X}(Z,B)$. Let us consider a morphism $f : A \to B$ in $\mathrm{L}(\mathcal{C}/_X)$. BY the right calculus of fractions in Remark 4.26, it corresponds to an object of $f_Z : \mathrm{colim}_{Z\to A, Z:(\mathrm{W}/_X)^{\mathrm{op}}} \mathrm{Hom}_{\mathrm{N}(\mathcal{C}/_X)}(Z,B)$. f_i is invertible in $\mathrm{L}(\mathcal{C})$ if and only if f_i is in W, since W is saturated. By definition of $\mathrm{W}/_X$, we conclude that f_i is in $\mathrm{W}/_X$. We deduce that f was invertible in the first place. For the second property of right approximability, we need to show that $\gamma/_X : \mathcal{C}/_X \to \mathrm{L}(\mathcal{C})/_X$ satisfies it. By Remark 4.16, it reduces to showing that $\mathcal{C} \to \mathrm{L}(\mathcal{C})$ satisfies it. We know, by Remark 4.18, that it does. Let us consider the following diagram:

$$\begin{array}{ccccc} & & \mathcal{C}/_X & \xrightarrow{\mathrm{can}_X} & \mathcal{C} \\ & \swarrow^{\gamma_X} & \downarrow{\scriptstyle \gamma/_X} & & \downarrow{\scriptstyle \gamma} \\ \mathrm{L}(\mathcal{C}/_X) & \xrightarrow[\varphi]{} & \mathrm{L}(\mathcal{C})/_X & \xrightarrow[\mathrm{can}_X]{} & \mathrm{L}(\mathcal{C}) \end{array}$$

γ is left exact by Remark 4.17. We deduce that $\gamma/_X$ is left exact. Since X is fibrant, $\mathcal{C}/_X$ has a natural structure of an ∞-category with weak equivalences and fibrations (fibrant objects in $\mathcal{C}/_X$ are also fibrant in $\mathcal{C}$). This implies that γ_X is left exact. By Theorem 4.13, φ is left exact. By Theorem 4.21, we conclude that φ is an equivalence. □

Lemma 4.28 ([nLa26c] 2.3)**.** Let $\mathcal{C}$ and $\mathcal{D}$ be two model categories. Let the following be a Quillen adjunction:

$$\mathcal{C} \underset{R}{\overset{L}{\rightleftarrows}} \mathcal{D}$$

Let us assume that for any morphism f in $\mathcal{D}$, f is a weak equivalence if and only if $R(f)$ is a weak equivalence. Then $L \dashv R$ is a Quillen equivalence if and only if, for any cofibrant object c, the unit of the adjunction $\eta_c : c \to RL(c)$ is a weak equivalence.

Proof. By definition of Quillen equivalence, we have to prove that η_c being a weak equivalence is equivalent to the following conditions:

(1) for any cofibrant object c, for any fibrant replacement

$$L(c) \overset{j}{\rightarrowtail} L(c)' \twoheadrightarrow *$$

the composite map:

$$c \xrightarrow{\eta_c} R(L(c)) \xrightarrow{R(j)} R(L(c)')$$

is a weak equivalence.

(2) for any fibrant object d, for any cofibrant replacement:

$$* \rightarrowtail R(d)' \overset{f}{\twoheadrightarrow} R(d)$$

the composite map

$$L(R(d)') \xrightarrow{L(f)} L(R(d)) \xrightarrow{\epsilon_d} d$$

is a weak equivalence.

Let us suppose that these two conditions hold. Since R preserves weak equivalences, $R(j)$ is a weak equivalence. We conclude that η is a weak equivalence by 2-out-of-3. Suppose that η is a weak equivalence for every cofibrant c. The first composite map is a weak equivalence since $R(j)$ is a weak equivalence. For the second condition we have to verify that $R(\epsilon_d \circ L(f))$ is a weak equivalence. By 2-out-of-3 it is equivalent to $R(\epsilon_d) \circ R(L(f)) \circ \eta$ being a weak equivalence. By the theory of adjunctions, that is the adjoint mate of $\epsilon_d \circ L(f)$. f is another adjoint mate of $\epsilon_d \circ L(f)$. Since the first is a weak equivalence, we conclude by uniqueness of adjoint mates. □

Lemma 4.29. Let $\mathcal{E}$ be a right proper model category. Let $\mathcal{E}_\infty \coloneqq \mathrm{N}(\mathcal{E})[\mathrm{W}^{-1}]$. Then $\mathcal{E}_\infty/_X \simeq \mathrm{L}(\mathrm{N}(\mathcal{E})/_X)$, for any X in $\mathcal{E}$.

Proof. Let X' be a fibrant replacement of X. By Theorem 4.27, we know that $\mathcal{E}_\infty/_{X'} \simeq \mathrm{L}(\mathcal{E}/_{X'})$. Under base change we obtain a diagram of the form:

$$\begin{array}{ccc} \mathrm{L}(\mathcal{E}/_{X'}) & \xrightarrow{\sim} & \mathcal{E}_\infty/_{X'} \\ \downarrow & & \downarrow \\ \mathrm{L}(\mathcal{E}/_X) & \longrightarrow & \mathcal{E}_\infty/X \end{array}$$

The top horizontal map is an equivalence by Theorem 4.27. We prove that the vertical maps are equivalences. The right vertical map is an equivalence since it is a base change of an equivalence in $\mathcal{E}_\infty$. Let $j : X \to X'$ be a map obtained by factoring $X \to *$ into a trivial cofibration and a fibration. It defines an adjunction:

$$\mathcal{E}/_X \underset{j^*}{\overset{j_!}{\rightleftarrows}} \mathcal{E}/_{X'} \quad (\perp)$$

We prove it is a Quillen equivalence. First, notice that it is a Quillen adjunction, since $j_!$ preserves both trivial fibrations and fibrations (it just forgets that morphisms lie over X'). For the same reason, a morphism is a weak equivalence in $\mathcal{E}/_{X'}$ if and only if it is a weak equivalence in $\mathcal{E}/_X$. We want to apply Lemma 4.28 to the opposite categories. We need to verify that, for any fibrant object $z : Z \twoheadrightarrow X'$, its counit ϵ_z is a weak equivalence. That is immediate by right properness: it is provided by the base change of j along z:

$$\begin{array}{ccc} j^*Z & \xrightarrow{\epsilon_z} & Z \\ \downarrow & \lrcorner & \downarrow z \\ X & \xrightarrow{j} & X' \end{array}$$

□

4.2. **Presentations of ∞-categories.**

Definition 4.30 ([GK17] 7.1, [nLa26d])**.** A combinatorial type-theoretic model category $\mathcal{M}$ is a model category that satisfies the following properties:

(1) $\mathcal{M}$ is locally cartesian closed
(2) $\mathcal{M}$ is combinatorial
(3) $\mathcal{M}$ is right proper
(4) cofibrations coincide with monomorphisms

Remark 4.31. In this definition, we slightly diverge from [GK17] since we do not require any simplicial structure. This choice is justified by Remark 2.9 and the fact that no result on univalence makes any use of the simplicial structure.

Definition 4.32 ([GK17] 1.2)**.** Let $\mathcal{P}$ be a presentable, locally cartesian closed ∞-category. Let $\mathrm{L} : \mathcal{P} \to \mathcal{C} \subseteq \mathcal{P}$ be an accessible localization with a right adjoint $G : \mathcal{C} \hookrightarrow \mathcal{P}$. For any morphism $f : T \to S$, we have the following commutative squares:

$$\begin{array}{ccc} \mathcal{P}/_S & \xrightarrow{\mathrm{L}/_S} & \mathcal{C}/_S \\ {\scriptstyle f_!}\uparrow & & \uparrow{\scriptstyle f_!} \\ \mathcal{P}/_T & \xrightarrow[\mathrm{L}/_T]{} & \mathcal{C}/_T \end{array} \qquad \begin{array}{ccc} \mathcal{P}/_S & \xleftarrow{G/_S} & \mathcal{C}/_S \\ {\scriptstyle f^*}\downarrow & & \downarrow{\scriptstyle f^*} \\ \mathcal{P}/_T & \xleftarrow[G/_T]{} & \mathcal{C}/_T \end{array}$$

We say that L is locally cartesian if the following square is also cartesian:

$$\begin{array}{ccc} \mathcal{P}/_S & \xrightarrow{\mathrm{L}/_S} & \mathcal{C}/_S \\ {\scriptstyle f^*}\downarrow & & \downarrow{\scriptstyle f^*} \\ \mathcal{P}/_T & \xrightarrow[\mathrm{L}/_t]{} & \mathcal{C}/_T \end{array}$$

Lemma 4.33 ([GK17] 1.3)**.** Let $\mathcal{P}$ be a presentable, locally cartesian closed ∞-category. Let $\mathrm{L} : \mathcal{P} \to \mathcal{C} \subseteq \mathcal{P}$ be an accessible localization with a right adjoint $G : \mathcal{C} \hookrightarrow \mathcal{P}$. The following are equivalent:

(1) L is locally cartesian
(2) for every $f : T \to S$ between local objects, $f^* : \mathcal{P}/_T \to \mathcal{P}/_S$ preserves local equivalences
(3) for every $f : T \to S$ between local objects, $f_* : \mathcal{P}/_S \to \mathcal{P}/_T$ preserves local objects

Proof. (1) $\Rightarrow$ (2) : By hypothesis, we have the following commutative diagram:

$$\begin{array}{ccccc} \mathcal{C}/_S & \xhookrightarrow{G/_S} & \mathcal{P}/_S & \xrightarrow{\mathrm{L}/_S} & \mathcal{C}/_S \\ {\scriptstyle f^*}\downarrow & & {\scriptstyle f^*}\downarrow & & \downarrow{\scriptstyle f^*} \\ \mathcal{C}/_T & \xhookrightarrow[G/_S]{} & \mathcal{P}/_T & \xrightarrow[\mathrm{L}/_t]{} & \mathcal{C}/_T \end{array}$$

Let α be a local equivalence in $\mathcal{P}/_S$. Then $\mathrm{L}/_S(\alpha)$ is an equivalence and so is $f^*\mathrm{L}/_S(\alpha) = L/_T(f^*\alpha)$, which implies that $f^*\alpha$ is a local equivalence in $\mathcal{P}/_T$. (2) $\Rightarrow$ (3) : Assume that f^* preserves local equivalences. Let $z : Z \to T$ be a local object of $\mathcal{P}/_T$. For any local equivalence $\alpha : p \to p'$ in $\mathcal{P}/_T$, there is a commutative square as follows:

$$\begin{array}{ccc} \mathrm{Hom}_{\mathcal{P}/_T}(f^*p', z) & \xrightarrow{\sim} & \mathrm{Hom}_{\mathcal{P}/_S}(p', f_*z) \\ {\scriptstyle f^*\alpha}\downarrow & & \downarrow{\scriptstyle \alpha} \\ \mathrm{Hom}_{\mathcal{P}/_T}(f^*p, z) & \xrightarrow{\sim} & \mathrm{Hom}_{\mathcal{P}/_S}(p, f_*z) \end{array}$$

The horizontal maps are the equivalences of the adjunction. The left vertical map is an equivalence since $f^*\alpha$ is a local equivalence by hypothesis. (3) $\Rightarrow$ (1) : We notice that (3) is equivalent to the following diagram being commutative:

$$\begin{array}{ccc} \mathcal{P}/_S & \xleftarrow{G/_S} & \mathcal{C}/_S \\ {\scriptstyle f_*}\uparrow & & \uparrow{\scriptstyle f_*} \\ \mathcal{P}/_T & \xleftarrow[G/_T]{} & \mathcal{C}/_T \end{array}$$

If we pass to left adjoints, we get exactly (1). □

Proposition 4.34 ([GK17] 1.5)**.** Let $\mathcal{C}$ be a presentable and locally cartesian closed ∞-category. Then the standard presentation $\mathrm{L} : \mathrm{Psh}(\mathcal{C}_k) \to \mathcal{C}$ is a locally cartesian localization.

Proof. We verify (2) of the previous lemma. Let $F : T \to S$ be a map between local objects. Let $k' \geq k$ be a cardinal such that both S and T are in $\mathcal{C}_{k'}$. We have the following diagram:

$$\begin{array}{ccccc} \mathcal{C}/_S & \overset{\jmath}{\hookrightarrow} & \mathrm{Psh}(\mathcal{C}_k)/_S & \hookrightarrow & \mathrm{Psh}(\mathcal{C}_{k'})/_S \\ & & \downarrow{\scriptstyle F^*} & & \downarrow{\scriptstyle F^*} \\ \mathcal{C}/_T & \overset{\jmath}{\hookrightarrow} & \mathrm{Psh}(\mathcal{C}_k)/_T & \hookrightarrow & \mathrm{Psh}(\mathcal{C}_{k'})/_T \end{array}$$

Since the horizontal functors are fully faithful, we can assume $k = k'$. F^* is a map between representable functors; therefore it is of the form: $\mathrm{Hom}_{\mathcal{C}_k}(-, f)$ for some arrow f in $\mathcal{C}_k$. We have to show that $F^*(\mathrm{colim}_k \mathrm{Hom}_{\mathcal{C}_{k/S}}(-, x_i) \to \mathrm{Hom}_{\mathcal{C}_{k/S}}(-, \mathrm{colim}_k x_i))$ is a local equivalence. We can construct the following diagram:

$$\begin{array}{ccccc} F^*\mathrm{colim}_k \mathrm{Hom}_{\mathcal{C}_{k/S}}(-, x_i) & \longrightarrow & \mathrm{colim}_k F^* \mathrm{Hom}_{\mathcal{C}_{k/S}}(-, x_i) & \xrightarrow{\sim} & \mathrm{Hom}_k \mathrm{Map}_{\mathcal{C}_{k/S}}(-, f^* x_i) \\ \downarrow & & & & \downarrow \\ F^*\mathrm{Hom}_{\mathcal{C}_{k/S}}(-, \mathrm{colim}_k x_i) & \xrightarrow{\sim} & \mathrm{Hom}_{\mathcal{C}_{k/S}}(-, f^* \mathrm{colim}_k x_i) & \longrightarrow & \mathrm{Hom}_{\mathcal{C}_{k/S}}(-, \mathrm{colim}_k f^* x_i) \end{array}$$

The maps displayed as equivalences are so because the Yoneda embedding preserves base change. The right vertical map is a local equivalence. The top left horizontal map is an equivalence since F^* is a left adjoint and therefore it preserves colimits. The left bottom horizontal map is an equivalence since $\mathcal{C}$ and $\mathcal{C}_k$ are locally cartesian closed. This implies that f^* preserves colimits. We conclude that the left vertical map is a local equivalence. □

Definition 4.35 ([nLa26b])**.** Let $\mathcal{M}$ be a model category. Let V be a class of morphisms in $\mathcal{M}$. The left Bousfield localization of $\mathcal{M}$ at V is the model category $V^{-1}\mathcal{M}$ defined as follows:

(1) as a category it is just $\mathcal{M}$
(2) its cofibrations remain the same
(3) its weak equivalences are V-local equivalences

These conditions define a new model structure with different fibrations (there are more weak equivalences and possibly more trivial cofibrations). We say that an arrow is a V-local fibration if it is a fibration in $V^{-1}\mathcal{M}$.

Remark 4.36. $\mathrm{N}(\mathcal{M}) = \mathrm{N}(V^{-1}\mathcal{M})$ extends to a localization: $\mathcal{C} \to V^{-1}\mathcal{C} := \mathrm{N}(\mathcal{M})[V^{-1}W]$.

Lemma 4.37 ([GK17] 7.2)**.** Let $\mathcal{M}$ be a model category in which cofibrations coincide with monomorphisms. If, for any fibration $f : X \twoheadrightarrow Y$ between fibrant objects, f^* preserves trivial cofibrations, then $\mathcal{M}$ is right proper.

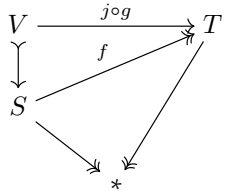
Proof. Let $g : V \twoheadrightarrow W$ be any fibration and let $w : W' \to W$ be a weak equivalence. We need to show that the map v in the following diagram is a weak equivalence:

$$\begin{array}{ccc} V' & \twoheadrightarrow & W' \\ {\scriptstyle v}\downarrow & \lrcorner & \downarrow{\scriptstyle w} \\ V & \underset{g}{\twoheadrightarrow} & W \end{array}$$

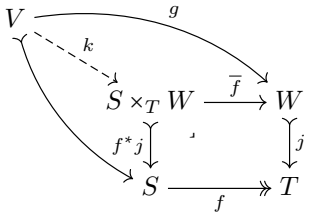
Let us consider a fibrant replacement of W:

$$W \overset{j}{\rightarrowtail} T \twoheadrightarrow *$$

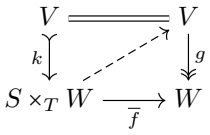
Let us factor $j \circ g$ into a trivial cofibration followed by a fibration:



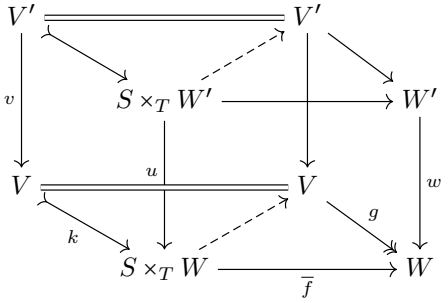
Let us consider the following pullback:



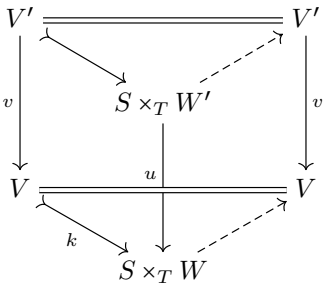
f^*j is a trivial cofibration by hypothesis. The map $V \to S$ is a trivial cofibration by definition. We deduce that k is a trivial cofibration by 2-out-of-3. The following lifting problem has a solution:



We can pull back this diagram along w:



Notice that u is a weak equivalence since it is the pullback of w along f. We factor w into a trivial cofibration followed by a trivial fibration. The pullback along f preserves both (the cofibration by hypothesis). By the pullback pasting lemma, the induced map is the pullback of w along f, and it is a composition of weak equivalences. We observe that v is a retract of u:



□

Lemma 4.38 ([GK17] 7.8)**.** Let $\mathcal{M}$ be a combinatorial type-theoretic model category. Let V be a class of morphisms of $\mathcal{M}$ that generates cofibrations and whose domains and codomains are small. Let $\mathrm{L} : \mathcal{M} \to V^{-1}\mathcal{M}$ be the left Bousfield localization of $\mathcal{M}$ at V. If the induced localization $\mathcal{C} := \mathrm{N}(\mathcal{M})[\mathrm{W}^{-1}]$, $\mathcal{C} \to V^{-1}\mathcal{C}$ is locally cartesian, then $V^{-1}\mathcal{M}$ is again a combinatorial type-theoretic model category.

Proof. Notice that $V^{-1}\mathcal{M}$ is a combinatorial, locally cartesian closed model category with cofibrations coinciding with monomorphisms, since it is a Bousfield localization. We are left to prove that it is right proper. By Lemma 4.37, we reduce to show that whenever $f : T \to S$ is a V-local fibration between V-local objects, f^* preserves local equivalences. Let us denote by $V^{-1}\mathrm{W}$ the class of V-local maps in $\mathcal{M}$. Let us denote by $V^{-1}\mathcal{C} := \mathrm{N}(V^{-1}\mathcal{M})[V^{-1}\mathrm{W}]$. Now we apply Lemma 4.29 and we get the following equivalence of ∞-categories:

$$\mathcal{C}/_S \simeq \mathrm{N}(\mathcal{M}/_S)[(W/_S)^{-1}]$$

Since S is V-local (it is needed because we do not know that $V^{-1}\mathcal{M}$ is right proper). By Theorem 4.27, we have the following equivalence:

$$(V^{-1}\mathcal{C})/_S \simeq \mathrm{N}((V/_S)^{-1}\mathcal{M}/_S)[V^{-1}(W/_S)^{-1}]$$

Now we consider the base change: $f^* : V^{-1}\mathcal{C}/_S \to V^{-1}\mathcal{C}/_T$. By hypothesis, $\mathcal{C} \to V^{-1}\mathcal{C}$ is locally cartesian. Therefore the following diagram is commutative:

$$\begin{array}{ccc} \mathcal{C}/_S & \xrightarrow{\mathrm{L}/_S} & V^{-1}\mathcal{C}/_S \\ {\scriptstyle f^*}\downarrow & & \downarrow{\scriptstyle f^*} \\ \mathcal{C}/_T & \xrightarrow[\mathrm{L}/_T]{} & V^{-1}C/_T \end{array}$$

Trivially, this implies that f^* preserves local equivalences (we look at which maps of the model categories are sent to isomorphisms when passing to the associated ∞-category). □

Remark 4.39. By Remark 2.9, any Cisinski model category $\mathcal{E}$ can be equipped with a simplicial structure. Since asking for such a structure is not restrictive we have omitted it until now. In the next result the simplicial structure considered is the one provided by the simplexes in Definition 2.2. This defines a mapping space for any object x:

$$\mathrm{Map}_{\mathcal{E}}(x, -)_n := \mathcal{E}(x \times \Delta^n, -)$$

Construction 4.40 ([GK17] 7.9)**.** Let $\mathcal{C}$ be a presentable ∞-category. We know that $\mathcal{C} \simeq \mathrm{Psh}(\mathcal{C}_k)$ for a large enough cardinal k, where $\mathcal{C}_k$ is the full subcategory of $\mathcal{C}$ spanned by k-compact objects. Moreover, if U is the collection of morphisms $\mathrm{colim}_i \mathrm{Hom}_{\mathcal{C}}(-, x_i) \to \mathrm{Hom}_{\mathcal{C}}(-, \mathrm{colim}_i x_i)$ for k-small diagrams $I \to \mathcal{C}_k$ we have that $\mathcal{C} \simeq U^{-1}\mathrm{Psh}(\mathcal{C}_k)$. Let us consider a fibrant simplicial category $\mathcal{D}$ such that $\mathrm{N}(\mathcal{D}) \simeq \mathcal{C}_k$. Let us consider $\mathrm{Pre}_{\Delta}(\mathcal{D}) := \mathrm{Fun}(\mathcal{D}^{\mathrm{op}}, \mathrm{s.Set})$ the 1-category of simplicial presheaves on $\mathcal{D}$ with the injective model structure. Let V be the smallest saturated class of morphisms generated by the maps

$$\mathrm{hocolim}_i \mathrm{Map}_{\mathcal{C}}(-, x_i) \to \mathrm{Map}_{\mathcal{C}}(-, \mathrm{hocolim}_i x_i)$$

in $\mathrm{Pre}_{\Delta}(\mathcal{D})$ corresponding to maps in U. Then $\mathcal{C} \simeq U^{-1}\mathrm{Psh}(\mathcal{C}_k) \simeq \mathrm{N}(V^{-1}\mathrm{Pre}_{\Delta}(\mathcal{D}))[\mathrm{W}^{-1}]$, where W is the class of weak equivalences in $\mathrm{Pre}_{\Delta}(\mathcal{D})$.

Theorem 4.41 ([GK17] 7.10)**.** Let $\mathcal{C}$ be a locally cartesian closed and presentable ∞-category (for example, a topos). Then the presentation $\mathcal{M} := V^{-1}\mathrm{Pre}_{\Delta}(\mathcal{D})$ of $\mathcal{C}$, produced in Construction 4.40, is a combinatorial type-theoretic model category.

Proof. Let us apply Lemma 4.38 to $\mathrm{Pre}_{\Delta}(\mathcal{D}) \to \mathcal{M}$. First, we need to verify that $\mathrm{Pre}_{\Delta}(\mathcal{D})$ is type-theoretic model category. Notice that it is clearly locally cartesian closed as a 1-category since it is a functor category with values in a locally cartesian closed category. Since we endowed $\mathrm{Pre}_{\Delta}(\mathcal{D})$ with the

injective model structure, we can verify that all properties of the model structure hold objectwise. That is immediate, since the 1-category of simplicial sets is a type-theoretic model topos. By Theorem 4.41, the Bousfield localization $\mathrm{Pre}_{\mathbb{\Delta}}(\mathcal{D}) \to \mathcal{M}$ induces the following right accessible localization at the level of ∞-categories: $\mathrm{Psh}(\mathcal{C}_k) \to \mathcal{C}$. This localization is locally cartesian by Proposition 4.34. We conclude that $\mathcal{M}$ is a combinatorial type-theoretic model category. □

Remark 4.42. Notice that the underlying 1-category of $\mathcal{M}$ is the same as that for $\mathrm{Pre}_{\mathbb{\Delta}}(\mathcal{D})$, which is a 1-topos. Therefore $\mathcal{M}$ is a right proper Cisinski model category.

4.3. **Mapping spaces for model categories.**

Remark 4.43 ([Cis19] 3.1.31)**.** Let A be any anima. There is a bijection

$$[*, A] \cong \pi_0(A)$$

Proposition 4.44 ([Cis19] 7.10.6)**.** Let $\mathcal{C}$ be a category. For any object X in $\mathcal{C}$, let $\mathrm{Map}(X, -) : \mathcal{C} \to \mathrm{An}$ be a functor. Assume that it satisfies the following conditions:

(1) it preserves limits
(2) for any object Y in $\mathcal{C}$, there is a natural bijection

$$\pi_0(\mathrm{Map}(X, Y)) \cong \mathrm{Hom}_{\mathrm{ho}(\mathcal{C})}(X, Y)$$

in Y

Then there is a canonical equivalence

$$\mathrm{Map}(X, Y) \simeq \mathrm{Hom}_{\mathcal{C}}(X, Y)$$

functorial in Y.

Proof. Let us consider $id_x \in \pi_0\mathrm{Map}_{\mathcal{C}}(x, x) \cong \mathrm{Hom}_{\mathrm{ho}(\mathcal{C})}(x, x)$. Let us pick a representative u of it. By Remark 4.43, we know it corresponds to a morphism $* \to \mathrm{Map}_{\mathcal{C}}(x, x)$. By the Yoneda lemma, there exists an essentially unique natural transformation of ∞-categories $\overline{u} : \mathrm{Hom}_{\mathcal{C}}(x, -) \to \mathrm{Map}_{\mathcal{C}}(x, -)$ that sends id_x to u. For any $f : x \to y$, if we apply π_0 to both sides, we obtain the following:

$$[id_x] \longmapsto [u] \longmapsto [id_x]$$

$$\begin{array}{ccccc}
\pi_0\mathrm{Hom}_{\mathcal{C}}(x, x) & \xrightarrow{\pi_0\overline{u}} & \pi_0\mathrm{Map}_{\mathcal{C}}(x, x) & \xrightarrow{\cong} & \mathrm{Hom}_{\mathrm{ho}(\mathcal{C})}(x, x) \\
\downarrow{\scriptstyle \pi_0 f_*} & & \downarrow{\scriptstyle \pi_0 f_*} & & \downarrow{\scriptstyle [f]_*} \\
\pi_0\mathrm{Hom}_{\mathcal{C}}(x, y) & \xrightarrow{\pi_0\overline{u}} & \pi_0\mathrm{Map}_{\mathcal{C}}(x, y) & \xrightarrow{\cong} & \mathrm{Hom}_{\mathrm{ho}(\mathcal{C})}(x, y)
\end{array}$$

$$[f] \longmapsto [f]$$

The composite horizontal maps are bijections by the 1-categorical Yoneda lemma. Let us consider any anima $a = \mathrm{colim}_{* \to a} * \simeq \mathrm{colim}_a *$. We have the following chain of bijections:

$$\begin{aligned}
&\mathrm{Hom}_{\mathrm{ho(An)}}(a, \mathrm{Hom}_{\mathcal{C}}(x, y)) \cong \pi_0\mathrm{Hom}_{\mathrm{An}}(a, \mathrm{Hom}_{\mathcal{C}}(x, y)) \cong \\
&\cong \pi_0 \mathrm{lim}_{a^{\mathrm{op}}} \mathrm{Hom}_{\mathrm{An}}(*, \mathrm{Hom}_{\mathcal{C}}(x, y)) \cong \pi_0\mathrm{Hom}_{\mathrm{An}}(*, \mathrm{lim}_{a^{\mathrm{op}}}\mathrm{Hom}_{\mathcal{C}}(x, y)) \cong \\
&\cong \pi_0\mathrm{Hom}_{\mathrm{An}}(*, \mathrm{Hom}_{\mathcal{C}}(x, \mathrm{lim}_{a^{\mathrm{op}}} y)) \cong \pi_0\mathrm{Hom}_{\mathrm{ho}\mathcal{C}}(x, \mathrm{lim}_{a^{\mathrm{op}}} y) \cong \\
&\cong \pi_0\mathrm{Map}_C(x, \mathrm{lim}_{a^{\mathrm{op}}} y) \cong \pi_0\mathrm{Map}_C(x, \mathrm{lim}_{a^{\mathrm{op}}} y) \cong \\
&\cong \pi_0 \mathrm{lim}_{a^{\mathrm{op}}} \mathrm{Map}_C(x, y) \cong \pi_0\mathrm{Hom}_{\mathrm{ho(An)}}(*, \mathrm{lim}_{a^{\mathrm{op}}}\mathrm{Map}_C(x, y)) \cong \\
&\cong \pi_0 \mathrm{lim}_{a^{\mathrm{op}}} \mathrm{Hom}_{\mathrm{ho(An)}}(*, \mathrm{Map}_C(x, y)) \cong \pi_0\mathrm{Hom}_{\mathrm{ho(An)}}(\mathrm{colim}_a *, \mathrm{Map}_C(x, y)) \cong \\
&\cong \pi_0\mathrm{Hom}_{\mathrm{An}}(a, \mathrm{Map}_C(x, y)) \cong \mathrm{Hom}_{\mathrm{ho(An)}}(a, \mathrm{Map}_C(x, y)).
\end{aligned}$$

By the 1-categorical Yoneda lemma, we deduce $\mathrm{Map}_C(x,y) \cong \mathrm{Hom}_C(x,y)$ in $\mathrm{ho}(\mathrm{An})$. This implies that they are equivalent in An. □

Definition 4.45 ([Cis19] 7.10.7)**.** Let X be an object in a locally small model category $\mathcal{C}$ with small limits. Let us denote by $\mathcal{C}_f$ the full category of fibrant objects in $\mathcal{C}$. A mapping space $\mathrm{Map}(X,-)$ is a functor $\mathcal{C}_f \to \mathrm{s.Set}_f$ with the following properties:

(1) $\mathrm{Map}(X,-)$ preserves homotopy limits
(2) there is a functorial bijection $\pi_0\mathrm{Map}(X,Y) \cong \mathrm{Hom}_{\mathrm{ho}(\mathcal{C})}(X,Y)$ for any Y in $\mathcal{C}$

Lemma 4.46 ([Cis19] 7.10.12-7.10.13)**.** Let $\mathrm{Map}(X,-)$ be a mapping space for a model category $\mathcal{C}$ with a class of weak equivalences W. Let us denote by $\mathrm{L}(\mathcal{C})$ the localization of $\mathrm{N}(\mathcal{C})$ at W. Then there exists a canonical equivalence $\mathbb{R}\mathrm{Map}(X,Y) \simeq \mathrm{Hom}_{\mathrm{L}(\mathcal{C})}(X,Y)$, functorial in Y.

4.4. **Essential Uniqueness theorem.**

Discussion 4.47. The remaining part of this chapter is to prove Theorem 4.61. We will state all results for tribes whenever it is possible. There is however a complication that makes Cisinski model categories a "preferred model" for the following reasons: Let $\mathcal{C}$ be a Cisinksi model category, then

(1) $\mathcal{C}$ is presentable.
(2) Let k be a noncountable regular cardinal. Let $\mathcal{C}_\infty$ be the localization of $\mathcal{C}$ at weak equivalences. Then k-compact ojects in $\mathcal{C}_\infty$ are the image under the localization functor of k-compact objects of $\mathcal{C}$.
(3) If we denote the localization of $\mathcal{C}$ at its weak equivalences with $\mathcal{C}_\infty$ then if $\mathcal{C}_\infty$ is a topos $X \mapsto (\mathcal{C}_\infty/_X)^{\simeq}_k$ preserves colimits.

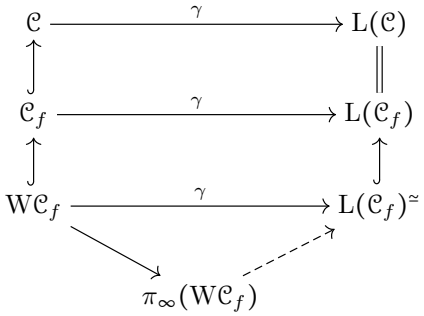
these results might not hold for a generic tribe, we therefore will resort to a different notion of smallness.

Definition 4.48. Let $\mathcal{C}$ be a π-tribe. Let $\pi : \mathcal{U}_\bullet \to \mathcal{U}$ be an univalent fibration for $\mathcal{C}$. We say that a morphism f is $\mathcal{U}$-small if f is a base change of π along some map.

Lemma 4.49. Let $(\mathcal{C}, \mathrm{Fib}, \mathrm{W})$ be a category with weak equivalences and fibrations. Consider the following ∞-category: $\mathrm{L}(\mathcal{C}) := \mathcal{C}[\mathrm{W}^{-1}]$. Let $\mathrm{W}\mathcal{C}_f$ be the subcategory of $\mathcal{C}$ spanned by weak equivalences between fibrant objects.

$$\pi_\infty \mathrm{W}\mathcal{C}_f \simeq \mathrm{L}(\mathcal{C})^{\simeq}$$

Proof. By Remark 4.17, the canonical functor $i : \mathrm{L}(\mathcal{C}_f) \to \mathrm{L}(\mathcal{C})$ is an equivalence. Let us construct the following diagram:



Let us consider the following map:

$$\mathrm{W}\mathcal{C}_f \longrightarrow \mathcal{C}_f$$

$\mathrm{W}\mathcal{C}_f$ is saturated in $\mathcal{C}_f$ since all objects considered are fibrant (analogously to Remark 4.26). We can apply Corollary 4.20: Let $\mathcal{C}$ be an ∞-category with weak equivalences W and fibrations Fib. Suppose W is saturated. Then the canonical functor $\mathrm{W} \to \mathrm{L}(\mathcal{C})^{\simeq}$ is final. We conclude that $\pi_\infty \mathrm{W}\mathcal{C}_f \simeq \mathrm{L}(\mathcal{C}_f)^{\simeq} \simeq \mathrm{L}(\mathcal{C})^{\simeq}$ thanks to Quillen's Theorem A. □

Notation 4.50. Let $\mathcal{T}_\infty$ be an ∞-topos. Let $\mathcal{T}$ be a π-tribe with a class of homotopy equivalences W. Assume that $\mathcal{T}_\infty \simeq \mathrm{N}(\mathcal{T})[\mathrm{W}^{-1}]$. Let $(\mathrm{WFib}/_X)^{\mathcal{U}}$ be the ∞-subcategory of $\mathrm{N}(\mathcal{T}/_X)$ spanned by weak equivalences between $\mathcal{U}$-small fibrations.

Remark 4.51. Let $\mathcal{T}$ be a π-tribe. Then $\mathrm{N}(\mathcal{T})$ is a category with weak equivalences and fibration as in Definition 4.6 were weak equivalences are homotopy equivalences and fibrations are the fibrations of the π-tribe.

Corollary 4.52. Under the assumptions of Notation 4.50 there is an equivalence:

$$\pi_\infty \mathrm{WFib}/_X \simeq (\mathcal{T}_\infty/_X)^{\simeq}$$

natural in X.

Proof. Apply Lemma 4.49 with $\mathcal{C} = \mathrm{N}((\mathcal{T}/_X)^\lambda)$. Naturality follows since homotopy equivalences are stable under pullback. □

Proposition 4.53. Let $\mathcal{T}$ be a π-tribe. Let $\pi : \mathcal{U}_\bullet \twoheadrightarrow \mathcal{U}$ be an univalent fibration in $\mathcal{T}$. Then there exists a bijection

$$[X, \mathcal{U}] \cong \pi_0\pi_\infty(\mathrm{WFib}/_X)^{\mathcal{U}}$$

Proof. Let us now consider $\int : [X, \mathcal{U}] \to \pi_0\pi_\infty \mathrm{WFib}/_X)^{\mathcal{U}}$ sending F to the base change of π along F:

$$\begin{array}{ccc} \int F & \longrightarrow & \mathcal{U}. \\ \downarrow & \lrcorner & \downarrow \pi \\ X & \xrightarrow[F]{} & \mathcal{U} \end{array}$$

The map considered is well defined: Let us consider the following diagram:

$$\begin{array}{ccc} & & \mathcal{U}^{\Delta^1} \\ & \overset{H}{\nearrow} & \downarrow \\ X & \xrightarrow[(F_0,F_1)]{} & \mathcal{U} \times \mathcal{U} \end{array}$$

By Proposition 2.15 and the definition of univalence we can assume $\mathcal{U}^{\Delta^1} = \mathrm{Eq}(\mathcal{U}_\lambda)$

$$\begin{array}{ccc} & & \mathrm{Eq}(\mathcal{U}_\bullet) \\ & \overset{H}{\nearrow} & \downarrow \\ X & \xrightarrow[(F_0,F_1)]{} & \mathcal{U} \times \mathcal{U} \end{array}$$

By Lemma 2.57 H corresponds to the following diagram:

$$\begin{array}{ccccc} X \times_{\mathcal{U}\times\mathcal{U}} \mathcal{U}_\bullet \times \mathcal{U} & & \xrightarrow{w} & & X \times_{\mathcal{U}\times\mathcal{U}} \mathcal{U} \times \mathcal{U}_\bullet \\ & {\scriptstyle p_0}\searrow & & \swarrow{\scriptstyle p_1} & \\ & & X & & \end{array}$$

plus the datum of two homotopy inverses l and r of w (left and right inverses) and two homotopies $H_l : w \circ l \Rightarrow id$ and $H_r : r \circ w \Rightarrow id$. If we consider the following diagram where all squares are pullbacks:

$$\begin{array}{ccccc} X \times_{\mathcal{U}\times\mathcal{U}} \mathcal{U}_\bullet \times \mathcal{U} & \longrightarrow & \mathcal{U}_\bullet \times \mathcal{U} & \longrightarrow & \mathcal{U}_\bullet \\ \downarrow{\scriptstyle p_0} & \lrcorner & \downarrow{\scriptstyle \pi\times id} & \lrcorner & \downarrow{\scriptstyle \pi} \\ X & \xrightarrow[(F_0,F_1)]{} & \mathcal{U} \times \mathcal{U} & \xrightarrow[pr_0]{} & \mathcal{U} \end{array}$$

we deduce that $X \times_{\mathcal{U}\times\mathcal{U}} \mathcal{U}_\bullet \times \mathcal{U} = \int F_0$ and that p_0 is a fibration. Analogously we deduce that $X \times_{\mathcal{U}\times\mathcal{U}} \mathcal{U} \times \mathcal{U}_\bullet = \int F_1$ and that p_1 is a fibration. Since both p_0 and p_1 are fibrations w is a weak

equivalence over X between $\int F_0$ and $\int F_1$ by Lemma 2.30. $\int$ is surjective as a map of sets by definition of $\mathcal{U}$-small morphism. Now let us consider the following data:

$$\begin{array}{ccc} \int F & \xrightarrow{\sim} & \int G \\ & \searrow \quad \swarrow & \\ & X & \end{array}$$

By Lemma 2.57, there is a dotted map in the following diagram:

$$\begin{array}{ccc} & & \mathrm{Eq}(\mathcal{U}_\bullet) \\ & \overset{w}{\nearrow} & \downarrow{\scriptstyle (s,t)} \\ X & \xrightarrow[(F,G)]{} & \mathcal{U}_\lambda \times \mathcal{U}_\lambda \end{array}$$

By definition of a univalent morphism, this is equivalent to F and G being right homotopic. By Lemma 2.25, we conclude that $F = G$ in $[X, \mathcal{U}]$, since $\mathcal{U}$ is fibrant. If we consider $\mathcal{C} := \mathcal{E}/_X$, then there exists a right calculus of fractions on $\mathrm{trivFib}/_X \hookrightarrow \mathcal{C}/_X$, since trivial fibrations are stable under composition and pullback. As in Remark 4.26, by Corollary 4.25, this calculus of fractions is also a calculus of fractions for the localization at W. It trivially restricts to $(\mathrm{WFib}/_X)^{\mathcal{U}}$. Suppose that $\int F \twoheadrightarrow X$ and $\int G \twoheadrightarrow X$ are in the same equivalence class in $\pi_0\pi_\infty \mathrm{W}(\mathrm{Fib}/_X)^\lambda$. $\mathrm{Hom}_{\pi_\infty(\mathrm{WFib}/_X)^{\mathcal{U}}}(\int F, \int G)$ is not empty by assumption. By the calculus of fractions previously mentioned, $\mathrm{Hom}_{\pi_\infty \mathrm{W}(\mathrm{Fib}/_X)^\lambda}(\int F, \int G) \simeq \mathrm{colim}_{z:((\mathrm{WFib}/_X)^{\mathcal{U}})^{\mathrm{op}}, z \to \int F} \mathrm{Hom}_{(\mathrm{WFib}/_X)^{\mathcal{U}}}(z, \int G)$. Since the right-hand side is not the empty space, there exists an element z in $(\mathrm{WFib}/_X)^{\mathcal{U}}$ such that $z \to \int F$ and $z \to \int G$. by the previous computation we conclude that there exists an $H : X \to \mathcal{U}$ such that:

(1) $z = \int H$
(2) $H \sim_{\mathrm{R}} G$
(3) $F \sim_{\mathrm{R}} H$

We conclude since $\sim_{\mathrm{R}}$ is an equivalence relation on fibrant objects. □

Remark 4.54. The previous result also holds for Cisinski model categories with minor changes.

Corollary 4.55. Let $\mathcal{C}$ be a Cisinski model category. Let $\pi : \mathcal{U}_\bullet \twoheadrightarrow \mathcal{U}$ be an univalent fibration in $\mathcal{C}$. Then there exists a bijection

$$\mathrm{Hom}_{\mathrm{ho}(\mathcal{C})}(X, \mathcal{U}) \cong [X, \mathcal{U}] \cong \pi_0\pi_\infty(\mathrm{WFib}/_X)^{\mathcal{U}}$$

Proof. The first bijection was proved in Lemma 2.33 (since X is cofibrant and $\mathcal{U}$ is fibrant) and the second bijection follows the same steps of the previous proof. By Lemma 2.30 $\mathrm{Eq}(\mathcal{U})$ represents also weak homotopy equivalences. □

Construction 4.56. Let $\mathcal{T}$ be a Cisinski model category such that $\mathrm{N}(T)[\mathrm{W}^{-1}] \simeq \mathcal{T}_\infty$ as in Notation 4.50. Let λ be a regular cardinal. Let $\pi : \mathcal{U}_\bullet \to \mathcal{U}_\lambda$ be a strict univalent universe for λ-small fibrations. Then we define $\mathrm{W}(\mathrm{Fib}/_X)^\lambda$ to be $(\mathrm{WFib}/_X)^{\mathcal{U}_\lambda}$. Then by Corollary 4.52 and Discussion 4.47 we have that $(\mathcal{T}_\infty/_X)^{\simeq}_\lambda \simeq \pi_\infty \mathrm{W}(\mathrm{Fib}/_X)^\lambda$ naturally in X in $\mathcal{T}$.

Discussion 4.57. The following theorem justifies the fact that some consider (2) as the actual definition of ∞-topos: an ∞-category satisfying descent.

Theorem 4.58 ([Lur03] 2.4.1)**.** Let $\mathcal{T}_\infty$ be an ∞-category. The following are equivalent:

(1) $\mathcal{T}_\infty$ is an ∞-topos
(2) there exists a cardinal k such that $\mathcal{T}_\infty$ is k-accessible and the Yoneda embedding $\mathcal{T}_\infty \to \mathrm{An}^{(\mathcal{T}_\infty)^{\mathrm{op}}_k}$ has a left exact left adjoint
(3) there exists an ∞-topos $\mathcal{P}$ such that $\mathcal{T}_\infty$ is a left exact localization of $\mathcal{P}$

Construction 4.59. Let us apply Proposition 4.44 to $\mathcal{C} := \mathcal{T}_\infty^{\mathrm{op}}$ and $\mathrm{Map}(-, \mathcal{U}_\lambda) : X \mapsto \pi_\infty \mathrm{W}(\mathrm{Fib}/_X)^\lambda \simeq (\mathcal{T}_\infty^\lambda/_X)^{\simeq}$. By Proposition 4.53, we know that $\pi_0 \mathrm{Map}(X, \mathcal{U}_\lambda) \cong [X, \mathcal{U}_\lambda] \cong \mathrm{Hom}_{\mathrm{ho}(\mathcal{T}_\infty)}(X, \mathcal{U}_\lambda)$.

Theorem 4.60. Let $\mathcal{T}_\infty$ be an ∞-topos. Let $\mathcal{T}$ be a Cisinski model category. Denote the weak homotopy equivalences of $\mathcal{T}$ with W. Suppose $\mathrm{N}(\mathcal{T})[\mathrm{W}^{-1}] \simeq \mathcal{T}_\infty$. Denote with $\mathrm{W}(\mathrm{Fib}/_X)^\lambda$ the subcategory of $\mathrm{N}(\mathcal{T}/_X)$ spanned by homotopy equivalences between λ-small fibrations. Suppose moreover that $\mathcal{T}$ has a strict univalent universe for λ-small fibrations $\pi : \mathcal{U}_{\lambda\bullet} \twoheadrightarrow \mathcal{U}_\lambda$. Then $\mathcal{T}_\infty(X, \mathcal{U}_\lambda) \simeq ((\mathcal{T}_\infty/_X)^\lambda)^{\simeq}$, functorially in X.

Proof. We can apply Proposition 4.44 to $\mathrm{W}(\mathrm{Fib}/_X)^\lambda \simeq ((\mathcal{T}_\infty/_X)^\lambda)^{\simeq}$ (Corollary 4.52) since the latter preserves colimits and $[X, \mathcal{U}_\lambda] \cong \pi_0\pi_\infty \mathrm{W}(\mathrm{Fib}/_X)^\lambda$ (Proposition 4.53). □

Theorem 4.61 (Essential uniqueness of universes for Cisinski model categories)**.** Let $\mathcal{T}_\infty$ be an ∞-topos. Let $\mathrm{Universes}^\lambda(\mathcal{T}_\infty)$ be the sub-groupoid of $\mathcal{T}_\infty^{\simeq}$ spanned by strict univalent universes for λ-small fibrations coming from a Cisinksi model category that presents $\mathcal{T}_\infty$. Then $\mathrm{Universes}^\lambda(\mathcal{T}_\infty)$ is a contractible space

Proof. It follows by the Yoneda lemma for ∞-catgories since all objects in $\mathrm{Universes}^\lambda(\mathcal{T}_\infty)$ represent the same functor thanks to Theorem 4.60. □

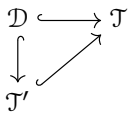
4.5. **Generalization to π-tribes.**

Notation 4.62. Throughout this subsection the term 'tribe' will denote π-tribes.

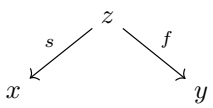
Remark 4.63. To generalize Theorem 4.61 to the context of π-tribes we need to prove that $X \mapsto \pi_\infty(\mathrm{WFib}/_X)^{\mathcal{U}}$ commutes with colimits.

Lemma 4.64. Let $\mathcal{T}$ be a tribe. Let $\mathcal{D}$ be a full subcategory of $\mathcal{T}$ that contains $*$, closed under pullbacks in $\mathcal{D}$ and such that for any object x in $\mathcal{D}$ there exists a path object of x that belongs to $\mathcal{D}$. Let W denote the homotopy equivalences of $\mathcal{T}$. Let $\mathrm{W}_{\mathcal{D}}$ denote homotopy equivalences in $\mathcal{D}$. Let L be the localization functor at homotopy equivalences. Then $\mathrm{L}(\mathcal{D}) \hookrightarrow \mathrm{L}(\mathcal{T})$ is full and faithful.

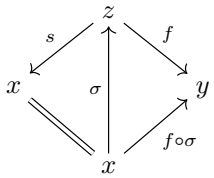
Proof. Throughout this proof we will omit to write the nerves of 1-categories. First notice that $\mathcal{D}$ has tribe structure induced by the one on $\mathcal{T}$. Let us consider $\mathcal{T}'$ the full subcategory of $\mathcal{T}$ spanned by object that lie in the image of $\mathrm{L}(\mathcal{D})$ under the localization. Since $\mathcal{T}'$ is closed under homotopy equivalences and contains $\mathcal{D}$ it has a natural structure of tribe induced by the one on $\mathcal{T}$. In the following diagram we notice that all functors considered are full and faithful:



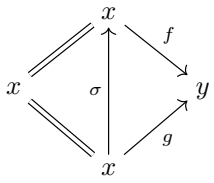
We prove that $\mathrm{hoL}(\mathcal{D}) \to \mathrm{hoL}(\mathcal{T})$ is full and faithful. Since both categories are tribes and, therefore categories with weak equivalences and fibrations, by Corollary 4.25 there is a right calculus of fractions for the localization L. For any x, y in $\mathcal{D}$ any arrow $x \to y$ in $\mathrm{hoL}(\mathcal{T})$ is equivalent to a couple of arrows in $\mathcal{T}$ $(s : z \xrightarrow{\sim} x, f : z \to y)$ where s is an homotopy equivalence.



since s is an homotopy equivalence between fibrant objects it has a strict section $\sigma : x \to z$. Then $(s, f) = (id, f \circ \sigma)$ in $\mathrm{Hom}_{\mathrm{hoL}(\mathcal{T})}(x, y)$ as proved by the following diagram:



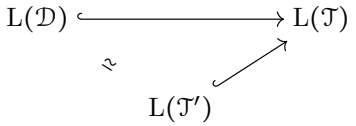
However $f \circ \sigma : x \to y$ is an "honest" arrow in $\mathcal{T}$ between objects in $\mathcal{D}$, therefore it belongs to $\mathrm{Hom}_{\mathrm{hoL}(\mathcal{D})}(x, y)$. We can also prove faithfulness for arrows of the form (id, f) for some f in $\mathcal{D}$. This is however trivial since whenever we have the following comparison:



where σ is an homotopy equivalence in $\mathcal{T}$, then σ is an homotopy equivalence in $\mathcal{D}$ since $\mathcal{D}$ is full and we can choose a path object that belongs to $\mathcal{D}$. This also works for $\mathrm{hoL}(\mathcal{D}) \hookrightarrow \mathrm{hoL}(\mathcal{T}')$ and $\mathrm{hoL}(\mathcal{T}') \hookrightarrow \mathrm{hoL}(\mathcal{T})$. Notice that, by definition $\mathrm{hoL}(\mathcal{D}) \hookrightarrow \mathrm{hoL}(\mathcal{T}')$ is also essentially surjective. Since all categories considered are tribes, all their localizations have finite limits and the localization functors are left exact. This also implies that $\mathrm{L}(\mathcal{D}) \to \mathrm{L}(\mathcal{T}')$ is left exact (all finite limits lift to finite limits in the above categories). By Theorem 4.21 we conclude that $\mathrm{L}(\mathcal{D}) \simeq \mathrm{L}(\mathcal{T}')$. Now we prove that $\mathrm{L}(\mathcal{T}') \to \mathrm{L}(\mathcal{T})$ is full and faithful. As already remarked both $\mathcal{T} \to \mathrm{L}(\mathcal{T})$ and $\mathcal{T}' \to \mathrm{L}(\mathcal{T}')$ admit a right calculus of fractions. Let x, y be two objects of $\mathcal{T}'$. By Remark 4.12 $\mathrm{L}(\mathcal{T})(x, y) \simeq \mathrm{colim}_{z:(\mathrm{W}/_x)^{\mathrm{op}}} \mathcal{T}(z, y)$ where $\mathrm{W}/_x$ is the following pullback:

$$\begin{array}{ccc} \mathrm{W}/_x & \longrightarrow & \mathrm{W} \\ \downarrow & \lrcorner & \downarrow \\ \mathcal{T}/_x & \longrightarrow & \mathcal{T} \end{array}$$

Now since $\mathcal{T}'$ is replete and x belongs to $\mathcal{T}'$ $\mathrm{W}/_x$ is corresponds to the weak equivalences over x in $\mathcal{T}'$. So the calculus of fractions on T' gives $\mathrm{L}(\mathcal{T}')(x, y) \simeq \mathrm{colim}_{z:(\mathrm{W}/_x)^{\mathrm{op}}} \mathcal{T}'(z, y) = \mathrm{colim}_{z:(\mathrm{W}/_x)^{\mathrm{op}}} \mathcal{T}(z, y)$ where the last equality holds since $\mathcal{T}'$ is full in $\mathcal{T}$. This proves the thesis:



□

Definition 4.65. In the assumptions of the previous lemma we say that $\mathcal{D}$ is a full sub-tribe of $\mathcal{T}$

Definition 4.66. Let $\mathcal{T}$ be a tribe. Let $\pi : \mathcal{U}_\bullet \twoheadrightarrow \mathcal{U}$ be an univalent map. Let $\mathcal{T}(X)$ be the full subcategory of fibrations over X. Let $\mathcal{T}(X)^{\mathcal{U}}$ be its full subcategory spanned by $\mathcal{U}$-small morphisms. $\mathcal{T}(X)$ has a natural structure of tribe. We say that $\mathcal{U}$ is a universe if $\mathcal{T}(X)^{\mathcal{U}}$ is a full sub-tribe of $\mathcal{T}(X)$.

Remark 4.67. In the appendix (Definition A.20) we will provide a more complete definition of a universe for a π-tribe. For the generality of our main goal we don't need any more requirements.

Corollary 4.68. Let $\mathcal{T}$ be a tribe. Let $\mathcal{U}$ be an universe. Let W be the class of homotopy equivalences of $\mathcal{T}$. Let $\mathcal{T}_\infty := \mathrm{N}(\mathcal{T})[\mathrm{W}^{-1}]$. Then the natural morphism

$$\pi_\infty(\mathrm{WFib}/_X)^{\mathcal{U}} \hookrightarrow (\mathcal{T}_\infty/_X)^{\simeq}$$

is full and faithful for any X in T.

Proof. By the previous lemma for $\mathcal{T} = \mathcal{T}(X)$ and $\mathcal{D} = \mathcal{T}(X)^{\mathcal{U}}$ we have an embedding:

$$\mathrm{L}(\mathcal{T}(X)^{\mathcal{U}}) \longrightarrow \mathrm{L}(\mathcal{T}(X)) \quad \simeq \quad \mathcal{T}_\infty/_X$$

where the right equivalence is given by Theorem 4.27. Since $\mathcal{T}(X)^{\mathcal{U}}$ is a tribe we can apply Corollary 4.52 to it. This gives an equivalence:

$$\pi_\infty(\mathrm{WFib}/_X)^{\mathcal{U}} \quad \simeq \quad \mathrm{L}(\mathcal{T}(X)^{\mathcal{U}})^{\simeq}$$

Since full and faithfulness is preserved when passing to groupoid cores this assembles to a fully faithful functor:

$$\pi_\infty(\mathrm{WFib}/_X)^{\mathcal{U}} \quad \simeq \quad \mathrm{L}(\mathcal{T}(X)^{\mathcal{U}})^{\simeq} \longrightarrow (\mathcal{T}_\infty/_X)^{\simeq}$$

as desired □

Theorem 4.69. Let $\mathcal{T}$ be a tribe. Let W be its class of homotopy equivalences. Let $\mathcal{U}$ be an universe for $\mathcal{T}$. Let $\mathcal{T}_\infty := \mathrm{N}(\mathcal{T})[\mathrm{W}^{-1}]$. If $\mathcal{T}_\infty$ is a topos then the functor $X \mapsto \pi_\infty(\mathrm{WFib}/_X)^{\mathcal{U}}$ preserves colimits.

Proof. Let $F : I \to \mathcal{T}_\infty$ be a diagram with colimit X in $\mathcal{T}_\infty$. Let $F(i) = X_i$. We have the following diagram of groupoids:

$$\begin{array}{ccc} (\mathcal{T}_\infty/_X)^{\simeq} & \xrightarrow{\sim} & \lim_I (\mathcal{T}_\infty/_{X_i})^{\simeq} \\ \uparrow & & \uparrow \\ \pi_\infty(\mathrm{WFib}/_X)^{\mathcal{U}} & & \lim_I \pi_\infty(\mathrm{WFib}/_{X_i})^{\mathcal{U}} \end{array}$$

the right vertical map is full and faithful since limits preserve full and faithfulness. The horizontal map is an equivalence by definition of topos. We claim that there exists a functor as the dotted one:

$$\begin{array}{ccc} (\mathcal{T}_\infty/_X)^{\simeq} & \xrightarrow{\sim} & \lim_I (\mathcal{T}_\infty/_{X_i})^{\simeq} \\ \uparrow & & \uparrow \\ \pi_\infty(\mathrm{WFib}/_X)^{\mathcal{U}} & \dashrightarrow & \lim_I \pi_\infty(\mathrm{WFib}/_{X_i})^{\mathcal{U}} \end{array}$$

Since T_∞ is the localization of a tribe, every pullback in T is sent to a pullback in $\mathcal{T}_\infty$. Let us consider an $\mathcal{U}$-small fibration $f : Y \twoheadrightarrow X$. If we pull back along all colimit inclusions we obtain the following diagram as follows:

$$\begin{array}{ccccc} Y_i & \longrightarrow & Y & \longrightarrow & \mathcal{U}_\bullet \\ {\scriptstyle f_i}\downarrow & \lrcorner & {\scriptstyle f}\downarrow & \lrcorner & \downarrow{\scriptstyle \pi} \\ X_i & \longrightarrow & X & \longrightarrow & \mathcal{U} \end{array}$$

where all squares are cartesian. Since colimits in $\mathcal{T}_\infty$ are effective we deduce that $f : Y \to X$ corresponds to the map induced by the functoriality of colim_I. Since the f_i's are small by construction the equivalence

$$(T_\infty/_X)^{\simeq} \simeq \lim_I (\mathcal{T}_\infty/_{X_i})^{\simeq}$$

restricts to a fully faithful functor

$$\pi_\infty(\mathrm{WFib}/_X)^{\mathcal{U}} \hookrightarrow \lim_I \pi_\infty(\mathrm{WFib}/_{X_i})^{\mathcal{U}}$$

We prove that the contructed functor is essentially surjective. Let us consider an I shaped diagram of cartesian squares as follows:

$$\begin{array}{ccc} Y_i & \longrightarrow & \mathcal{U}_\bullet \\ {\scriptstyle f_i}\downarrow & \lrcorner & \downarrow{\scriptstyle \pi} \\ X_i & \longrightarrow & \mathcal{U} \end{array}$$

By universality of colimits in $\mathcal{T}_\infty$ this assembles to a morphism $Y = \mathrm{colim}_I Y_i \to X$ such that all squares in the following diagram are cartesian (π^* preserves colimits).

$$\begin{array}{ccccc} Y_i & \longrightarrow & Y & \dashrightarrow & \mathcal{U}_\bullet \\ {\scriptstyle f_i}\downarrow & \lrcorner & \downarrow & & \downarrow{\scriptstyle \pi} \\ X_i & \longrightarrow & X & \dashrightarrow & \mathcal{U} \end{array}$$

This proves essential surjectivity. □

Definition 4.70 ([CH25] Definition 2.45)**.** Let S be a class of morphisms in a topos $\mathcal{T}$. We say that S is a local class if the following conditions are satisfied:

(1) it is closed under base change
(2) it is closed under coproducts
(3) given a pullback square in $\mathcal{T}$ of the form:

$$\begin{array}{ccc} Y' & \xrightarrow{g'} & Y \\ {\scriptstyle f'}\downarrow & \lrcorner & \downarrow{\scriptstyle f} \\ X' & \xrightarrow[g]{} & X \end{array}$$

where f' is in S and g is an effective epimorphism, then f is in S.

Proposition 4.71 ([CH25] Proposition 2.46)**.** Let S be a class of morphisms in a topos $\mathcal{T}$ that is stable under base change. Let An and Cat denote respectively the wide categories of animae and categories. The following conditions are equivalent:

(1) S is a local class
(2) the functor $\mathcal{T}^{\mathrm{op}} \to \mathrm{Cat}$ that sends an object x to the full subcategory of $\mathcal{T}/_x$ spanned by the morphisms in S preserves limits
(3) the functor $\mathcal{T}^{\mathrm{op}} \to \mathrm{An}$ that sends an object x to the full subcategory of $(\mathcal{T}/_x)^{\simeq}$ spanned by the morphisms in S preserves limits.

Remark 4.72. By virtue of the previous proposition we deduce that $\mathcal{U}$-small morphisms in a tribe span a local class of morphisms in $\mathcal{T}_\infty$.

Theorem 4.73 (Essential uniqueness for tribes)**.** Let $\mathcal{T}$ be a tribe with an universe $\pi : \mathcal{U}_\bullet \twoheadrightarrow \mathcal{U}$. Let W be the class of homotopy equivalences in $\mathcal{T}$. Let $\mathcal{T}_\infty := \mathrm{N}(\mathcal{T})[\mathrm{W}^{-1}]$ be an ∞-topos. Then for any x in $\mathcal{T}$ there is a canonical equivalence:

$$\mathcal{T}_\infty(x, \mathcal{U}) \simeq \pi_\infty(\mathrm{WFib}/_x)^{\mathcal{U}}$$

Proof. The proof is analogous to the one of Theorem 4.60 using Theorem 4.69 to apply the mapping space criteria Proposition 4.44. □

Definition 4.74 ([CH25] Definition 2.47)**.** Let S be a local class of morphisms in a topos $\mathcal{T}$. A morphism $f_S : X_S \to Y_S$ classifies S if and only if for every object X in $\mathcal{T}$, the map

$$\mathcal{T}(X, X_S) \to (T/_X)^{\simeq}, \qquad g \mapsto g^*(f_S)$$

is a monomorphism of animae whose image is given by the full subanima spanned by the morphisms in S. In this situation we say that X_S classifies S.

Example 4.75. Let $\mathcal{T}$ be a tribe with a strict univalent universe $\mathcal{U}$, then $\mathcal{U}$ classifies the local class of $\mathcal{U}$-small morphisms in $\mathcal{T}_\infty$.

Corollary 4.76 ([CH25])**.** Let S be a local class in a topos $\mathcal{T}$. Then there exists a classifying object for S if and only if the full subcategory in $\mathcal{T}/_X$ spanned by S is small for every object X in $\mathcal{T}$.

Remark 4.77. While a strict univalent universe $\mathcal{U}$ is a classifying object for $\mathcal{U}$-small fibrations the converse is clearly false. It however work up to canonical equivalence.

4.6. **Essential uniqueness of Π and Σ.**

Discussion 4.78. Let $\mathcal{T}$ be a π-tribe. Let $A \twoheadrightarrow \Gamma$ be a fibration. Let $\mathcal{T}_\infty$ be the localization of $\mathcal{T}$ at its homotopy equivalences. Π and Σ are respectively the right and left adjoint of $- \times_\Gamma A : \mathcal{T}/_\Gamma \to \mathcal{T}/_A$. The aim of this subsection is to prove that these constructions descend to $\mathcal{T}_\infty$: $- \times_\Gamma A$ already induces the base change functor $\mathcal{T}_\infty/_\Gamma \to \mathcal{T}_\infty/_A$ since all pullbacks in T_∞ can be strictified to strict pullbacks in T. Adjoints of ∞-categories are unique up to a contractible space of choices so showing that Π induces a right adjoint $\mathcal{T}_\infty/_A \to \mathcal{T}_\infty/_\Gamma$ would prove that all choices of Π lead to a contractible space of choices. Dually for Σ.

Proposition 4.79. Let Let $\mathcal{T}$ be a π-tribe. Let $A \twoheadrightarrow \Gamma$ be a fibration. Let $\mathcal{T}_\infty$ be the localization of $\mathcal{T}$ at its homotopy equivalences. Then for any $X \twoheadrightarrow \Gamma$, for any $Y \twoheadrightarrow A$ in $\mathcal{T}$ there is a canonical equivalence of spaces:

$$(\mathcal{T}_\infty)/_\Gamma(X, \Pi_A Y) \simeq (\mathcal{T}_\infty)/_A(X \times_\Gamma A, Y)$$

natural in X and Y

Proof. The adjunction on $\mathcal{T}$ induces a natural map $(\mathcal{T}_\infty)/_\Gamma(X, \Pi_A Y) \to (\mathcal{T}_\infty)/_A(X \times_\Gamma A, Y)$. Let us denote with $\mathcal{T}(\Gamma)$ and $\mathcal{T}(A)$ respectively the full subcategories of $\mathcal{T}/_\Gamma$ and $\mathcal{T}/_A$ spanned by fibrations. Since T is a tribe, the localization of T at homotopy equivalences admits a right calculus of fractions (as in Remark 4.26):

$$(\mathcal{T}_\infty)/_\Gamma(X, \Pi_A Y) \simeq \mathrm{colim}_{z : \mathrm{W}(X)^{\mathrm{op}}} \mathcal{T}(\Gamma)(z, \Pi_A Y)$$

where $\mathrm{W}(X)$ is the subcategory of $\mathcal{T}/_X$ spanned by trivial fibrations. This equivalence is clearly compatible with the natural morphism induced by the adjunction. Thanks to the adjunction in T we have the following isomorphism:

$$\mathcal{T}(\Gamma)(z, \Pi_A Y) \cong \mathcal{T}(A)(z \times_\Gamma A, Y)$$

Let $W_A(X)$ be the subcategory of $T/_{A \times_\Gamma X}$ spanned by trivial fibrations. Notice that $- \times_\Gamma A : \mathrm{W}(X) \to \mathrm{W}_A(X)$ has a right adjoint and is therefore cofinal, this provides an equivalence:

$$\mathrm{colim}_{z : \mathrm{W}(X)^{\mathrm{op}}} \mathcal{T}(\Gamma)(z, \Pi_A Y) \simeq \mathrm{colim}_{z' : \mathrm{W}_A(X)^{\mathrm{op}}} \mathcal{T}(A)(z', Y)$$

Since $\mathcal{T}(A)$ admits a calculus of fractions at $\mathrm{W}_A(X)$ we deduce that the right-hand side is equivalent to $(\mathcal{T}_\infty)/_A(X \times_\Gamma A, Y)$. All these equivalences are compatible with the natural map constructed at the beginning:

$$\begin{array}{ccc} (\mathcal{T}_\infty)/_\Gamma(X, \Pi_A Y) & \longrightarrow & (\mathcal{T}_\infty)/_A(X \times_\Gamma A, Y) \\ \wr & & \wr \\ \mathrm{colim}_{z : \mathrm{W}(X)^{\mathrm{op}}} \mathcal{T}(\Gamma)(z, \Pi_A Y) & \simeq & \mathrm{colim}_{z' : \mathrm{W}_A(X)^{\mathrm{op}}} \mathcal{T}(A)(z', Y) \end{array}$$

We conclude that it is an equivalence. □

Remark 4.80. An analogous result holds for Σ by replacing the use of a right calculus of fractions with a left calculus of fractions.

Appendix A. Natural models for homotopy type theory and model indepence for tribes

Discussion A.1. We will provide a comparison between the definition of *natural model for homotopy type theory* in the sense of [Awo17] and the already discussed definition of π-tribe. We would also discuss the notion of a tribe with universe types in the first language.

A.1. Natural models for homotopy type theory and tribes.

Definition A.2 ([Awo17] Definition 1)**.** A natural model for homotopy type theory (natural model for HoTT) is a couple $(\mathcal{C}, p : \mathbb{F}_\bullet \to \mathbb{F})$ where $\mathcal{C}$ is a 1-category and p is a locally representable morphism of (possibly large) presheaves on $\mathcal{C}$.

Remark A.3. The previous definition is equivalent to a 1-category endowed with a class of morphisms, called fibrations, that are stable under pullbacks and compositions.

Notation A.4. Let $(\mathcal{C}, p : \mathbb{F}_\bullet \to \mathbb{F})$ be a natural model for HoTT. Let the following square be a pullback square:

$$\begin{array}{ccc} \mathbb{F}_\bullet \times_{\mathbb{F}} \mathbb{F}_\bullet & \longrightarrow & \mathbb{F}_\bullet \\ \downarrow & \lrcorner & \downarrow p \\ \mathbb{F}_\bullet & \xrightarrow[p]{} & \mathbb{F} \end{array}$$

We denote with pr any of the composites $\mathbb{F}_\bullet \times_{\mathbb{F}} \mathbb{F}_\bullet \to \mathbb{F}$.

Definition A.5. A natural model for HoTT with intentional identity types is a natural model for HoTT $(\mathcal{C}, p)$ such that there is a pullback of the form:

$$\begin{array}{ccc} \mathrm{Path} & \xrightarrow{\mathrm{i}} & \mathbb{F}_\bullet \\ {\scriptstyle (s,t)}\downarrow & \lrcorner & \downarrow p \\ \mathbb{F}_\bullet \times_{\mathbb{F}} \mathbb{F}_\bullet & \xrightarrow[\mathrm{Po}]{} & \mathbb{F} \end{array}$$

and a map Id as in the following diagram:

$$\begin{array}{ccc} \mathbb{F}_\bullet & \xrightarrow{\mathrm{Id}} & \mathrm{Path} \\ & {\scriptstyle \Delta}\searrow & \downarrow {\scriptstyle (s,t)} \\ & & \mathbb{F}_\bullet \times_{\mathbb{F}} \mathbb{F}_\bullet \end{array}$$

such that for any x in $\mathcal{C}$ $\mathrm{Id} \times id_{\mathcal{C}(-,x)}$ has the left lifting property against p.

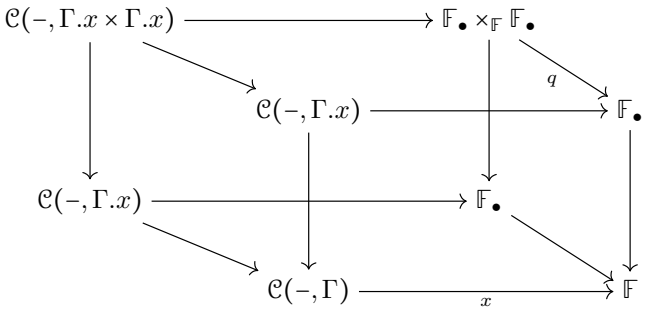
Proposition A.6 ([Awo17] Proposition 20)**.** Let $(\mathcal{C}, p : \mathbb{F}_\bullet \to \mathbb{F})$ be a natural model for HoTT with intentional identity types. Then it models the type theoretic rules for identity types.

Proposition A.7. Let $(\mathcal{C}, p : \mathbb{F}_\bullet \to \mathbb{F})$ be a natural model for HoTT with intentional identity types. Then, for any type $x : \Gamma$, there exists a type $\mathrm{Path}(\Gamma.x) : \Gamma.x.x$ that is a path object for $\Gamma.x$ when seen as an object of $\mathcal{C}$.

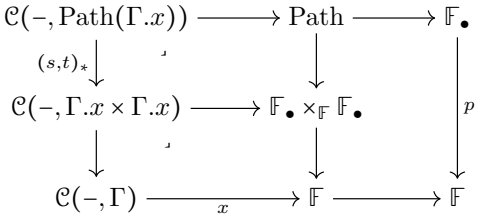
Proof. Let us first examine the following configuration of pullback squares:

$$\begin{array}{ccccc} (*) & \longrightarrow & \mathbb{F}_\bullet \times_{\mathbb{F}} \mathbb{F}_\bullet & \longrightarrow & \mathbb{F}_\bullet \\ \downarrow & \lrcorner & {\scriptstyle q}\downarrow & \lrcorner & \downarrow \\ & & \mathbb{F}_\bullet & \longrightarrow & \mathbb{F} \\ & & \downarrow p & & \\ \mathcal{C}(-,\Gamma) & \xrightarrow[x]{} & \mathbb{F} & & \end{array}$$

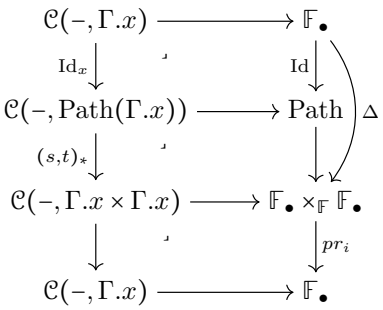
Since $(*)$ is obtained as the pullback of locally representable maps it must be of the form $\mathcal{C}(-,d)$ for some d in $\mathcal{C}$. By pullback pasting we obtain the following diagram whose faces are pullback squares:



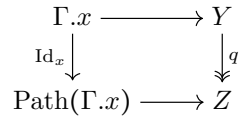
We notice that the initial object of the diagram must be $(*)$ by uniqueness of pullbacks. Let us now consider the following diagram where all squares are cartesian:



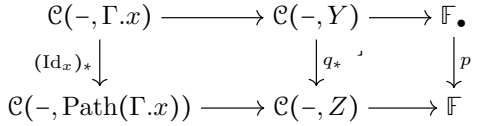
We claim that $\mathrm{Path}(\Gamma.x)$ is the path object of $\Gamma.x$. We prove that it factors $\Delta:\Gamma.x\to\Gamma.X\times\Gamma.x$. Let us consider the following diagram where all squares are cartesian:



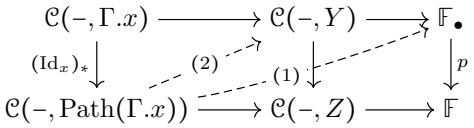
for $i=0,1$. Since the total right vertical map is the identity, so is the total vertical left map. By the universal property of the product this implies the claim. We are left to prove that Id_x has the left lifting property against fibrations. Let us consider a square in $\mathcal{C}$:



where q is a fibration. By definition of fibration we obtain a square of large presheaves on $\mathcal{C}$, as follows:



where the right square is cartesian. The axioms on Id_x provide us with a lift as (1) in the following diagram:

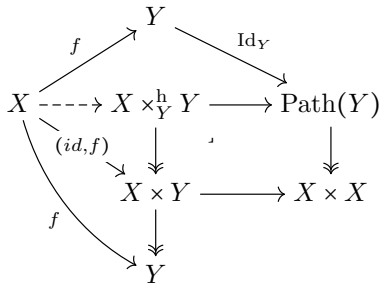


Since the right square is cartesian we obtain a lift as (2) due to the universal property of pullbacks. We conclude by the Yoneda lemma. □

Remark A.8. Let $x : \Gamma$ be a type in context Γ. We will denote with an uppercase letter the corresponding context $\Gamma.x = X$.

Lemma A.9. Let C be a natural model for HoTT with intentional identity types. Let $f : X \to Y$ be a morphism in $\mathcal{C}$. Then f can be factored in a fibration followed by a section of an equivalence fibration.

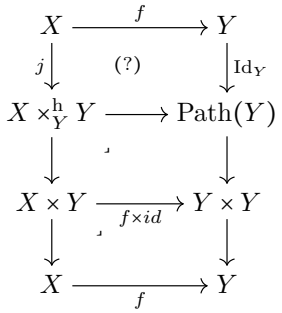
Proof. As it was already remarked in various sections, the notion of path object leads to a definition of homotopy pullback (as the one illustrated in section 2). We can therefore construct the following diagram:



which provides the desired factorization. □

Lemma A.10. Under the assumptions of the previous lemma, j is anodyne map.

Proof. Let us consider the following diagram:



the middle and bottom square are both pullbacks and so is the total diagram. By pullback pasting lemmas the top square is also a pullback square. Any base change of Id_Y is also a base change of $\mathbb{F}_\bullet \to \mathrm{Path}$ so it has the left lifting property against fibrations. □

Remark A.11. We notice that in the previous proof, the definition of $\mathbb{F}_\bullet \to \mathrm{Path}$ implies that anodyne maps are stable under base change. This proves that a natural model for HoTT with intentional identities and whose objects are fibrant is a tribe. We will therefore use the latter term for both.

Notation A.12. As done in the previous sections, we will omit the datum of $p : \mathbb{F}_\bullet \to \mathbb{F}$ in the context of tribes. To enhance natural models of HoTT to π-types we need to ask that all fibrations are exponentiable.

A.2. Π and Σ types in tribes.

Construction A.13 ([Awo17] Section 2)**.** Let $f : B \to A$ be an exponentiable morphism in a 1-category with finite limits $\mathcal{C}$. Let us denote with a the map $A \to *$. Define $b \coloneqq a \circ f$. Then we can define an endofunctor on $\mathcal{C}$ given by the following composite: $P_f \coloneqq a_! \circ f_* \circ b^*$. We also write $P_f(X) = \Sigma_{a:A} X^{B_a}$ and we call it *polynomial functor associated to* f.

Lemma A.14. There is a natural bijection between maps $g : Y \to \Sigma_{a:A} X^{B_a}$ and pairs of maps $(g_1 : Y \to A, Y \times_A B \to X)$ as indicated by the following diagram:

$$\begin{array}{ccccc} X & \xleftarrow{g_2} & Y \times_A B & \longrightarrow & B \\ & & \downarrow & \lrcorner & \downarrow f \\ & & Y & \xrightarrow[g_1]{} & A \end{array}$$

Proof. Let us compose g with the projection $\pi_1 : \Sigma_{a:A} X^{B_a} \to A$ to get $g_1 : Y \to A$. We can se $g : X \to f_* b^* X$ as a morphism in $\mathcal{C}/_A$. Under the adjunctions f_* and b^* we obtain a morphism $g_2 : b_! f^* Y \to X$ which is a morphism $Y \times_A B \to X$ in $\mathcal{C}$. This assignment is reveresible and natural in Y. □

Construction A.15. Let $\mathcal{C}$ be a π-tribe. Let us consider the polynomial functor associated to $p : \mathbb{F}_\bullet \to \mathbb{F}$ in $\mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathrm{Set})$ $\Sigma_{A:\mathbb{F}} X^A \coloneqq P_p(\mathcal{C}(-, X))$. The previous lemma implies that any morphism $\mathcal{C}(-, \Gamma) \to \Sigma_{A:\mathbb{F}} \mathbb{F}^A$ corresponds to a diagram of the form:

$$\begin{array}{ccccc} \mathbb{F} & \xleftarrow{B} & \mathcal{C}(-, \Gamma.A) & \longrightarrow & \mathbb{F}_\bullet \\ & & \downarrow & \lrcorner & \downarrow p \\ & & \mathcal{C}(-, \Gamma) & \xrightarrow[A]{} & \mathbb{F} \end{array}$$

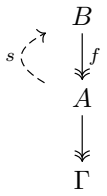
Since p is locally representable we can elaborate further:

$$\begin{array}{ccccc} \mathbb{F}_\bullet & \longleftarrow & \mathcal{C}(-, B) & & \\ {\scriptstyle p}\downarrow & \llcorner & \downarrow & & \\ \mathbb{F} & \xleftarrow{B} & \mathcal{C}(-, A) & \longrightarrow & \mathbb{F}_\bullet \\ & & \downarrow & \lrcorner & \downarrow p \\ & & \mathcal{C}(-, \Gamma) & \xrightarrow[A]{} & \mathbb{F} \end{array}$$

This is equivalent to the followng diagram in $\mathcal{C}$ where all arrows are fibrations:

$$\begin{array}{c} B \\ \downarrow \\ A \\ \downarrow \\ \Gamma \end{array}$$

Remark A.16. Analogously, $\Sigma_{A:\mathbb{F}}(\mathbb{F}_\bullet)^A$ represents diagrams in $\mathcal{C}$ of the form:



where all vertical maps are fibrations and s is a section of f. By functoriality of P_p there is a canonical map $P_p(p) : \Sigma_{A:\mathbb{F}}(\mathbb{F}_\bullet)^A \to \Sigma_{A:\mathbb{F}}\mathbb{F}^A$ which, on diagrams, corresponds to forgetting the section.

Proposition A.17 ([Awo17] Proposition 8)**.** Let $\mathcal{C}$ be a natural model for HoTT. Let $P_p(p)$ the map defined above. Then the type theoretic rules for Π-types are modeled by maps:

$$\Sigma_{A:\mathbb{F}}(\mathbb{F}_\bullet)^A \xrightarrow{\lambda} \mathbb{F}_\bullet$$

$$\Sigma_{A:\mathbb{F}}(\mathbb{F})^A \xrightarrow[\Pi]{} \mathbb{F}$$

such that the following square is a pullback:

$$\begin{array}{ccc} \Sigma_{A:\mathbb{F}}(\mathbb{F}_\bullet)^A & \xrightarrow{\lambda} & \mathbb{F}_\bullet \\ {\scriptstyle P_p(p)}\downarrow & \lrcorner & \downarrow{\scriptstyle p} \\ \Sigma_{A:\mathbb{F}}(\mathbb{F})^A & \xrightarrow[\Pi]{} & \mathbb{F} \end{array}$$

Construction A.18. To model Σ-types we need to replace $\Sigma_{A:\mathbb{F}}\mathbb{F}_\bullet^A$ with a different presheaf since Σ-types have a diferent introduction rule. Let $\mathcal{C}$ be a π-tribe. Let $\hat{\mathcal{C}} := \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathrm{Set})$. We construct $\Sigma_{A:\mathbb{F}}\Sigma_{B:\mathbb{F}^A}\Sigma_{a:A}B(a)$ as follows: $\Sigma_{A:\mathbb{F}}\mathbb{F}^A$ gives a map to $\mathbb{F}$ by construction. If we pull it back along p we obtain the following cartesian square:

$$\begin{array}{ccc} \Sigma_{(A,B):\Sigma_{A:\mathbb{F}}\mathbb{F}^A} A & \longrightarrow & \mathbb{F}_\bullet \\ \downarrow & \lrcorner & \downarrow{\scriptstyle p} \\ \Sigma_{A:\mathbb{F}}\mathbb{F}^A & \xrightarrow[\mathrm{can}]{} & \mathbb{F} \end{array}$$

We claim that for any presheaf X there exists is a map $\mathrm{ev}_X : P_p(X) \times_{\mathbb{F}} \mathbb{F}_\bullet \to X$ natural in X. Denote $\mathbb{F} \to *$ with f and $\mathbb{F}_\bullet \to *$ with $f_\bullet$. $P_p(X) = f_! p_* f_\bullet^* X$ can be seen as an object $p_* f_\bullet^* X \to \mathbb{F}$ in $\mathcal{C}/_{\mathbb{F}}$. So the co-unit of $p^* \dashv p_*$ gives a map $\epsilon_{\mathbb{F}_\bullet \times X} : \mathbb{F}_\bullet \times_{\mathbb{F}} p_* f_\bullet^*(X) = p^* p_*(X \times_{\mathbb{F}} \mathbb{F}_\bullet) \to X \times_{\mathbb{F}} \mathbb{F}_\bullet$. By post composition with the projection to X we obtain the desired map. If we pull back $\mathrm{ev}_{\mathbb{F}}$ along p we obtain $\Sigma_{A:\mathbb{F}}\Sigma_{B:\mathbb{F}^A}\Sigma_{a:A}B(a)$ as in the following diagram:

$$\begin{array}{ccc} \Sigma_{A:\mathbb{F}}\Sigma_{B:\mathbb{F}^A}\Sigma_{a:A}B(a) & \longrightarrow & \mathbb{F}_\bullet \\ \downarrow & \lrcorner & \downarrow{\scriptstyle p} \\ \Sigma_{(A,B):\Sigma_{A:\mathbb{F}}\mathbb{F}^A} A & \xrightarrow[\mathrm{ev}_{\mathbb{F}}]{} & \mathbb{F} \\ \downarrow & & \\ \Sigma_{A:\mathbb{F}}\mathbb{F}^A & & \end{array}$$

Proposition A.19 ([Awo17] Proposition 12)**.** Let $\mathcal{C}$ be a natural model for HoTT. Let $P_p(p)$ the map defined above. Then the type theoretic rules for Σ-types are modeled by maps:

$$\Sigma_{A:\mathbb{F}}\Sigma_{B:\mathbb{F}^A}\Sigma_{a:A}B(a) \xrightarrow{\text{pair}} \mathbb{F}_\bullet$$

$$\Sigma_{A:\mathbb{F}}(\mathbb{F})^A \xrightarrow[\Sigma]{} \mathbb{F}$$

such that the following square is a pullback:

$$\begin{array}{ccc} \Sigma_{A:\mathbb{F}}\Sigma_{B:\mathbb{F}^A}\Sigma_{a:A}B(a) & \xrightarrow{\text{pair}} & \mathbb{F}_\bullet \\ {\scriptstyle \text{proj}}\downarrow & \lrcorner & \downarrow{\scriptstyle p} \\ \Sigma_{A:\mathbb{F}}(\mathbb{F})^A & \xrightarrow[\Sigma]{} & \mathbb{F} \end{array}$$

A.3. **Tribes with universes.**

Definition A.20. Let $\mathcal{C}$ be a π-tribe. We say that $\mathcal{C}$ is a tribe with universes (or that it satisfies the axiom of univalence) if any map $\mathcal{C}(-,X) \to \mathbb{F}$ factors through a $\mathcal{C}(-,\mathcal{U})$ for a certain $\mathcal{U}$ in $\mathcal{C}$ nwith the following properties:

(1) Let $\pi : \mathcal{U}_\bullet \twoheadrightarrow \mathcal{U}$ be the fibation associated with $\mathcal{C}(-,\mathcal{U}) \to \mathbb{F}$ then π is univalent.
(2) $\mathcal{U}$-small morphisms are closed under composition, Σ-types (when they exist), Π-types and identity-types.

We say that fibration $\pi : \mathcal{U}_\bullet \twoheadrightarrow \mathcal{U}$ is a universe if it is univalent and satisfiest the property (3).

Remark A.21. Let us denote with $\mathcal{C}_\infty$ the localization of $\mathcal{C}$ at homotopy equivalences. Since every morphism is small relatively to at least one universe, the family of categories $\pi_\infty(\mathrm{WFib}/_X)^{\mathcal{U}} f \simeq (\mathcal{C}_\infty/_X)^{\simeq}_{\mathcal{U}} \hookrightarrow (\mathcal{C}_\infty/_X)^{\simeq}$ defined before is a cofinal chain of subcategories of $(\mathcal{C}_\infty/_X)^{\simeq}$.

Question A.22. A complication to adapt Theorem 4.61 to this generality arises because the notion of being $\mathcal{U}$-small is intrinsic of the model $\mathcal{C}$ while being λ-compact in the localized topos is not. We propose a possible solution.

Solution A.23. Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be π-tribes that present the same ∞-category $\mathcal{C}_\infty$. Let $\mathcal{U}_1^\bullet \twoheadrightarrow \mathcal{U}_1$ be a universe for $\mathcal{C}_1$. Then there exists a universe in $\mathcal{C}_2$ of the form $\overline{\pi} : \overline{\mathcal{U}_1^\bullet} \twoheadrightarrow \mathcal{U}_1$.

Proof. We illustrate what has been shown yet. By Remark A.21 there exists an universe $\mathcal{U}_2$ of $\mathcal{C}_2$ such that $(\mathcal{C}_\infty/_X)^{\simeq}_{\mathcal{U}_1} \hookrightarrow (\mathcal{C}_\infty/_X)^{\simeq}_{\mathcal{U}_2}$. By Lemma 4.49 we can construct the following diagram:

$$\begin{array}{ccccc} \mathcal{C}_\infty(X,\mathcal{U}_1) & \xrightarrow{\sim} & \pi_\infty(\mathrm{WFib}/_X)^{\mathcal{U}_1} & \xrightarrow{\sim} & (\mathcal{C}_\infty/_X)^{\simeq}_{\mathcal{U}_1} \\ \downarrow & & \downarrow & & \downarrow \\ \mathcal{C}_\infty(X,\mathcal{U}_2) & \xrightarrow{\sim} & \pi_\infty(\mathrm{WFib}/_X)^{\mathcal{U}_2} & \xrightarrow{\sim} & (\mathcal{C}_\infty/_X)^{\simeq}_{\mathcal{U}_2} \end{array}$$

If we consider $X = \mathcal{U}_1$ and we take $id_{\mathcal{U}_1}$, this defines a $\mathcal{U}_2$-small fibration $\overline{\pi} : \overline{\mathcal{U}_1^\bullet} \twoheadrightarrow \mathcal{U}_1$ in $\mathcal{C}_2$. Since $\mathcal{U}_2$ is strict univalent this defines, up to homotopy, a map $\mathcal{U}_1 \to \mathcal{U}_2$ in $\mathcal{C}_2$. Such a map represents the leftmost vertical map in the previous diagram. f is a monomorphism since under $\mathcal{C}(X,-)$ it corresponds to the inclusion $(\mathcal{C}_\infty)^{\simeq}_{\mathcal{U}_1} \hookrightarrow (\mathcal{C}_\infty)^{\simeq}_{\mathcal{U}_2}$. By Theorem 2.70 f being an homotopy embedding is equivalent to $\overline{\pi} : \overline{\mathcal{U}_1} \twoheadrightarrow \mathcal{U}$ being univalent. We believe that this condition is equivalent to f being an homotopy emmbedding in $\mathcal{C}_2$. What needs to be done is to prove that $\overline{\pi}$ satisfies condition (3) of Definition A.20. □

Acknowledgments

I would like to thank some people that have been important for the development of my thesis and for my years as a master student.
I thank my family for the support I received (both economical and not) that has allowed me to study in Regensburg as an ALGANT student. I thank my advisor, Denis-Charles Cisinski for revitalizing my interest in mathematics with unique courses and for giving me a topic that piqued my interest for the thesis. I thank him also for the sharp insights that have guided me while gathering my ideas for the main proofs.
I thank my friends from Milan for following my same path through the study sessions and have thought me how to live like a self-sustaining human being. I thank Frank and Rishab for bringing a joyful atmosphere to the life in Regensburg (and for the respect towards my sister). I thank Andrea for being a motivating figure that pushed me to pursue my mathematic interests with more dedication. I thank Francesca for the crazy nights in Milan that have shaped my memories of home. I thank Davide for helping me correct the grammar of this paper and give it a bit of his style.

References

[AP26] Daniël Apol and Maximilian Petrowitsch. Elementary ∞-toposes from type theory, 2026. 14, 17, 20

[Awo17] Steve Awodey. Natural models of homotopy type theory, 2017. 4, 62, 65, 66, 67

[CCNW26] Denis-Charles Cisinski, Bastiaan Cnossen, Kim Nguyen, and Tashi Walde. Formalization of higher categories. Lecture notes, Universität Regensburg, 2026. 42, 43, 45

[CH25] Bastiaan Cnossen and Marc Hoyois. Higher topos theory, 2025. Lecture notes based on a course by Marc Hoyois at the University of Regensburg. 60, 61

[Cis02] Denis-Charles Cisinski. Théories homotopiques dans les topos. *Journal of Pure and Applied Algebra*, 174(1):43–82, 2002. 7, 11, 12

[Cis14] Denis-Charles Cisinski. Univalent universes for elegant models of homotopy types, 2014. 5, 36, 37

[Cis19] Denis-Charles Cisinski. *Higher Categories and Homotopical Algebra*, volume 180 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, 2019. 3, 5, 11, 13, 42, 44, 45, 46, 53, 54

[Gir71] Jean Giraud. *Cohomologie non abélienne*, volume 179 of *Die Grundlehren der mathematischen Wissenschaften*. Springer-Verlag, Berlin, Heidelberg, New York, 1971. 3, 40

[GK17] David Gepner and Joachim Kock. Univalence in locally cartesian closed infinity-categories, 2017. 4, 48, 49, 50, 52

[Joy17] Andre Joyal. Notes on clans and tribes, 2017. 14

[KLV18] Chris Kapulkin, Peter LeFanu Lumsdaine, and Vladimir Voevodsky. Univalence in simplicial sets, 2018. 35, 36, 37

[Lur03] Jacob Lurie. On infinity topoi, 2003. 56

[nLa26a] nLab authors. combinatorial model category. https://ncatlab.org/nlab/show/combinatorial+model+category, July 2026. Revision 84. 5

[nLa26b] nLab authors. left Bousfield localization of model categories. https://ncatlab.org/nlab/show/left+Bousfield+localization+of+model+categories, July 2026. Revision 75. 50

[nLa26c] nLab authors. Quillen equivalence. https://ncatlab.org/nlab/show/Quillen+equivalence, July 2026. Revision 39. 47

[nLa26d] nLab authors. type-theoretic model category. https://ncatlab.org/nlab/show/type-theoretic+model+category, July 2026. Revision 23. 48

[Qui67] Daniel Gray Quillen. *Homotopical algebra*. Lecture Notes in Mathematics, No. 43. Springer-Verlag, Berlin, 1967. 13

[Rie14] Emily Riehl. *Categorical Homotopy Theory*, volume 24 of *New Mathematical Monographs*. Cambridge University Press, 2014. 5

[Shu15] Michael Shulman. The univalence axiom for elegant reedy presheaves, 2015. 35, 36

[Shu19] Michael Shulman. All $(\infty, 1)$-toposes have strict univalent universes, 2019. 2, 3, 21, 23, 28, 29, 31, 35